Department of Physics & Electronics · Institute of Excellence in Higher Education, Bhopal

Lecture Notes in Physics · B.Sc. II Semester

Thermodynamics

Concise treatment of Thermodynamics

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A Note to the Reader

These notes have one purpose: to make thermodynamics feel like a *conversation with nature*, not a list of formulae to memorise. Every derivation is preceded by a short motivation, so the mathematics arrives after the idea — never before it.

The treatment covers the full B.Sc. II syllabus without unnecessary detour. Read these as a map; keep a good textbook beside you (*Garg, Bansal & Ghosh* works well) for context and depth. Every algebraic step is written out in full — slow down wherever it feels unclear, and skim freely wherever it feels obvious.

If you spot a misprint or an explanation that could be sharper, do write in at `am.phy.in@gmail.com`. Student feedback is the only way these notes improve.

— *Amit Vishwakarma*

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Contents

CHAPTER 1 **The Zeroth Law of Thermodynamics** 12

Statement 12

1.1.1 Thermal Walls: Diathermic and Adiabatic 12

1.1.2 Schematic Illustration 13

Mathematical Formulation 13

1.2.1 Equilibrium Constraints 13

1.2.2 The Cancellation Argument 14

1.2.3 The Empirical Temperature 14

An Illustrative Example: The Paramagnetic van der Waals Gas 14

1.3.1 Setup 14

1.3.2 The Equilibrium Condition 15

1.3.3 Identifying the Empirical Temperature 15

CHAPTER 2 **The Ideal Gas** 16

The Empirical Gas Laws 16

The Ideal Gas Equation of State 17

2.2.1 Microscopic Form: Boltzmann's Constant 18

Work Done by an Ideal Gas 18

2.3.1 Setting Up: The Piston-Cylinder System 18

2.3.2 Work in a Finite Process 20

Kinetic Theory of Ideal Gases 22

2.4.1 Setup and Assumptions 22

2.4.2 Pressure from Molecular Collisions 23

2.4.3 Temperature as a Measure of Kinetic Energy 24

The Equipartition of Energy 24

2.5.1 Consistency with the Kinetic Theory Result 24

2.5.2 Degrees of Freedom by Molecule Type 25

The Maxwell–Boltzmann Velocity Distribution 26

2.6.1 The Boltzmann Factor 26

2.6.2 Distribution of One Velocity Component 26

2.6.3 The Three-Dimensional Velocity Distribution 27

2.6.4 The Speed Distribution $F(v)$ 27

2.6.5 Characteristic Speeds 28

2.6.6 Momentum and Energy Distributions 29

Transport Phenomena 30
2.7.1 The Mean Free Path 30
2.7.2 The Three Transport Coefficients 32

CHAPTER 3 **Real Gases** 36
Andrews' Experiment on CO_2 36
Van der Waals Gas Equation 37
3.2.1 Correction for Finite Molecular Volume 37
3.2.2 Correction for Intermolecular Attraction 38
Critical Constants of the van der Waals Gas 39
3.3.1 The Critical Point — What We Are Looking For 39
3.3.2 The Mathematical Conditions at the Critical Point 40
3.3.3 The van der Waals Equation 40
3.3.4 The Critical Compressibility Factor 42
3.3.5 Reducing the van der Waals Equation 43
Characteristic Temperatures of a vdW gas 43
Virial Equation of State 46

CHAPTER 4 **First Law of Thermodynamics** 47
Point Function and Path Function 47
Internal Energy: The 1st Thermodynamic Potential 48
4.2.1 Joule Expansion 49
The First Law of Thermodynamics 49
4.3.1 Mechanical (pressure–volume) work 49
4.3.2 Generalized work 50
4.3.3 Chemical work and chemical potential 50
Applications of the First Law of Thermodynamics 51
4.4.1 Heat Capacity (C) 51
4.4.2 Adiabatic Process: Derivation of $PV^\gamma = \text{constant}$ 53
4.4.3 Relation Between Adiabatic and Isothermal Elasticity 54

CHAPTER 5 **Second Law of Thermodynamics** 55
Heat Engine 55
5.1.1 Working Principle 55
5.1.2 Energy Balance — First Law Applied to a Cycle 56
5.1.3 Thermal Efficiency 56
Refrigerator 56
5.2.1 Working Principle 56
5.2.2 Energy Balance 57
5.2.3 Coefficient of Performance 57
Second Law of Thermodynamics 57
5.3.1 Equivalence of the Two Statements 58
Reversible and Irreversible Processes 59
5.4.1 Reversible Process 60
5.4.2 Irreversible Process 60

Carnot Engine 61
5.5.1 Carnot's Theorem 62
5.5.2 Thermodynamic Temperature Scale 63
5.5.3 Unattainability of Absolute Zero 65
The Carnot Cycle 65
5.6.1 Net Work Done 67
5.6.2 Efficiency of the Carnot Cycle 67
Otto Cycle (Petrol Engine) 68
5.7.1 Net Work Done 69
5.7.2 Efficiency of the Otto Cycle 69
Diesel Cycle (Diesel Engine) 70
5.8.1 Efficiency of the Diesel Cycle 71
5.8.2 Comparison: Otto vs. Diesel Cycle 71

CHAPTER 6 **Entropy** 73
Clausius Theorem 73
Consequences of Clausius Theorem 74
6.2.1 Reversible Cyclic Process 74
6.2.2 Irreversible Cyclic Process 74
6.2.3 Path Independence for Reversible Processes 75
Entropy 75
6.3.1 Definition of Entropy 75
Entropy Change for Irreversible Processes 75
Entropy Change for an Isolated System 76
6.5.1 Principle of Increase of Entropy 77
6.5.2 Reversible and Irreversible Processes 77
6.5.3 Entropy and Equilibrium 77
First Law Revisited: Entropy Representation 78
6.6.1 Significance of the Entropy Formulation 78

CHAPTER 7 **Thermodynamic Potentials** 79
The Fundamental Relation 79
7.1.1 Extensivity and the Euler Relation 79
7.1.2 The Gibbs–Duhem Relation 80
7.1.3 The Legendre Transform: Changing Natural Variables 80
Internal Energy U 80
7.2.1 Definition and Physical Content 80
7.2.2 First Maxwell Relation 81
Enthalpy H 82
7.3.1 Motivation and Definition 82
7.3.2 Enthalpy and Heat at Constant Pressure 82
7.3.3 Second Maxwell Relation 82
Helmholtz Free Energy F 83
7.4.1 Motivation and Definition 83
7.4.2 Third Maxwell Relation 83

Gibbs Free Energy G 85
7.5.1 Motivation and Definition 85
7.5.2 The Work Interpretation 85
7.5.3 Fourth Maxwell Relation 85
The Maxwell Relations: Unified Summary 87

CHAPTER 8 **Thermodynamic relations** 88
The Gibbs–Helmholtz Equation 88
$C_P - C_V$ Relations 89
The $T\,dS$ Equations 89
8.3.1 Entropy as a Function of T and V 90
8.3.2 Entropy as a Function of T and P 90
8.3.3 Entropy as a Function of P and V 91
Thermodynamic Stability Conditions 91
Clapeyron Equation 92
8.5.1 Clausius–Clapeyron Equation (Liquid–Vapour Transition) . . . 93
Joule–Thomson Expansion 94
8.6.1 Isoenthalpic Nature of Joule–Thomson Expansion 94
8.6.2 Joule–Thomson Coefficient 95
8.6.3 Ideal Gas Case 96
8.6.4 Joule–Thomson Coefficient for van der Waals 96
Cooling by Adiabatic Demagnetization 97
8.7.1 The Quest for Ultralow Temperatures 97
8.7.2 Thermodynamics of a Magnetic System 98
8.7.3 The Adiabatic Condition 99
8.7.4 Curie's Law and the Cooling Formula 100
8.7.5 The S–T Diagram 101
8.7.6 Practical Aspects and Limitations 101

CHAPTER 9 **The Third Law of Thermodynamics** 103
The Need for an Absolute Entropy Reference 103
Statements of the Third Law 103
9.2.1 Planck's Statement 104
9.2.2 Nernst Heat Theorem (1906) 104
9.2.3 The Unattainability Principle (Simon, 1927) 104
Heat Capacity Must Vanish at Absolute Zero 105
Debye Theory and the Low-Temperature Behaviour of Solids 105
9.4.1 The Debye Model 105
9.4.2 TikZ: Debye Heat Capacity 106
Residual Entropy and Frozen Disorder 106
9.5.1 Ice: A Classic Example (Pauling, 1935) 107
9.5.2 Carbon Monoxide: Orientational Disorder 107
Adiabatic Demagnetisation and Reaching $T \approx 0$ 107
9.6.1 Entropy of a Spin System 107
9.6.2 The Demagnetisation Cycle 108

Third Law and Chemical Equilibrium 109
9.7.1 The van 't Hoff Equation and the Integration Constant 109
Third-Law Entropy and Spectroscopic Entropy 109
Behaviour of Other Thermodynamic Quantities at $T = 0$ 110
9.9.1 Thermal Expansion Vanishes 110
9.9.2 Thermoelectric Effects Vanish 110
9.9.3 Isothermal Compressibility 110
The Third Law in Quantum Field Theory and Cosmology 111
9.10.1 Black Hole Thermodynamics 111
9.10.2 The Cosmological Significance of the Third Law 111
Summary 111

CHAPTER 10 **Calorimetry and Thermometry** 113
Fourier's Law of Heat Conduction 113
10.1.1 Vector Form 113
10.1.2 The Heat Equation 113
10.1.3 Thermal Resistance 114
Specific Heat and Calorimetry 115
10.2.1 Water Equivalent 115
10.2.2 Bomb Calorimetry 115
Searle's Method: Thermal Conductivity of Good Conductors 115
10.3.1 Apparatus and Principle 115
10.3.2 Derivation 116
Lee's Disc Method: Poor Conductors 116
Newton's Law of Cooling and its Limitations 116
10.5.1 Solution and Exponential Decay 117
10.5.2 Stefan–Boltzmann Correction at High Temperatures 117
Thermal Waves and Periodic Heating 117
Thermometers: Principles and Types 118
10.7.1 Liquid-in-Glass Thermometers 118
10.7.2 Constant-Volume Gas Thermometer (CVGT) 118
Electrical Thermometry 118
10.8.1 Platinum Resistance Thermometer (PRT) 118
10.8.2 Thermistors 119
10.8.3 Noise Thermometry (Johnson–Nyquist) 120
The Seebeck Effect and Thermocouples 120
10.9.1 Seebeck Coefficient 120
10.9.2 Microscopic Origin: Mott Formula 120
10.9.3 Standard Thermocouple Types 121
The Peltier Effect 121
The Thomson Effect 122
10.11.1 Kelvin's Relations 122
Figure of Merit and Thermoelectric Efficiency 122
Onsager Reciprocity and the Unified Transport Theory 123
10.13.1 Generalised Forces and Fluxes 123

10.13.2 Thermoelectric Transport Matrix 123
10.13.3 Physical Interpretation 124
The International Temperature Scale (ITS-90) 124
Summary 125

CHAPTER 11 **Black Body Radiation** 126
Basic Definitions 126
Conservation of Radiant Energy 127
Kirchhoff's Law of Thermal Radiation 127
11.3.1 Statement of Kirchhoff's Law 128
11.3.2 Implication of the Law 128
11.3.3 Proof 128
11.3.4 Black Body 129
Wien's Displacement Law 130
Stefan–Boltzmann Law 130
Rayleigh–Jeans Law 131
11.6.1 Derivation 132
11.6.2 Ultraviolet Catastrophe 132
Planck's Radiation Law 132
11.7.1 Planck's Quantum Hypothesis 133
11.7.2 Average Energy of an Oscillator 133
11.7.3 Planck's Radiation Law 134
11.7.4 Derivation of Classical Laws from Planck's Radiation Law . . . 134

CHAPTER A **The Polytropic Process** 137

CHAPTER B **Equations of State** 138
Virial Equation 138
van der Waals Equation 139
Berthelot Equation 140
Clausius Equation 140
Redlich–Kwong Equation 141
Soave–Redlich–Kwong Equation 141
Peng–Robinson Equation 142
Hard-Sphere Equation 142
Benedict–Webb–Rubin Equation 143
Radiation Equation of State 143
Degenerate Fermi Gas 144
Bose Gas Equation of State 144

CHAPTER C **Spherical Coordinate System** 146

List of Figures

1.1 Left: A and B are each in equilibrium with C, but separated by an adiabatic wall. Right: the zeroth law guarantees that A and B are also in mutual equilibrium once brought into diathermic contact. 13

1.2 A paramagnetic van der Waals gas: the mechanical and magnetic degrees of freedom define two subsystems that must share a common empirical temperature. 15

2.1 Piston–cylinder system. A gas at pressure P pushes a frictionless piston of cross-sectional area A. An outward displacement dl produces a volume change $dV = A\,dl$ and work $\delta W = P\,dV$. 19

2.2 A gas being compressed by an external force F applied to a piston of area A. An outward displacement dl changes the volume by $dV = A\,dl$. 19

2.3 P–V diagrams for the six standard thermodynamic processes. The shaded areas equal the work done *by* the gas in each case. The dashed curve in (f) indicates that free expansion is not quasi-static and has no well-defined path on the P–V diagram. 20

2.4 A single molecule of mass m bouncing back and forth in a cubical box of side L. The x-component of velocity reverses at each wall collision. The momentum transferred to the right wall per collision is $2mv_x$, and the round-trip time is $2L/v_x$. 22

2.5 Degrees of freedom and corresponding average energies per molecule from the equipartition theorem. Each quadratic degree of freedom contributes $\frac{1}{2}k_BT$. The diatomic molecule has five active modes at moderate temperatures (three translational, two rotational) and seven at high temperatures when vibration is activated. 25

2.6 Maxwell–Boltzmann speed distribution $F(v)$ at three temperatures. As temperature rises, the distribution broadens and its peak shifts to higher speeds. The three characteristic speeds v_{mp}, $\langle v\rangle$, and v_{rms} (shown for the medium-T curve) satisfy $v_{mp} < \langle v\rangle < v_{rms}$. 28

2.7 Collision cylinder used to derive the mean free path. A molecule of diameter d moving with speed v sweeps out a cylindrical volume $\pi d^2 v\,dt$ in time dt. Any molecule whose centre lies within this cylinder suffers a collision. The collision frequency is $f_{\text{coll}} = \sqrt{2}\,n\pi d^2\langle v\rangle$ after correcting for the relative motion of target molecules. 30

2.8 Survival equation: the fraction of molecules that travel a distance x without collision decays exponentially with decay length λ (the mean free path). At $x = \lambda$, a fraction $1/e \approx 37\%$ of molecules remain uncollided. 32

2.9 Molecular mechanism of viscosity. In a gas with a velocity gradient du/dz, molecules crossing an imaginary horizontal plane at z arrive from one mean free path above $(z + \lambda)$ or below $(z - \lambda)$. They carry more (or less) x-momentum than the local average, creating a net shear stress $\tau = \eta\, du/dz$. 33

3.1 Andrew's Isotherms of CO_2 36

5.1 Schematic of an idealised heat engine operating between a hot reservoir at T_H and a cold reservoir at T_C. 56
5.2 Schematic of an idealised refrigerator. Work W is supplied to extract heat Q_C from the cold reservoir and deliver Q_H to the hot reservoir. . . 57
5.3 A Clausius-violating engine combined with a normal heat engine produces a Kelvin–Planck-violating engine. 58
5.4 A Kelvin–Planck-violating engine driving a refrigerator produces a Clausius-violating device. 59
5.5 A Carnot engine operating between hot reservoir T_H and cold reservoir T_C. 62
5.6 Proof of Carnot's theorem: an arbitrary engine $\overline{\text{CE}}$ (acting as a heat engine) combined with a Carnot engine CE (running in reverse as a refrigerator). 62
5.7 Three Carnot engines connected in series to derive the functional form of Carnot efficiency. 64

6.1 Illustration for Clausius theorem using a Carnot engine 73

8.1 The porous experiment 94
8.2 Entropy–temperature diagram for adiabatic demagnetization. The upper curve $(H = 0)$ lies above the lower curve $(H = H_i)$ because entropy is higher when the field is absent (disordered spins). **Step 1** $(A \to B)$: isothermal magnetisation at $T = T_i$ reduces entropy as heat flows to the bath. **Step 3** $(B \to C)$: adiabatic demagnetisation at constant entropy S_B moves the system left to a lower temperature T_f. Repeating the cycle reaches progressively lower temperatures. 101

9.1 Debye heat capacity as a function of reduced temperature T/Θ_D. At low T the curve follows the T^3 law (dashed); at high T it saturates at the classical Dulong–Petit value $3Nk_B$ (dotted). 106
9.2 Entropy–temperature diagram for adiabatic demagnetisation. Step **1** $(A \to B)$: isothermal magnetisation reduces entropy at constant T_1 (field switched on). Step **2** $(B \to C)$: adiabatic demagnetisation reduces temperature to $T_2 \ll T_1$ at constant entropy (field switched off). Repeated cycles bring the system progressively closer to $T = 0$ but never reach it. 108

10.1 Composite wall: three layers in series. Each layer has its own conductivity k_i and thermal resistance $R_i = L_i/k_iA$. 114
10.2 Searle's apparatus. Thermocouples at positions $T_1, \ldots, T_4$ monitor the rod; inlet and outlet water temperatures θ_1, θ_2 measure heat removal. . 115
10.3 Lee's disc apparatus for poor conductors. The bad-conductor sample is sandwiched between the steam chest and the brass disc. 116

10.4 Exponential decay of excess temperature under Newton's law of cooling. The time constant $\tau = 1/K$ is the e-folding time. 117
10.5 Four-wire resistance measurement. Current is supplied through the outer pair; voltage is sensed through the inner pair with negligible current, so lead resistances do not appear in the measurement. 119
10.6 Resistance vs temperature for an NTC thermistor ($B = 3950\,\mathrm{K}$, $R_0 = 10\,\mathrm{k\Omega}$ at $25\,^\circ\mathrm{C}$) compared with a Pt100 PRT (scaled for visibility). The thermistor's exponential sensitivity is immediately apparent. 119
10.7 Thermocouple circuit. The net emf depends on both the material combination and the temperature difference between junctions. 120
10.8 The four transport laws as manifestations of the single Onsager matrix. The reciprocity $L_{12} = L_{21}$ (centre) is equivalent to Kelvin's relation $\Pi = ST$. 124

List of Tables

2.1 The three primary transport phenomena. 32

4.1 Examples of generalized work terms 50

7.1 The four thermodynamic potentials, natural variables, and differentials (fixed N implied for brevity in the last column). 81

7.2 The four Maxwell relations. 87

8.1 Cooling techniques and the temperature ranges they access. 97

A.1 Polytropic process 137

Chapter 1

The Zeroth Law of Thermodynamics

Before we state the zeroth law, it is worth pausing to ask a deceptively simple question: *what does it mean for two systems to be in thermal equilibrium?*

When two systems are brought into physical contact — say, a hot block of metal placed on a cold one — we observe that their measurable properties (pressure, volume, magnetisation, ...) change with time, and then, after a sufficiently long wait, they stop changing. We say the combined system has reached **thermal equilibrium**. The zeroth law makes a profound assertion about this state.

1.1 Statement

Zeroth Law If two systems A and B are each separately in thermal equilibrium with a third system C, then A and B are also in thermal equilibrium with each other.
The name "zeroth" is historically curious: this law was recognised *after* the first and second laws had already been formulated and numbered, yet it is logically prior to both. Hence it was retroactively assigned the number zero.

The content of the law may seem almost trivially obvious, but its power lies in what it *implies*: the existence of a well-defined state function — the **empirical temperature** Θ — that takes the same value for any two systems in mutual equilibrium. In other words, the zeroth law is the logical foundation upon which the very concept of temperature rests.

1.1.1 Thermal Walls: Diathermic and Adiabatic

The zeroth law implicitly introduces two idealised types of boundary:

- **Diathermic wall:** A boundary that allows thermal interaction between systems. Two systems separated by a diathermic wall will eventually reach thermal equilibrium with one another.
- **Adiabatic wall:** A boundary that prevents all thermal interaction. Systems separated by an adiabatic wall do not influence each other's thermal state, no matter how long they remain in contact.

The zeroth law can be restated in this language: if A and C are separated by a diathermic wall and reach equilibrium, and if B and C are similarly separated by a diathermic wall and reach equilibrium, then placing A and B in thermal contact

(removing an adiabatic wall between them) produces *no further change* — they are already in equilibrium.

1.1.2 Schematic Illustration

The following diagram illustrates the logical content of the zeroth law. On the left, systems A and B are each in diathermic contact with C but are separated from each other by an adiabatic wall. On the right, the adiabatic wall is replaced by a diathermic one — the zeroth law asserts that equilibrium between A and B is immediately established.

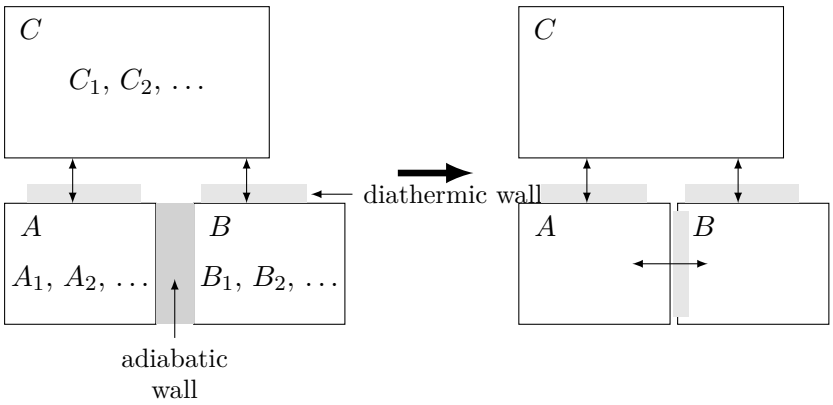

Figure 1.1. Left: A and B are each in equilibrium with C, but separated by an adiabatic wall. Right: the zeroth law guarantees that A and B are also in mutual equilibrium once brought into diathermic contact.

1.2 Mathematical Formulation

We now make the content of the zeroth law precise. Let the equilibrium state of systems A, B, and C be described by the coordinate sets $\{A_1, A_2, \ldots\}$, $\{B_1, B_2, \ldots\}$, and $\{C_1, C_2, \ldots\}$ respectively.

1.2.1 Equilibrium Constraints

The equilibrium of A and C imposes a functional constraint among their coordinates:

$$f_{AC}(A_1, A_2, \ldots;\ C_1, C_2, \ldots) = 0. \tag{1.1}$$

Similarly, the equilibrium of B and C imposes:

$$f_{BC}(B_1, B_2, \ldots;\ C_1, C_2, \ldots) = 0. \tag{1.2}$$

Each of these equations can be solved for the coordinate C_1, yielding:

$$C_1 = F_{AC}(A_1, A_2, \ldots;\ C_2, C_3, \ldots), \tag{1.3}$$

$$C_1 = F_{BC}(B_1, B_2, \ldots;\ C_2, C_3, \ldots). \tag{1.4}$$

Since both expressions equal C_1, the condition that C is simultaneously in equilibrium with both A and B requires:

$$F_{AC}(A_1, A_2, \ldots;\ C_2, \ldots) = F_{BC}(B_1, B_2, \ldots;\ C_2, \ldots). \tag{1.5}$$

1.2.2 The Cancellation Argument

Here is where the zeroth law does its essential work. Equation (1.5) involves the coordinates of all three systems. However, the zeroth law asserts that A and B are *also* in mutual equilibrium — a condition that involves *only* the coordinates of A and B, with no reference to C.

This is only possible if the coordinates of C *cancel out* from equation (1.5), leaving a relation purely between $\{A_i\}$ and $\{B_i\}$. In other words, each side of (1.5) must be expressible as a function of its own system's coordinates alone:

$$\Theta_A(A_1, A_2, \ldots) = \Theta_B(B_1, B_2, \ldots). \tag{1.6}$$

1.2.3 The Empirical Temperature

Equation (1.6) is the central result. It states that the condition for thermal equilibrium between any two systems can always be written as the equality of a *single scalar function* of each system's coordinates. This function is the **empirical temperature**:

$$\boxed{\Theta \;=\; \Theta(X_1,\, X_2,\, \ldots)} \tag{1.7}$$

where $\{X_i\}$ are the thermodynamic coordinates of the system in question.

A few remarks are in order:

1. The function Θ is called *empirical* because its precise form depends on the choice of thermometric substance and procedure. Different choices yield different scales, all of which are valid as long as they agree on which states are in equilibrium.

2. The set of all states satisfying $\Theta = \text{const.}$ defines an **isotherm** of the system. Two systems in thermal equilibrium lie on the same isotherm.

3. The function $\Theta(X_1, X_2, \ldots)$ is called the **equation of state** of the system. Specifying Θ specifies the thermal character of the system completely.

4. The zeroth law does *not* tell us what Θ is numerically — it only guarantees that such a function *exists*. The construction of a universal temperature scale requires the second law (Carnot's theorem), which yields the absolute Kelvin scale T.

1.3 An Illustrative Example: The Paramagnetic van der Waals Gas

1.3.1 Setup

To see the zeroth law in action, consider a single physical system with two distinguishable subsystems:

- **System A (mechanical):** a gas with volume V and pressure P, obeying a modified equation of state that accounts for intermolecular forces.

- **System B (magnetic):** the same gas, treated as a paramagnet with magnetisation M in an applied magnetic field B.

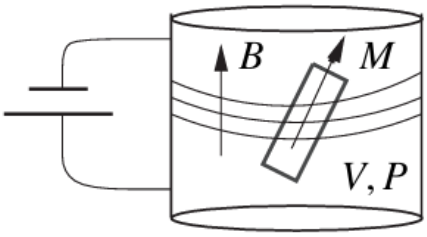

Figure 1.2. A paramagnetic van der Waals gas: the mechanical and magnetic degrees of freedom define two subsystems that must share a common empirical temperature.

1.3.2 The Equilibrium Condition

Empirically, one finds that the equilibrium constraint between the mechanical and magnetic coordinates takes the form:

$$\frac{1}{N}\left(P + \frac{a}{V^2}\right)(V - b) \;\propto\; \frac{NB}{M}. \tag{1.8}$$

Here a accounts for the attractive interactions between molecules and b for their finite volume — familiar from the van der Waals equation. The right-hand side is the ratio B/M that appears in the magnetic Curie relation. The proportionality (1.8) asserts that these two very different physical quantities track one another in equilibrium: a reflection of the zeroth law in action.

1.3.3 Identifying the Empirical Temperature

Choosing the proportionality constant to match the conventional Kelvin scale, we write:

$$\underbrace{\frac{1}{Nk_B}\left(P + \frac{a}{V^2}\right)(V - b)}_{\Theta_A} \;=\; \underbrace{\frac{N\mu_B^2 B}{3k_B M}}_{\Theta_B} \;=\; \Theta. \tag{1.9}$$

Each side depends on the coordinates of only one subsystem, confirming the structure of equation (1.6). Setting $\Theta = T$ (the Kelvin temperature) immediately recovers two well-known equations of state:

$$\boxed{\begin{cases} \left(P + \dfrac{a}{V^2}\right)(V - b) = Nk_BT & \text{(van der Waals gas)} \\ M = \dfrac{N\mu_B^2 B}{3k_BT} & \text{(Curie paramagnet)} \end{cases}} \tag{1.10}$$

This is a beautiful illustration of the zeroth law: the two equations in (1.10) are not independent postulates but are *linked* by the requirement that a single empirical temperature Θ must simultaneously characterise the thermal state of both subsystems. The zeroth law is what allows us to say that the T appearing in the van der Waals equation is the *same* T as in the Curie law.

Chapter 2

The Ideal Gas

The ideal gas is the simplest — and arguably the most important — model in classical thermodynamics. Despite being an idealisation, it describes the behaviour of real dilute gases with remarkable accuracy, and its equation of state serves as the starting point for almost every thermodynamic calculation at the undergraduate level.

The model rests on three physical assumptions:

- **Point masses:** The molecules of an ideal gas are treated as structureless point particles whose individual volumes are negligible compared to the volume of the container.
- **No intermolecular forces:** Molecules interact with one another only at the instant of collision. Between collisions, there are no attractive or repulsive forces.
- **Perfectly elastic collisions:** All collisions — whether molecule–molecule or molecule–wall — conserve both kinetic energy and momentum. No energy is lost to internal degrees of freedom during a collision.

These three assumptions together define a gas whose macroscopic behaviour is governed entirely by temperature, pressure, and volume — with no residual dependence on the chemical identity of the molecules.

2.1 The Empirical Gas Laws

Before arriving at the general equation of state, it is instructive to survey the individual gas laws discovered empirically over two centuries of careful experiment. Each law isolates the relationship between two thermodynamic variables while holding the third fixed.

Boyle's Law (1662)

At constant temperature, the pressure of a fixed mass of gas is inversely proportional to its volume:

$$P \propto \frac{1}{V} \qquad (T,\, n = \text{const.})$$

$$PV = \text{const.} \tag{2.1}$$

Physically, compressing a gas into a smaller volume increases the rate at which molecules strike the container walls, and hence the pressure.

Charles's Law (1787)

At constant pressure, the volume of a fixed mass of gas is directly proportional to its absolute temperature:

$$V \propto T \qquad (P,\, n = \text{const.})$$

$$\frac{V}{T} = \text{const.} \tag{2.2}$$

The temperature here must be measured on the absolute (Kelvin) scale. This law was historically important in establishing the concept of absolute zero as the temperature at which an ideal gas would contract to zero volume.

Gay-Lussac's Law (1808)

At constant volume, the pressure of a fixed mass of gas is directly proportional to its absolute temperature:

$$P \propto T \qquad (V,\, n = \text{const.})$$

$$\frac{P}{T} = \text{const.} \tag{2.3}$$

Avogadro's Law (1811)

At fixed temperature and pressure, the volume occupied by a gas is directly proportional to the number of moles n:

$$V \propto n \qquad (T,\, P = \text{const.})$$

$$\frac{V}{n} = \text{const.} \tag{2.4}$$

This law carries a remarkable implication: equal volumes of *any* ideal gas, under identical conditions of temperature and pressure, contain the same number of molecules — regardless of their chemical identity. At standard temperature and pressure (STP: $T = 273.15\,\text{K}$, $P = 1\,\text{atm}$), one mole of any ideal gas occupies $22.4\,\text{L}$.

2.2 The Ideal Gas Equation of State

The four empirical laws can be unified into a single proportionality:

$$PV \propto nT. \tag{2.5}$$

Introducing a universal proportionality constant R, we arrive at the **macroscopic equation of state**:

$$\boxed{PV = nRT} \tag{2.6}$$

The constant R is the **universal gas constant**:

$$R = 8.314\ \text{J}\,\text{mol}^{-1}\,\text{K}^{-1}.$$

It is called "universal" because it takes the same numerical value for every ideal gas — a direct consequence of Avogadro's law.

2.2.1 Microscopic Form: Boltzmann's Constant

Avogadro's Number

The number of constituent particles in exactly one mole of any substance is **Avogadro's number**:

$$\boxed{N_A = 6.02214076 \times 10^{23}\ \mathrm{mol}^{-1}.}$$

If a sample contains n moles, the total number of molecules is:

$$N = nN_A \implies n = \frac{N}{N_A}. \tag{2.7}$$

Substituting into (2.6):

$$PV = \frac{N}{N_A} RT = N \cdot \frac{R}{N_A} \cdot T. \tag{2.8}$$

The combination R/N_A is itself a fundamental constant, the **Boltzmann constant**:

$$k_B = \frac{R}{N_A} = 1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}. \tag{2.9}$$

The ideal gas equation therefore has an equivalent **microscopic form**:

$$\boxed{PV = Nk_BT.} \tag{2.10}$$

Dividing both sides by V and defining the **number density** $\bar{n} = N/V$, we obtain a third useful form:

$$\boxed{P = \bar{n}\, k_BT.} \tag{2.11}$$

Equation (2.10) is more than a rewriting of (2.6). It connects a macroscopic quantity (pressure P) directly to the count of microscopic entities (N), anticipating the kinetic theory derivation in Section 2.4.

2.3 Work Done by an Ideal Gas

Energy enters and leaves a thermodynamic system in two forms: heat and work. We examine the mechanical work done when a gas changes its volume.

2.3.1 Setting Up: The Piston-Cylinder System

Consider a gas enclosed in a cylinder fitted with a frictionless, massless piston of cross-sectional area A. The gas exerts a pressure P on the piston, producing a force:

$$F = P \cdot A. \tag{2.12}$$

When the piston is displaced by an infinitesimal amount dl (measured outward from the gas), the work done **by the gas** on the piston is:

$$\delta W = F \cdot dl = PA\, dl. \tag{2.13}$$

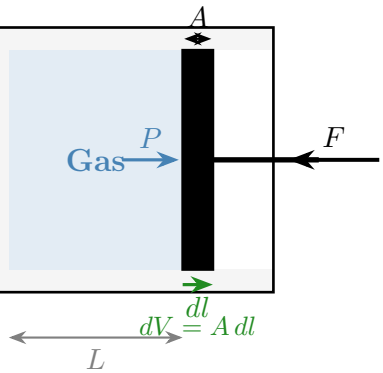

Figure 2.1. Piston–cylinder system. A gas at pressure P pushes a frictionless piston of cross-sectional area A. An outward displacement dl produces a volume change $dV = A\,dl$ and work $\delta W = P\,dV$.

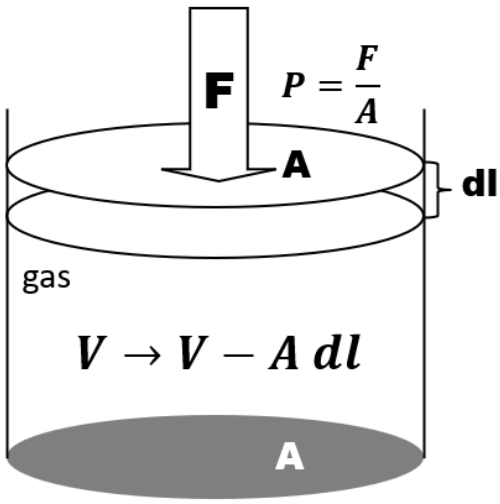

Figure 2.2. A gas being compressed by an external force F applied to a piston of area A. An outward displacement dl changes the volume by $dV = A\,dl$.

Since the change in volume is $dV = A\,dl$, this becomes:

$$\delta W = P\,dV. \tag{2.14}$$

Sign Convention

- **Expansion** ($dV > 0$): The gas pushes the piston outward. Work done *by* the gas is *positive*: $\delta W > 0$.
- **Compression** ($dV < 0$): An external agent pushes the piston inward. Work done *by* the gas is *negative*: $\delta W < 0$.

Throughout these notes, W denotes work done *by* the system.

2.3.2 Work in a Finite Process

For a finite volume change from V_1 to V_2, the total work done by the gas is:

$$W = \int_{V_1}^{V_2} P\, dV. \tag{2.15}$$

This integral is only well-defined for quasi-static processes — processes slow enough that the system remains in thermodynamic equilibrium at every intermediate stage, so that P is a well-defined single-valued function of V throughout the process.

We now evaluate (2.15) for the standard thermodynamic processes.

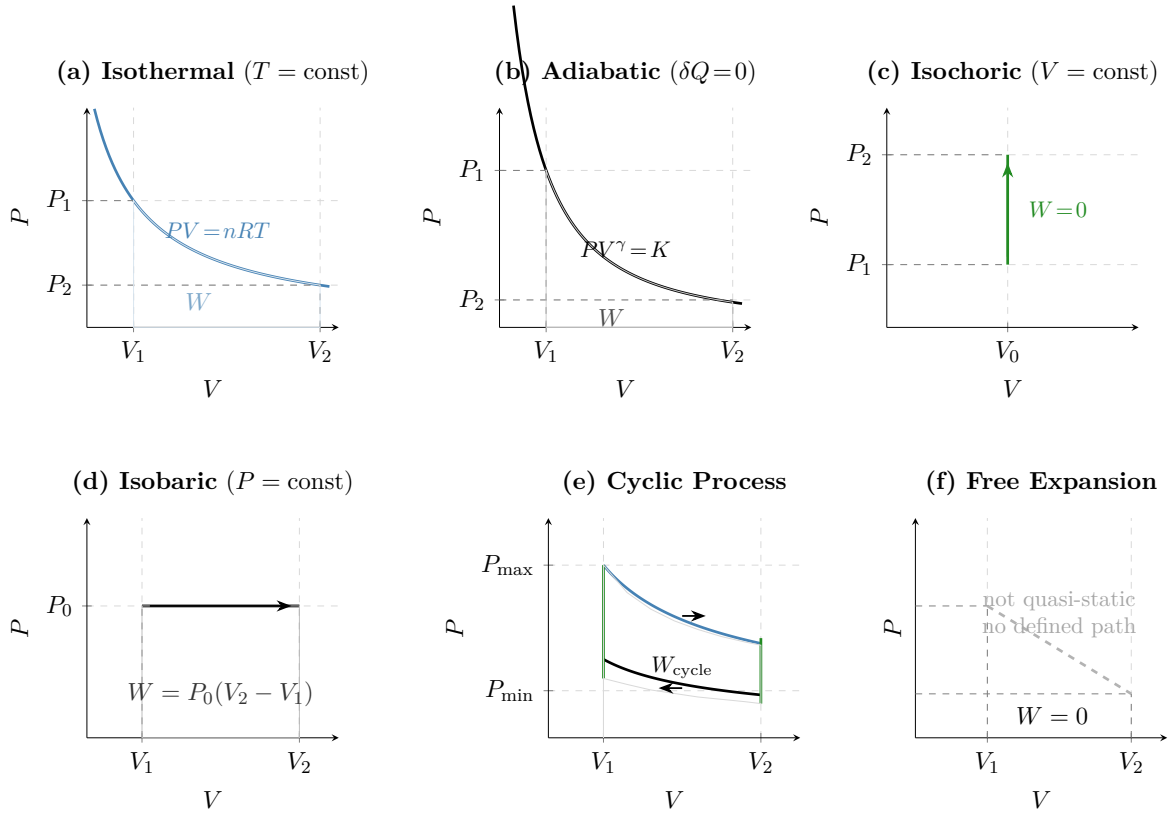

Figure 2.3. P–V diagrams for the six standard thermodynamic processes. The shaded areas equal the work done *by* the gas in each case. The dashed curve in (f) indicates that free expansion is not quasi-static and has no well-defined path on the P–V diagram.

Isothermal Process (T = const.)

For an ideal gas at constant temperature, the equation of state gives $P = nRT/V$. Substituting:

$$W = \int_{V_1}^{V_2} \frac{nRT}{V}\, dV = nRT \int_{V_1}^{V_2} \frac{dV}{V} = nRT \Big[\ln V\Big]_{V_1}^{V_2}. \tag{2.16}$$

$$\boxed{W_{\text{iso}} = nRT \ln\left(\frac{V_2}{V_1}\right).} \tag{2.17}$$

For an expansion ($V_2 > V_1$), $W > 0$, as expected.

Adiabatic Process ($\delta Q = 0$)

In an adiabatic process no heat is exchanged. One can show (see Chapter 4) that the equation of state along an adiabat is:

$$PV^\gamma = K = \text{const.}, \qquad \gamma = \frac{C_P}{C_V}.$$

Therefore $P = KV^{-\gamma}$, and the work integral becomes:

$$\begin{aligned} W &= \int_{V_1}^{V_2} KV^{-\gamma}\, dV = K\left[\frac{V^{-\gamma+1}}{-\gamma+1}\right]_{V_1}^{V_2} = \frac{K}{1-\gamma}\left(V_2^{1-\gamma} - V_1^{1-\gamma}\right) \\ &= \frac{1}{1-\gamma}\left(KV_2^{1-\gamma} - KV_1^{1-\gamma}\right) = \frac{1}{1-\gamma}\left(P_2V_2 - P_1V_1\right). \end{aligned} \tag{2.18}$$

$$\boxed{W_{\text{ad}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1}.} \tag{2.19}$$

Since no heat enters the system, this work is done entirely at the expense of the internal energy.

Isochoric Process ($V = \text{const.}$)

At constant volume, $dV = 0$ identically, so:

$$W_{\text{isoch}} = \int_{V_1}^{V_2} P\, dV = 0. \tag{2.20}$$

No mechanical work is performed. Any heat supplied to the system increases its internal energy alone.

Isobaric Process ($P = \text{const.}$)

At constant pressure, P comes outside the integral:

$$W_{\text{isob}} = P\int_{V_1}^{V_2} dV = P(V_2 - V_1). \tag{2.21}$$

Using $PV = nRT$, we have $P(V_2 - V_1) = nR(T_2 - T_1)$, giving the equivalent form:

$$\boxed{W_{\text{isob}} = nR(T_2 - T_1).} \tag{2.22}$$

Cyclic Process

In a cyclic process the system is returned to its initial state. Since all state functions return to their original values, the net work done over one complete cycle is:

$$W_{\text{cycle}} = \oint P\, dV, \tag{2.23}$$

which equals the *area enclosed* by the cycle on a P–V diagram. The sign depends on the direction of traversal: clockwise gives $W > 0$ (net work output); anticlockwise gives $W < 0$ (net work input).

Free Expansion

In free expansion, a gas expands into a vacuum through a partition that is suddenly removed. Since the external pressure is zero ($P_{\text{ext}} = 0$), no force is exerted on the gas molecules by the surroundings:

$$W_{\text{free}} = \int P_{\text{ext}}\, dV = 0. \tag{2.24}$$

No work is done, despite the volume increasing. This is an inherently irreversible process and is not quasi-static.

Problem 2.3.1. Polytropic Process Calculate the work done by a gas following the polytropic path $PV^x = C$, where C and x are constants. (Refer Appendix A.)

2.4 Kinetic Theory of Ideal Gases

The macroscopic gas laws tell us *what* an ideal gas does. Kinetic theory tells us *why*. It derives the equation of state from first principles by modelling the gas as a large number of molecules in ceaseless random motion.

2.4.1 Setup and Assumptions

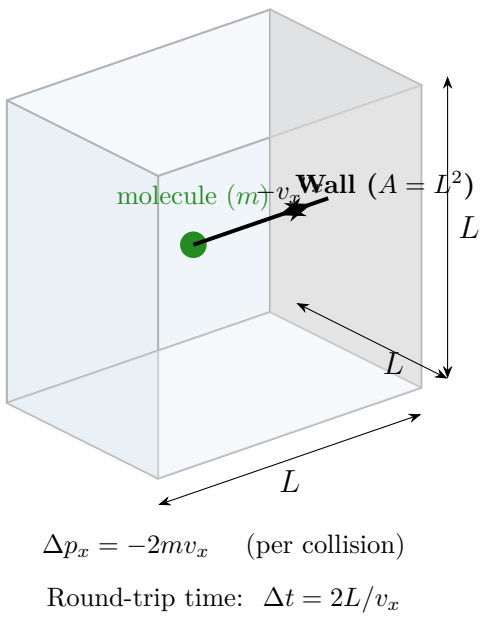

Figure 2.4. A single molecule of mass m bouncing back and forth in a cubical box of side L. The x-component of velocity reverses at each wall collision. The momentum transferred to the right wall per collision is $2mv_x$, and the round-trip time is $2L/v_x$.

Consider N identical molecules, each of mass m, enclosed in a cubical box of side L. The volume of the box is $V = L^3$. A molecule moves with velocity $\vec{v} = (v_x, v_y, v_z)$. We assume:

- The molecules are hard spheres that undergo perfectly elastic collisions with the walls.

- Intermolecular collisions do not affect the calculation of pressure (they redistribute velocities but do not change the overall momentum transferred to the walls).
- The motion is isotropic: no direction in space is preferred.

2.4.2 Pressure from Molecular Collisions

Momentum change at a single collision

Focus on one molecule moving with velocity component v_x in the x-direction. When it collides elastically with the wall perpendicular to the x-axis, only v_x reverses:

$$v_x \longrightarrow -v_x.$$

The change in the molecule's x-momentum is:

$$\Delta p_x = m(-v_x) - m(v_x) = -2mv_x. \tag{2.25}$$

By Newton's third law, the momentum transferred *to the wall* per collision is $+2mv_x$.

Collision frequency for one molecule

After bouncing off the right wall, the molecule travels across the box (length L) and back before hitting the same wall again. The round-trip time is:

$$\Delta t = \frac{2L}{v_x}. \tag{2.26}$$

The force exerted on the wall by this single molecule is:

$$f_x = \frac{\Delta p_x}{\Delta t} = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}. \tag{2.27}$$

Total force from all N molecules

Summing over all N molecules:

$$F_x = \sum_{i=1}^{N} \frac{mv_{xi}^2}{L} = \frac{m}{L}\sum_{i=1}^{N} v_{xi}^2. \tag{2.28}$$

The pressure on the wall of area $A = L^2$ is:

$$P = \frac{F_x}{L^2} = \frac{m}{L^3}\sum_{i=1}^{N} v_{xi}^2 = \frac{m}{V}\sum_{i=1}^{N} v_{xi}^2. \tag{2.29}$$

Invoking isotropy

Since molecular motion is isotropic, the average squared velocity is equally distributed among the three directions:

$$\langle v_x^2\rangle = \langle v_y^2\rangle = \langle v_z^2\rangle = \frac{1}{3}\langle v^2\rangle, \tag{2.30}$$

where $\langle v^2 \rangle = \langle v_x^2 + v_y^2 + v_z^2 \rangle$ is the mean-square speed. Substituting $\sum_i v_{xi}^2 = N\langle v_x^2 \rangle = \frac{1}{3}N\langle v^2 \rangle$:

$$\boxed{PV = \frac{1}{3}\, mN\langle v^2 \rangle.} \tag{2.31}$$

Defining the total mass $M = Nm$, the gas density $\rho = M/V$, and the root-mean-square speed $v_{rms} = \sqrt{\langle v^2 \rangle}$, we obtain the compact form:

$$\boxed{P = \frac{1}{3}\, \rho\, v_{rms}^2.} \tag{2.32}$$

2.4.3 Temperature as a Measure of Kinetic Energy

Comparing the kinetic-theory result (2.31) with the microscopic ideal gas law $PV = Nk_BT$:

$$\frac{1}{3}\, mN\langle v^2 \rangle = Nk_BT \implies \frac{1}{2}\, m\langle v^2 \rangle = \frac{3}{2}\, k_BT. \tag{2.33}$$

The left side is the average translational kinetic energy of one molecule:

$$\boxed{\langle \kappa \rangle = \frac{1}{2}\, mv_{rms}^2 = \frac{3}{2}\, k_BT.} \tag{2.34}$$

Insight

The Physical Meaning of Temperature Equation (2.34) reveals what temperature *is* at the molecular level: it is a direct measure of the average translational kinetic energy of the molecules. A hotter gas is simply one whose molecules are moving faster, on average.

2.5 The Equipartition of Energy

Equipartition Theorem For a system in thermal equilibrium at temperature T, each independent quadratic degree of freedom in the Hamiltonian contributes, on average, an energy of $\frac{1}{2}k_BT$ to the total internal energy.
A "quadratic degree of freedom" is any generalised coordinate q or momentum p that appears in the Hamiltonian as αq^2 or αp^2. Translational and rotational kinetic energies are examples; vibrational modes contribute *two* such terms (one kinetic, one potential).

2.5.1 Consistency with the Kinetic Theory Result

A monatomic ideal gas molecule can only translate — there are no rotational or vibrational degrees of freedom. Its three translational momenta (p_x, p_y, p_z) contribute three quadratic terms to the Hamiltonian, so the equipartition theorem predicts:

$$\langle \kappa \rangle = 3 \times \frac{1}{2}k_BT = \frac{3}{2}k_BT. \tag{2.35}$$

This is precisely the result derived in (2.34) from kinetic theory — a satisfying consistency check.

2.5.2 Degrees of Freedom by Molecule Type

- **Monatomic gas** ($f = 3$): Pure translational motion.

$$u = \frac{3}{2}k_B T. \tag{2.36}$$

- **Diatomic gas** ($f = 5$ at moderate temperatures): Three translational modes plus two rotational modes. Rotation about the bond axis is excluded because the moment of inertia about that axis is negligibly small.

$$u = \frac{5}{2}k_B T. \tag{2.37}$$

At high temperatures, vibrational modes become active. Each vibration adds *two* quadratic degrees of freedom (one kinetic, one potential), raising f to 7:

$$u = \frac{7}{2}k_B T \quad \text{(with vibration active)}. \tag{2.38}$$

- **Polyatomic gas**: In general, a non-linear molecule has 3 translational + 3 rotational degrees of freedom, plus however many vibrational modes are thermally accessible. A linear polyatomic molecule has only 2 rotational degrees of freedom.

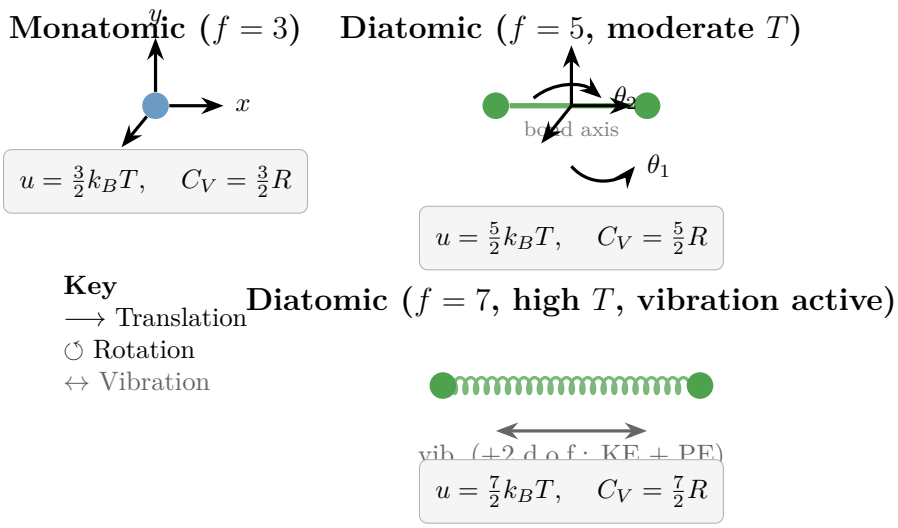

Figure 2.5. Degrees of freedom and corresponding average energies per molecule from the equipartition theorem. Each quadratic degree of freedom contributes $\frac{1}{2}k_B T$. The diatomic molecule has five active modes at moderate temperatures (three translational, two rotational) and seven at high temperatures when vibration is activated.

Insight

Frozen Degrees of Freedom Quantum mechanics imposes minimum energy thresholds on rotational and vibrational modes. At sufficiently low temperatures, the thermal energy $k_B T$ is insufficient to excite these modes, and they are said to be *frozen out* — they contribute nothing to the heat capacity. This is why the heat capacity of diatomic hydrogen, for example, rises in discrete steps as the

temperature is increased, as first rotational and then vibrational modes become thermally accessible.

Exercise 2.5.1

Exercise A non-linear triatomic molecule (e.g. H_2O) has 3 translational and 3 rotational degrees of freedom; assume vibrations are frozen. Calculate the total internal energy of 1 mole of this gas and hence its molar heat capacity at constant volume C_V.

A fuller discussion of degrees of freedom and their connection to heat capacities is given in Chapter 4.

2.6 The Maxwell–Boltzmann Velocity Distribution

The kinetic theory result $\langle \kappa \rangle = \frac{3}{2} k_B T$ tells us the *average* speed, but says nothing about how molecular speeds are distributed around that average. The Maxwell–Boltzmann distribution answers this question quantitatively.

2.6.1 The Boltzmann Factor

Insight

Maxwell–Boltzmann Statistics A cornerstone result of statistical mechanics states that, for a system in thermal equilibrium at temperature T, the probability of a molecule occupying a state of energy ϵ is proportional to the **Boltzmann factor**:

$$P(\epsilon) \propto e^{-\epsilon/k_B T}.$$

In the discrete case, if n_i molecules occupy a state of energy ϵ_i, then:

$$n_i = \frac{N}{Z} e^{-\epsilon_i/k_B T}, \qquad Z = \sum_j e^{-\epsilon_j/k_B T}, \tag{2.39}$$

where Z is the *partition function* and the sum runs over all accessible states. In the continuous case, the probability density is:

$$f(\epsilon)\, d\epsilon = A\, e^{-\epsilon/k_B T}\, d\epsilon, \tag{2.40}$$

where A is a normalisation constant chosen so that $\int_0^\infty f(\epsilon)\, d\epsilon = 1$.

2.6.2 Distribution of One Velocity Component

The kinetic energy associated with motion along the x-axis alone is $\epsilon_x = \frac{1}{2} m v_x^2$. By the Boltzmann factor, the probability density for the x-component of velocity is:

$$f(v_x)\, dv_x = A \exp\left(-\frac{m v_x^2}{2 k_B T}\right) dv_x. \tag{2.41}$$

Determining the normalisation constant A

The normalisation condition requires:

$$\int_{-\infty}^{+\infty} f(v_x)\, dv_x = 1 \implies A \int_{-\infty}^{+\infty} e^{-mv_x^2/2k_BT}\, dv_x = 1. \tag{2.42}$$

This is a standard Gaussian integral. With the substitution $a = m/2k_BT$:

$$\int_{-\infty}^{+\infty} e^{-av_x^2}\, dv_x = \sqrt{\frac{\pi}{a}} = \sqrt{\frac{2\pi k_BT}{m}}. \tag{2.43}$$

Therefore:

$$A = \sqrt{\frac{m}{2\pi k_BT}}. \tag{2.44}$$

The normalised one-dimensional distribution is:

$$\boxed{f(v_x)\, dv_x = \left(\frac{m}{2\pi k_BT}\right)^{1/2} \exp\left(-\frac{mv_x^2}{2k_BT}\right) dv_x.} \tag{2.45}$$

This is a Gaussian centred at $v_x = 0$, reflecting the fact that no direction of motion is preferred.

2.6.3 The Three-Dimensional Velocity Distribution

Since the gas is isotropic, the motions in the x, y, and z directions are statistically independent. The joint probability density for the velocity vector $\vec{v} = (v_x, v_y, v_z)$ is the product of three independent one-dimensional distributions:

$$\begin{aligned} f(\vec{v})\, d^3v &= f(v_x)\, f(v_y)\, f(v_z)\; dv_x\, dv_y\, dv_z \\ &= \left(\frac{m}{2\pi k_BT}\right)^{3/2} \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_BT}\right) dv_x\, dv_y\, dv_z. \end{aligned} \tag{2.46}$$

Since $v_x^2 + v_y^2 + v_z^2 = v^2$:

$$f(\vec{v})\, d^3v = \left(\frac{m}{2\pi k_BT}\right)^{3/2} \exp\left(-\frac{mv^2}{2k_BT}\right) d^3v. \tag{2.47}$$

2.6.4 The Speed Distribution $F(v)$

In many applications we do not care about the direction of $\vec{v}$, only its magnitude $v = |\vec{v}|$. We therefore integrate over all directions in velocity space.

Change to spherical coordinates in velocity space

The volume element in spherical coordinates is:

$$d^3v = v^2 \sin\theta\; dv\, d\theta\, d\phi. \tag{2.48}$$

Integrating over all solid angles:

$$\int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\phi = [-\cos\theta]_0^{\pi} \cdot 2\pi = 2 \cdot 2\pi = 4\pi. \tag{2.49}$$

The speed distribution

The probability of a molecule having speed between v and $v + dv$ is:

$$F(v)\,dv = 4\pi v^2 f(v)\,dv. \tag{2.50}$$

Substituting the expression for $f(v)$:

$$\boxed{F(v)\,dv = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) dv.} \tag{2.51}$$

The factor v^2 arises from the spherical shell in velocity space and represents the increasing number of states available at higher speeds; the Boltzmann factor $e^{-mv^2/2k_BT}$ suppresses the distribution at large speeds. Their competition produces a peaked distribution with a well-defined most probable speed.

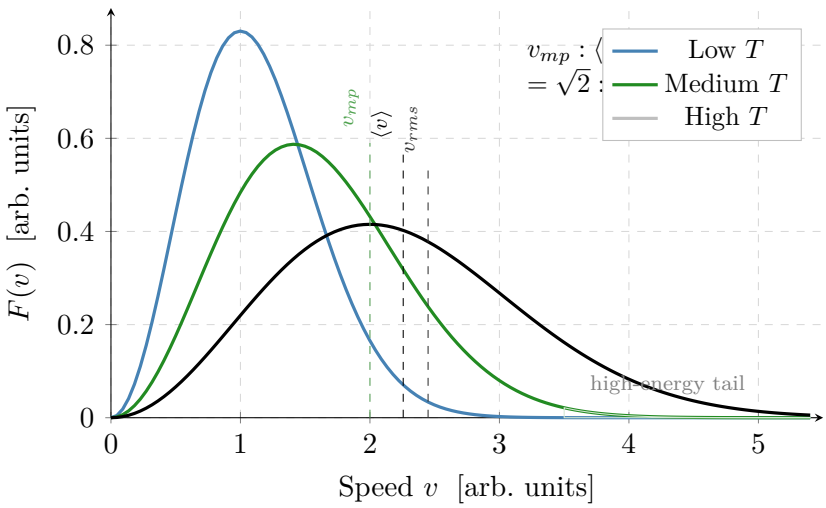

Figure 2.6. Maxwell–Boltzmann speed distribution $F(v)$ at three temperatures. As temperature rises, the distribution broadens and its peak shifts to higher speeds. The three characteristic speeds v_{mp}, $\langle v \rangle$, and v_{rms} (shown for the medium-T curve) satisfy $v_{mp} < \langle v \rangle < v_{rms}$.

2.6.5 Characteristic Speeds

Three speeds characterise the Maxwell–Boltzmann distribution.

Most probable speed v_{mp}

This is the speed at which $F(v)$ is maximum. Setting $dF/dv = 0$:

$$\frac{d}{dv}\left[v^2\, e^{-mv^2/2k_BT}\right] = 0. \tag{2.52}$$

Applying the product rule:

$$2v\, e^{-mv^2/2k_BT} + v^2 \cdot e^{-mv^2/2k_BT} \cdot \left(-\frac{mv}{k_BT}\right) = 0. \tag{2.53}$$

Dividing through by $v\, e^{-mv^2/2k_BT}$ (which is non-zero for $v > 0$):

$$2 - \frac{mv_{mp}^2}{k_BT} = 0 \implies \boxed{v_{mp} = \sqrt{\frac{2k_BT}{m}}.} \tag{2.54}$$

Mean speed $\langle v \rangle$

The mean speed is the first moment of the distribution:

$$\langle v \rangle = \int_0^\infty v\, F(v)\, dv = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3\, e^{-mv^2/2k_B T}\, dv. \tag{2.55}$$

Using the standard result $\int_0^\infty x^3 e^{-ax^2} dx = 1/(2a^2)$ with $a = m/2k_B T$:

$$\langle v \rangle = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \cdot \frac{1}{2} \left(\frac{2k_B T}{m} \right)^2 = \sqrt{\frac{8k_B T}{\pi m}}. \tag{2.56}$$

$$\boxed{\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}}.} \tag{2.57}$$

Root-mean-square speed v_{rms}

The mean-square speed is the second moment:

$$\langle v^2 \rangle = \int_0^\infty v^2\, F(v)\, dv = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^4\, e^{-mv^2/2k_B T}\, dv. \tag{2.58}$$

Using $\int_0^\infty x^4 e^{-ax^2} dx = \frac{3\sqrt{\pi}}{8a^{5/2}}$:

$$\langle v^2 \rangle = \frac{3k_B T}{m}. \tag{2.59}$$

$$\boxed{v_{rms} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}}.} \tag{2.60}$$

Hierarchy of characteristic speeds

All three speeds are proportional to $\sqrt{k_B T/m}$, but with different numerical prefactors. Their ordering is:

$$\boxed{v_{mp} \;:\; \langle v \rangle \;:\; v_{rms} = \sqrt{2} \;:\; \sqrt{\frac{8}{\pi}} \;:\; \sqrt{3} \approx 1.414 \;:\; 1.596 \;:\; 1.732.} \tag{2.61}$$

In particular, $v_{mp} < \langle v \rangle < v_{rms}$.

2.6.6 Momentum and Energy Distributions

Momentum distribution

Substituting $p = mv$ (so $v = p/m$, $dv = dp/m$) into (2.51):

$$F(p)\, dp = 4\pi \left(\frac{1}{2\pi m k_B T} \right)^{3/2} p^2 \exp\left(-\frac{p^2}{2m k_B T} \right) dp. \tag{2.62}$$

Energy distribution

The translational kinetic energy is $\epsilon = \frac{1}{2}mv^2$, so $v = \sqrt{2\epsilon/m}$ and $dv = (1/2)\sqrt{2/m\epsilon}\,d\epsilon$. Substituting into (2.51) and simplifying:

$$F(\epsilon)\,d\epsilon = \frac{2}{\sqrt{\pi}}\,\frac{\sqrt{\epsilon}}{(k_BT)^{3/2}}\,e^{-\epsilon/k_BT}\,d\epsilon. \tag{2.63}$$

The mean energy follows from the first moment:

$$\langle\epsilon\rangle = \int_0^\infty \epsilon\,F(\epsilon)\,d\epsilon = \frac{3}{2}\,k_BT, \tag{2.64}$$

in perfect agreement with the equipartition theorem.

2.7 Transport Phenomena

So far we have studied gases in equilibrium. When a gas is subjected to gradients — of velocity, temperature, or concentration — molecules transport momentum, energy, or mass across those gradients. This section derives the three principal transport coefficients from kinetic theory.

2.7.1 The Mean Free Path

The mean free path λ is the average distance a molecule travels between successive collisions. It is the fundamental length scale governing all transport phenomena.

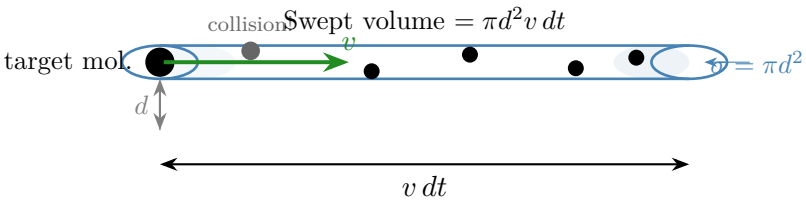

Figure 2.7. Collision cylinder used to derive the mean free path. A molecule of diameter d moving with speed v sweeps out a cylindrical volume $\pi d^2 v\,dt$ in time dt. Any molecule whose centre lies within this cylinder suffers a collision. The collision frequency is $f_{\text{coll}} = \sqrt{2}\,n\pi d^2\langle v\rangle$ after correcting for the relative motion of target molecules.

Collision cross-section

Model molecules as hard spheres of diameter d. A collision occurs whenever the centres of two molecules come within a distance d of each other. One can therefore treat one molecule as having an effective radius d while all others are point masses. The effective cross-sectional area for a collision is:

$$\sigma = \pi d^2. \tag{2.65}$$

Collision rate for a stationary background

In a time interval dt, a molecule moving at speed v sweeps out a cylinder of volume:

$$dV_{\text{cyl}} = \sigma v\, dt = \pi d^2 v\, dt. \tag{2.66}$$

The number of molecules (number density n) within this cylinder is:

$$dN = n\, \pi d^2 v\, dt. \tag{2.67}$$

The collision frequency — collisions per unit time — is:

$$f_{\text{coll}} = \frac{dN}{dt} = n\pi d^2 v. \tag{2.68}$$

Correcting for relative motion

The above assumes that target molecules are stationary, which overestimates λ. In reality every molecule is moving. The relevant speed is the *mean relative speed* v_{rel} between two molecules. For any two velocity vectors $\vec{v}_1$ and $\vec{v}_2$:

$$v_{\text{rel}}^2 = |\vec{v}_1 - \vec{v}_2|^2 = v_1^2 + v_2^2 - 2\vec{v}_1 \cdot \vec{v}_2. \tag{2.69}$$

Averaging over the Maxwell–Boltzmann distribution:

1. The cross term vanishes: $\langle \vec{v}_1 \cdot \vec{v}_2 \rangle = \langle \vec{v}_1 \rangle \cdot \langle \vec{v}_2 \rangle = 0$ (the average velocity in any direction is zero).
2. For identical molecules: $\langle v_1^2 \rangle = \langle v_2^2 \rangle = \langle v^2 \rangle$.

Therefore $\langle v_{\text{rel}}^2 \rangle = 2\langle v^2 \rangle$, and for a Maxwellian distribution:

$$v_{\text{rel}} = \sqrt{2}\, \langle v \rangle. \tag{2.70}$$

Replacing v with $\sqrt{2}\, \langle v \rangle$ in the collision frequency:

$$f_{\text{coll}} = \sqrt{2}\, n\pi d^2 \langle v \rangle. \tag{2.71}$$

Mean free path

The mean free path is the average distance covered between collisions:

$$\lambda = \frac{\langle v \rangle}{f_{\text{coll}}} = \frac{\langle v \rangle}{\sqrt{2}\, n\pi d^2 \langle v \rangle}. \tag{2.72}$$

$$\boxed{\lambda = \frac{1}{\sqrt{2}\, n\pi d^2}.} \tag{2.73}$$

Using $n = P/k_B T$ from the ideal gas law:

$$\lambda = \frac{k_B T}{\sqrt{2}\, \pi d^2 P}. \tag{2.74}$$

Thus λ increases with temperature (faster molecules, same density) and decreases with pressure (higher density, more collisions). At atmospheric pressure, $\lambda \sim 10^{-7}$ m for typical gases.

Survival Equation The probability that a molecule travels a distance x without suffering a collision decays exponentially:

$$N(x) = N(0) \exp\left(-\frac{x}{\lambda}\right). \tag{2.75}$$

This is the molecular analogue of radioactive decay, with λ playing the role of the decay length.

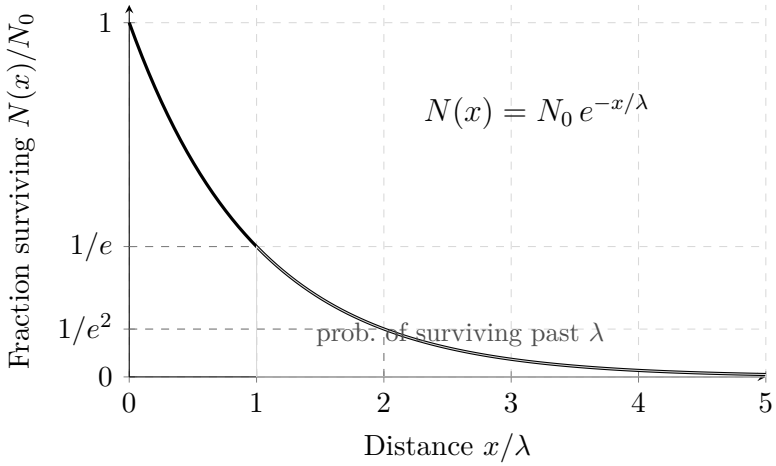

Figure 2.8. Survival equation: the fraction of molecules that travel a distance x without collision decays exponentially with decay length λ (the mean free path). At $x = \lambda$, a fraction $1/e \approx 37\%$ of molecules remain uncollided.

2.7.2 The Three Transport Coefficients

When a gas is not in equilibrium, random molecular motion acts to erase spatial gradients. The three scenarios below differ only in *what* is being transported.

Table 2.1. The three primary transport phenomena.

Phenomenon	Quantity transported	Driving gradient	Coefficient
Viscosity	Momentum ($m\langle v\rangle$)	Velocity du/dz	Viscosity η
Thermal conduction	Heat energy	Temperature dT/dz	Conductivity K
Diffusion	Mass / molecules	Concentration dn/dz	Diffusion coeff. D

Viscosity: Transport of Momentum

Consider a gas flowing in the x-direction with a velocity gradient du/dz along the z-axis. At height z, the flow speed is $u(z)$. Molecules crossing an imaginary horizontal plane at z originate, on average, from one mean free path above or below it:

- Molecules from **above** $(z + \lambda)$ carry x-momentum $m\left(u + \lambda \frac{du}{dz}\right)$ per molecule.

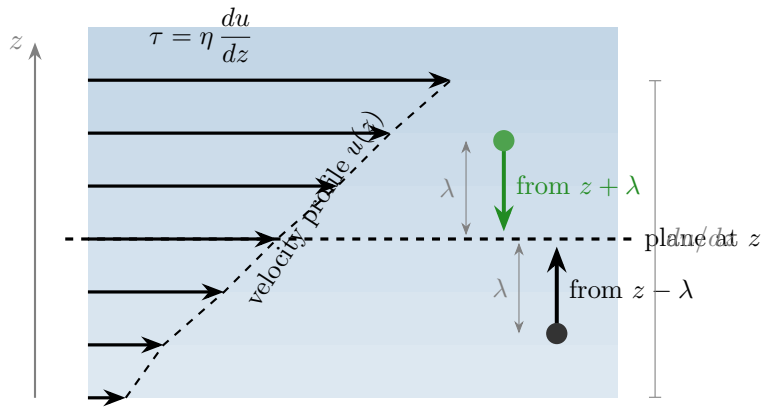

Figure 2.9. Molecular mechanism of viscosity. In a gas with a velocity gradient du/dz, molecules crossing an imaginary horizontal plane at z arrive from one mean free path above $(z + \lambda)$ or below $(z - \lambda)$. They carry more (or less) x-momentum than the local average, creating a net shear stress $\tau = \eta\, du/dz$.

- Molecules from **below** $(z - \lambda)$ carry x-momentum $m\left(u - \lambda \frac{du}{dz}\right)$ per molecule.

Treating molecular motion as distributed equally among six directions, the flux of molecules crossing the plane from each side is $\frac{1}{6}n\bar{v}$ per unit area per unit time (where $\bar{v} = \langle v \rangle$). The net downward momentum flux (shear stress) is:

$$\begin{aligned}\tau &= \frac{1}{6}nm\bar{v}\left(u + \lambda\frac{du}{dz}\right) - \frac{1}{6}nm\bar{v}\left(u - \lambda\frac{du}{dz}\right)\\ &= \frac{1}{6}nm\bar{v}\cdot 2\lambda\frac{du}{dz} = \frac{1}{3}nm\bar{v}\lambda\frac{du}{dz}.\end{aligned} \tag{2.76}$$

Newton's law of viscosity states $\tau = \eta\, du/dz$. Comparing:

$$\boxed{\eta = \frac{1}{3}nm\bar{v}\lambda.} \tag{2.77}$$

Properties:

1. *Pressure independence:* Since $\lambda \propto 1/n$, the product $n\lambda$ is constant, and η is independent of pressure. This counterintuitive result was first predicted by Maxwell and later confirmed experimentally.

2. *Temperature dependence:* Since $\bar{v} \propto \sqrt{T}$ and λ is pressure-independent but weakly temperature-dependent, $\eta \propto \sqrt{T}$. Gas viscosity *increases* with temperature — the opposite of liquids, where viscosity decreases with temperature.

Thermal Conduction: Transport of Energy

Consider a temperature gradient dT/dz. The average energy carried by a molecule is $\epsilon = C_v T$, where C_v is the molecular heat capacity. A molecule arriving from height $z + \lambda$ carries energy $C_v(T + \lambda\, dT/dz)$; one from $z - \lambda$ carries $C_v(T - \lambda\, dT/dz)$. The net heat flux is:

$$\begin{aligned}q &= \frac{1}{6}n\bar{v}\left[C_v\left(T + \lambda\tfrac{dT}{dz}\right) - C_v\left(T - \lambda\tfrac{dT}{dz}\right)\right]\\ &= \frac{1}{6}n\bar{v}\cdot 2C_v\lambda\frac{dT}{dz} = \frac{1}{3}n\bar{v}\lambda C_v\frac{dT}{dz}.\end{aligned} \tag{2.78}$$

The negative sign (heat flows from hot to cold) is accounted for by the direction convention. Comparing with Fourier's law $q = -K\,dT/dz$:

$$\boxed{K = \frac{1}{3}\,n\bar{v}\lambda C_v.} \tag{2.79}$$

Relationship between K and η:
Since $\eta = \frac{1}{3}nm\bar{v}\lambda$, we can write:

$$K = \eta\,\frac{C_v}{m}. \tag{2.80}$$

Defining the specific heat capacity (per unit mass) $c_v = C_v/m$:

$$\boxed{K = \eta\,c_v.} \tag{2.81}$$

Gases with high viscosity are therefore typically good thermal conductors. Both K and η are independent of pressure by the same $n\lambda = \text{const.}$ argument.

Diffusion: Transport of Mass

Consider a concentration gradient dn/dz of a particular molecular species. Molecules arriving from $z + \lambda$ come from a region of density $n + \lambda\,dn/dz$; those from $z - \lambda$ come from $n - \lambda\,dn/dz$. The net particle flux (number of molecules crossing unit area per unit time) is:

$$J = \frac{1}{6}\bar{v}\Big[\Big(n - \lambda\tfrac{dn}{dz}\Big) - \Big(n + \lambda\tfrac{dn}{dz}\Big)\Big] = -\frac{1}{3}\bar{v}\lambda\,\frac{dn}{dz}. \tag{2.82}$$

Comparing with Fick's first law $J = -D\,dn/dz$:

$$\boxed{D = \frac{1}{3}\,\bar{v}\lambda.} \tag{2.83}$$

The three coefficients are linked by a simple chain of relations:

$$D = \frac{\eta}{\rho}, \qquad K = \rho\,c_v\,D, \tag{2.84}$$

where $\rho = nm$ is the mass density. These relations confirm that viscosity, thermal conduction, and diffusion are three aspects of the same underlying molecular transport process.

Historical Note

Einstein and the Diffusion Coefficient (1905) The kinetic-theory expression $D = \frac{1}{3}\bar{v}\lambda$ is a result of equilibrium statistical mechanics. In 1905, **Albert Einstein** approached diffusion from a different angle, considering the *non-equilibrium* random walk of a Brownian particle in a fluid.
Einstein derived a relation connecting the diffusion coefficient to purely equilibrium quantities: the particle's mobility μ (its drift speed per unit applied force) and

the thermal energy $k_B T$:

$$D = \mu\, k_B T. \tag{2.85}$$

This **Einstein–Smoluchowski relation** is a profound example of the *fluctuation-dissipation theorem*: a system's response to an external perturbation (mobility) is related to its spontaneous thermal fluctuations (diffusion). It also provided the first quantitative method for measuring Avogadro's number from macroscopic observations of Brownian motion — one of the decisive experimental confirmations of the atomic hypothesis.
The kinetic-theory result $D = \frac{1}{3}\bar{v}\lambda$ emerges as a special case of Einstein's framework when the molecule's mobility is evaluated from its mean free path and average speed.

Insight

From Microscopic Chaos to Macroscopic Order The mean free path λ is the bridge between molecular chaos and macroscopic order. The same quantity that sets the scale for the *disorder* of molecular motion (through the speed distribution) also determines the three macroscopic *order-restoring* mechanisms — viscosity, conduction, and diffusion — that drive a gas back towards equilibrium whenever it is perturbed.

Chapter 3

Real Gases

3.1 Andrews' Experiment on CO_2

In an attempt to liquefy CO_2 at different temperatures, Thomas Andrews conducted experiments in 1869. He applied pressure while keeping Temperature fixed, and observe how volume changes with pressure.

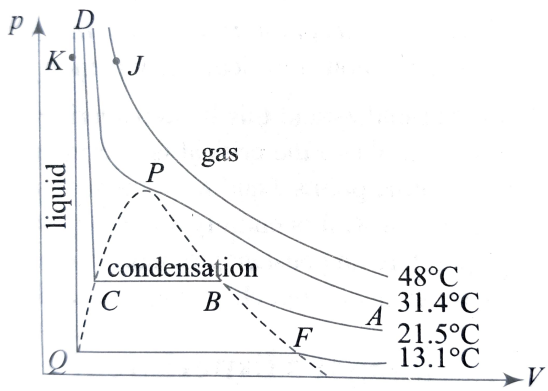

Figure 3.1. Andrew's Isotherms of CO_2

key observations:

- At high temperatures, the isotherms resemble those predicted by the ideal gas law ($PV =$ Constant, Rectangular Hyperbola).
- At lower temperatures, the isotherms exhibit a horizontal portion CB, where pressure remains constant over a range of volumes. (sudden shrink in gas volume, Condensation of gas into liquid), This horizontal region CB corresponds to the coexistence of liquid and gaseous phases.
- As the temperature increases, the horizontal portion of the isotherm shortens and finally disappears at a particular temperature (Point P) called the **critical temperature** $T_c = 31.4^0C$.

Experiment 3.1.1

At the critical point P:
The pressure and volume at this point are known as the **critical pressure**

P_c and **critical volume** V_c, respectively.

- **Critical Temperature (T_c):** The temperature above which a gas cannot be liquefied by pressure alone.
- **Critical Pressure (p_c):** The pressure required for liquefaction at T_c.
- **Critical Volume (V_c):** The volume occupied by 1 mole at p_c and T_c.

- At 13.4°C, CO_2 begins to liquefy at approximately 50 atm (Point P). During liquefaction, pressure remains constant (Line BC) until the gas is entirely liquid. The steep curve CD and QK shows that liquids are highly incompressible.
- The critical point (T_c, P_c, V_c) is a point of inflection (where curve goes concave to convex).

The critical point corresponds to a point of inflection on the critical isotherm:

$$\left(\frac{\partial P}{\partial V}\right)_T = 0, \tag{3.1}$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0. \tag{3.2}$$

3.2 Van der Waals Gas Equation

The ideal gas equation of state assumes that gas molecules occupy negligible volume and exert no forces on one another.

In real gases, both assumptions fail. van der Waals introduced corrections to account for the finite size of molecules and intermolecular attractions.

3.2.1 Correction for Finite Molecular Volume

Real gas molecules have a finite size and therefore exclude a certain volume from being available for molecular motion. Lets calculate the volume not availble to a molecule in Total Volume V,

When two hard spherical molecules of radius r approach each other, the center of one molecule cannot come within a distance $2r$ of the center of the other. Hence, the unavailable volume around each molecule is a sphere of radius $2r$.

The non-available volume at per two molecule location is therefore

$$v'_{\text{non-available}} = \frac{4}{3}\pi(2r)^3 = 8\left(\frac{4}{3}\pi r^3\right). \tag{3.3}$$

Since this non-available volume is shared equally between two molecules, the effective non-available volume per molecule becomes

$$v_{\text{non-available}} = \frac{v'_{non-available}}{2} = 4\left(\frac{4}{3}\pi r^3\right). \tag{3.4}$$

If there are n mole (gas of N molecules, $N = nN_A$) of molecule are in total Volume V, than non-available Volume to a molecule will be

$$V_{non-available} = 4nN_A\left(\frac{4}{3}\pi r^3\right) \tag{3.5}$$

In the ideal gas scene, gas in volume V (number of moles $= n$) has a volume of V available for molecular motion. Here, the volume available for molecular motion will be reduced by $V_{non-available}$. where

$$V_{\text{available}} = V - V_{non-available}. \tag{3.6}$$

Accordingly, the volume V in the ideal gas equation must be replaced by $V - 4nN_A\left(\frac{4}{3}\pi r^3\right)$.

Ideal gas equation now reduced to

$$PV_{available} = nRT \tag{3.7}$$

$$P\left[V - 4nN_A\left(\frac{4}{3}\pi r^3\right)\right] = nRT \tag{3.8}$$

or we can introduce a parameter b such that

$$P\left(V - nb\right) = nRT \tag{3.9}$$

where b is the volume not available for a molecule for motion in Volume V, that contains 1 mole of gas.

$$\boxed{b = 4N_A\left(\frac{4}{3}\pi r^3\right) = \frac{1}{n}V_{non-available}} \tag{3.10}$$

3.2.2 Correction for Intermolecular Attraction

In a real gas, attractive forces act between molecules. A molecule approaching the container wall experiences an inward pull due to neighboring molecules, resulting in a reduction of momentum transferred to the wall. Consequently, the observed pressure P is less than the pressure P_{ideal} that would be exerted in the absence of intermolecular attractions.

The magnitude of this pressure reduction depends on the number of interacting molecular pairs density. In Gas of N molecules, one molecule can make pair with $(N-1)$ other molecules. hence total of N molecules can make $N(N-1)/2 \propto N^2$ pairs.

number of interacting pairs per unit volume

$$\text{pair density} \propto \frac{N^2}{V^2} \tag{3.11}$$

Since pressure is determined by local molecular interactions near the container wall and intermolecular attraction is a pairwise effect, the reduction in pressure is proportional to the local density of interacting molecular pairs, which varies as the square of molecular density

$$\Delta P \propto \frac{N^2}{V^2} \tag{3.12}$$

$$P_{ideal} - P = \frac{an^2}{V^2} \tag{3.13}$$

where a is a constant characteristic of the strength of intermolecular attraction. P_{ideal} is given by Eq. 3.9

$$P_{ideal} = \frac{nRT}{(V - nb)} \tag{3.14}$$

By equation 3.14 and 3.14 :

$$\boxed{\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT.} \tag{3.15}$$

- The constant a measures the strength of intermolecular attraction. and Hence Larger values of a correspond to gases that are more easily liquefied.
- The constant b represents the effective molecular volume.

Historical Note

Johannes Diderik van der Waals His equation of state was the first to bridge the gap between gases and liquids, establishing the principle of the "continuity of states".

Limitations of van der Waals Equation
Despite its success, the van der Waals equation has limitations:

- It fails to accurately predict behavior near the critical point.
- Quantitative agreement with experiments is limited.
- It does not account for molecular shape or quantum effects.

Nevertheless, it represents a major improvement over the ideal gas model.

3.3 Critical Constants of the van der Waals Gas

3.3.1 The Critical Point — What We Are Looking For

Every real gas, when compressed isothermally at temperatures below a certain threshold, undergoes a phase transition: the gas liquefies. Above a particular temperature, however, no amount of pressure can produce a distinct liquid phase — the gas passes continuously into a dense fluid. The boundary between these two regimes is the **critical point**, characterised by three quantities:

Symbol	Name	Physical meaning
T_c	Critical temperature	Highest T at which liquid and gas can coexist
P_c	Critical pressure	Pressure at the critical point
V_c	Critical molar volume	Molar volume at the critical point

At the critical point the isotherm in the P–V plane has a *horizontal inflection*: the curve is simultaneously flat and has zero curvature. This gives us two conditions that, together with the equation of state itself, form a system of three equations in the three unknowns T_c, P_c, V_c.

3.3.2 The Mathematical Conditions at the Critical Point

For any isotherm $P = P(V_m)$ at constant T, the critical point satisfies:

$$\left(\frac{\partial P}{\partial V_m}\right)_T \bigg|_c = 0 \tag{3.16}$$

$$\left(\frac{\partial^2 P}{\partial V_m^2}\right)_T \bigg|_c = 0 \tag{3.17}$$

where the subscript c denotes evaluation at the critical point. The first condition says the isotherm has zero slope (the familiar flat top of the critical isotherm); the second says this is an inflection point, not merely a local extremum.

3.3.3 The van der Waals Equation

Recall the van der Waals equation of state for one mole of gas:

$$\boxed{P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}} \tag{3.18}$$

We now apply conditions (3.16) and (3.17) to this expression.

First Derivative

Differentiating (3.18) with respect to V_m at constant T:

$$\begin{aligned} \frac{\partial P}{\partial V_m} &= \frac{\partial}{\partial V_m}\left[\frac{RT}{V_m - b} - \frac{a}{V_m^2}\right] \\ &= -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3} \end{aligned} \tag{3.19}$$

Setting this equal to zero at the critical point (T_c, V_c):

$$\frac{RT_c}{(V_c - b)^2} = \frac{2a}{V_c^3} \tag{I}$$

Second Derivative

Differentiating (3.19) once more:

$$\frac{\partial^2 P}{\partial V_m^2} = \frac{\partial}{\partial V_m}\left[-\frac{RT}{(V_m-b)^2} + \frac{2a}{V_m^3}\right]$$
$$= \frac{2RT}{(V_m-b)^3} - \frac{6a}{V_m^4} \tag{3.20}$$

Setting this equal to zero at the critical point:

$$\frac{2RT_c}{(V_c-b)^3} = \frac{6a}{V_c^4} \tag{II}$$

Critical Volume V_c

Divide equation (II) by equation (I) term by term:

$$\frac{\dfrac{2RT_c}{(V_c-b)^3}}{\dfrac{RT_c}{(V_c-b)^2}} = \frac{\dfrac{6a}{V_c^4}}{\dfrac{2a}{V_c^3}} \tag{3.21}$$

$$\frac{2}{V_c-b} = \frac{3}{V_c} \tag{3.22}$$

implies

$$\boxed{V_c = 3b} \tag{3.23}$$

The critical volume is exactly three times the excluded volume parameter b. This is a clean, parameter-free result: the entire volume occupied at the critical point is set by the finite size of the molecules alone.

the Critical Temperature T_c

Substitute $V_c = 3b$ into equation (I):

$$\frac{RT_c}{(3b-b)^2} = \frac{2a}{(3b)^3} \tag{3.24}$$

$$\frac{RT_c}{4b^2} = \frac{2a}{27b^3} \tag{3.25}$$

Solving for T_c:

$$RT_c = \frac{2a}{27b^3} \times 4b^2 = \frac{8a}{27b} \tag{3.26}$$

$$\boxed{T_c = \frac{8a}{27Rb}} \tag{3.27}$$

The critical temperature rises with stronger intermolecular attraction (larger a) and decreases with larger molecular size (larger b) — both trends are physically intuitive.

the Critical Pressure P_c

Substitute $V_c = 3b$ and $T_c = 8a/27Rb$ directly into the van der Waals equation (3.18):

$$\begin{aligned} P_c &= \frac{RT_c}{V_c - b} - \frac{a}{V_c^2} \\ &= \frac{R \cdot \dfrac{8a}{27Rb}}{3b - b} - \frac{a}{(3b)^2} \end{aligned} \tag{3.28}$$

$$\boxed{P_c = \frac{a}{27b^2}} \tag{3.29}$$

a and b in terms of Critical constants

These three results allow us to express the van der Waals constants a and b in terms of directly measurable quantities:

$$a = 3P_cV_c^2 = \frac{27R^2T_c^2}{64P_c}, \qquad b = \frac{V_c}{3} = \frac{RT_c}{8P_c} \tag{3.30}$$

3.3.4 The Critical Compressibility Factor

A particularly revealing combination of the three critical constants is the **critical compressibility factor**:

$$Z_c = \frac{P_cV_c}{RT_c} = \frac{\dfrac{a}{27b^2} \cdot 3b}{R \cdot \dfrac{8a}{27Rb}} \tag{3.31}$$

Simplifying:

$$Z_c = \frac{\dfrac{3a}{27b}}{\dfrac{8a}{27b}} = \frac{3}{8} \tag{3.32}$$

$$\boxed{Z_c = \frac{P_cV_c}{RT_c} = \frac{3}{8} = 0.375} \tag{3.33}$$

Insight

The van der Waals equation predicts $Z_c = 3/8 = 0.375$ for *every* substance — a universal constant, independent of a and b. Experimentally, real gases give Z_c in the range 0.23–0.29, somewhat below this value. The discrepancy arises because the van der Waals equation oversimplifies the repulsive core and the density dependence of the attractive term. Nevertheless, the prediction of a universal Z_c is a profound qualitative success and motivates the *Law of Corresponding States* discussed in the next section.

3.3.5 Reducing the van der Waals Equation

Using T_c, P_c, V_c as natural units, define the **reduced variables**:

$$\pi = \frac{P}{P_c}, \qquad \phi = \frac{V_m}{V_c}, \qquad \tau = \frac{T}{T_c} \tag{3.34}$$

Substitute $P = \pi P_c$, $V_m = \phi V_c$, $T = \tau T_c$ into the van der Waals equation and use the expressions (3.23)–(3.29):

$$\begin{aligned}
\pi P_c &= \frac{R\tau T_c}{\phi V_c - b} - \frac{a}{(\phi V_c)^2} \\
\frac{\pi a}{27b^2} &= \frac{\tau \cdot \dfrac{8a}{27b}}{3\phi b - b} - \frac{a}{9\phi^2 b^2} \\
\frac{\pi}{27b^2} &= \frac{8\tau}{27b(3\phi - 1)} - \frac{1}{9\phi^2 b^2}
\end{aligned} \tag{3.35}$$

Multiplying through by $27b^2$:

$$\pi = \frac{8\tau b}{b(3\phi - 1)} - \frac{3}{\phi^2} = \frac{8\tau}{3\phi - 1} - \frac{3}{\phi^2} \tag{3.36}$$

$$\boxed{\left(\pi + \frac{3}{\phi^2}\right)(3\phi - 1) = 8\tau} \tag{3.37}$$

Insight

The Law of Corresponding States. Equation (3.37) contains *no material constants* — a, b, and R have all cancelled. This means that *all* van der Waals gases obey the same equation when expressed in reduced variables. Two gases at the same reduced temperature τ and reduced pressure π will have the same reduced volume ϕ and the same compressibility factor Z. This remarkable prediction — the Law of Corresponding States — is approximately obeyed by real non-polar gases, and forms the basis of generalised correlations used throughout chemical engineering.

3.4 Characteristic Temperatures of a vdW gas

Critical Temperature

Definition 3.4.1. The *critical temperature* T_c of a gas is the highest temperature above which the gas cannot be liquefied by the application of pressure alone.

For a van der Waals gas, the critical temperature is obtained from the critical point conditions

$$\left(\frac{\partial P}{\partial V}\right)_T = 0, \qquad \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0,$$

and is given by

$$T_c = \frac{8a}{27Rb}.$$

Boyle Temperature

Definition 3.4.2. The *Boyle temperature* T_B is the temperature at which a real gas obeys Boyle's law most closely over an extended range of pressures.

At this temperature, the second virial coefficient vanishes:

$$B(T_B) = 0.$$

For a van der Waals gas,

$$B(T) = b - \frac{a}{RT},$$

which gives

$$T_B = \frac{a}{Rb}.$$

Temperature of Inversion

Definition 3.4.3. The *temperature of inversion* T_i is the temperature at which the Joule–Thomson coefficient

$$\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_H$$

changes sign.

For a van der Waals gas, the Joule–Thomson coefficient is

$$\mu_{JT} = \frac{1}{C_P}\left(\frac{2a}{RT} - b\right).$$

Setting $\mu_{JT} = 0$, the (maximum) inversion temperature is

$$T_i = \frac{2a}{Rb}.$$

Relationship Between the Temperatures

Using the above expressions, we obtain

$$T_i = 2T_B, \qquad T_c = \frac{8}{27}T_B,$$

and hence

$$T_i = \frac{27}{4}T_c.$$

Exercise 3.4.1

Important Excercise

(a) Determine the SI units for the van der Waals constants a and b based on the equation 3.15.

(b) Calculate Critical constants using Eq. 3.1 and 3.2 for Van der waals Equation of state.

Answer:

$$V_c = 3b, \tag{3.38}$$

$$P_c = \frac{a}{27b^2}, \tag{3.39}$$

$$T_c = \frac{8a}{27Rb}. \tag{3.40}$$

(c) Determine the work performed by a gas following a van der Waals equation during isothermal expansion.

(d) Using Moler Volume $V_m = \frac{V}{n}$, Find the Value of Compressibility factor $Z = \frac{PV_m}{RT}$ for the vdW Equation of state and expand in terms of Negative power of V_m.

(e) Solve $\frac{\delta Z}{\delta \rho}|_{\rho=0,T=T_B} = 0$ where $\rho = V_m^{-1}$: molar density for vdW equation of state. where T_B is known as Boyle Temperature.

answer:

$$T_B = \frac{a}{Rb} \tag{3.41}$$

(f) for dilute gas ($\rho \xrightarrow[\text{dilute gas}]{\text{high} \quad V_m} 0$), proof that vdW Equation of state reduces to:

$$PV = nRT_B \tag{3.42}$$

(g) Determine the temperature (T_i called inversion Temperature) at which the condition

$$T\left(\frac{\partial V}{\partial T}\right)_P = V$$

is satisfied.

answer:

$$T_i = \frac{2a}{Rb} \tag{3.43}$$

(h) differerence of heat capacities is given by:

$$\boxed{C_P = C_V + P\left(\frac{\partial V}{\partial T}\right)_P} \tag{3.44}$$

calculate $C_P - C_V$ for van-der waals gas.

(i) Try all the above calculations for other Equation of states given in the Appendix **??**

3.5 Virial Equation of State

Historical Note

Kamerling Onnes
The same guy famous for discovering superconductivity, Also studied the behaviour of real gases. He proposed an idea of write real gas equation as virial equation.

A more general description of real gases is provided by the virial equation of state:

$$Z = \frac{PV_m}{RT} = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \cdots \tag{3.45}$$

where $B(T)$, $C(T)$, etc., are temperature-dependent virial coefficients. These coefficient represents contributions from two-body, three-body and higher-order intermolecular interactions.

since, Molar density $\rho = \frac{n}{V} = V_m^{-1}$,
Equation 3.45 takes the following form:

$$Z = \frac{P}{\rho RT} = 1 + B(T)\rho + C(T)\rho^2 + ... \tag{3.46}$$

- The second coefficient B is particularly important; it represents pairwise molecular interactions.
- The **Boyle Temperature** is defined as the temperature at which $B = 0$, where the gas obeys Boyle's law over a wide range of pressure.
- factor $Z = \frac{PV}{nRT} = \frac{PV_m}{RT} = \frac{P}{\rho RT}$ is called **compressibility factor**. For Ideal gas, Z = 1. And for real gases it deviates from 1.

Chapter 4

First Law of Thermodynamics

4.1 Point Function and Path Function

In thermodynamics, physical quantities are classified according to whether their values depend only on the state of the system or also on the manner in which the process is carried out. This leads to the concepts of *point functions* and *path functions*.

Point Function (State Function)

A **point function**, also called a *state function*, is a quantity whose value depends only on the state of the system and not on the path followed to reach that state.

If a system changes from an initial state A to a final state B, the change in a point function is uniquely determined by these two states:

$$\Delta X = X(B) - X(A), \tag{4.1}$$

and is independent of the intermediate steps of the process.

Mathematically, the differential of a point function is an *exact differential*, denoted by dX.

Examples of point functions:

Internal energy (U), Entropy (S), Enthalpy (H), Pressure (P), Volume (V), Temperature (T)

For a cyclic process,

$$\oint dX = 0, \tag{4.2}$$

which is a defining property of point functions.

Path Function

A **path function** is a quantity whose value depends on the path taken during a process and not solely on the initial and final states.

For a process from state A to state B, the value of a path function differs for different paths connecting the same states.

The differential of a path function is an *inexact differential* and is represented by symbols such as δQ or δW.

Examples of path functions:

- Heat (Q)

- Work (W)

For a cyclic process,

$$\oint \delta Q \neq 0, \qquad \oint \delta W \neq 0, \tag{4.3}$$

in general.

Insight

System possess State functions of it Point functions represent properties stored in the system and characterize its equilibrium state. Path functions, on the other hand, represent modes of energy transfer across the system boundary during a process.
Thus, a system *possesses* internal energy or entropy, but it does not *possess* heat or work.

4.2 Internal Energy: The 1st Thermodynamic Potential

Internal energy is the total energy contained within a thermodynamic system. It is a State Function, meaning its value is determined solely by the current state of the system and not by how the system reached that state.

From a microscopic perspective, a system (such as a gas) consists of molecules in constant, random motion. The total energy of these molecules can be broken down into several components based on their Degrees of Freedom:

Translational Kinetic Energy (K_{trans}): Energy due to the linear motion of the center of mass of molecules. For the ith molecule:

$$K_{trans,i} = \frac{1}{2}mv_i^2$$

Rotational Kinetic Energy (K_{rot}): Energy associated with the rotation of molecules about their axes:

$$K_{rot,i} = \frac{1}{2}I\omega_i^2$$

Vibrational Energy (E_{vib}): Energy from the internal oscillation of atoms within a molecule (possessing both kinetic and potential components):

$$E_{vib,i} = K_{vib} + U_{vib}$$

Intermolecular Potential Energy (U_{pot}): Energy due to the forces of attraction (Van der Waals forces) or repulsion between molecules.

The **Internal Energy (U)** is the sum of all these microscopic forms of energy for all N particles in the system:

$$U = \sum_{i=1}^{N} \left(\frac{1}{2}mv_i^2 + \frac{1}{2}I\omega_i^2 + E_{vib,i} \right) + U_{potential} + \ldots \tag{4.4}$$

For an **Ideal Gas**, we assume that intermolecular forces are negligible ($U_{potential} \approx 0$). Thus, the internal energy of an ideal gas is purely a sum of the kinetic energies of its molecules.

4.2.1 Joule Expansion

In this experiment, a gas expands into a vacuum within a thermally insulated container. During the expansion, the Volume (V) increases and the Pressure (P) decreases. Despite the massive changes in P and V, the **Temperature (T)** of the gas remains constant (isothermal process).

Argument

Since the state changed (P and V altered) but the internal energy remained constant (as no work was done and no heat was exchanged), Joule concluded that U does not depend on volume or pressure. Instead, it is linked to the variable that did not change: Temperature.

Conclusion

For an ideal gas, the internal energy is a function of **absolute temperature only**:

$$\boxed{U = U(T)} \tag{4.5}$$

This implies that as long as the temperature of an ideal gas is constant, its internal energy remains unchanged, regardless of how the pressure or volume varies.

4.3 The First Law of Thermodynamics

The First Law of Thermodynamics is a statement of the conservation of energy.

> The heat δQ supplied to a system is used partly to increase the internal energy of the system and partly to perform work.

In finite form, the First Law is written as

$$Q = \Delta U + W \tag{4.6}$$

where W is the work done by the system.

In differential form,

$$\delta Q = dU + \delta W \tag{4.7}$$

or, equivalently, the more convenient form,

$$dU = \delta Q - \delta W \tag{4.8}$$

4.3.1 Mechanical (pressure–volume) work

If a gas exerts pressure P on its surroundings and its volume changes from V to $V + dV$, the infinitesimal work done by the system is

$$\delta W = P\,dV \tag{4.9}$$

Hence, the First Law becomes

$$\boxed{dU = \delta Q - P\,dV} \tag{4.10}$$

4.3.2 Generalized work

In a general system, work may be performed by forces other than pressure. Suppose generalized forces J_i act on the system and produce corresponding infinitesimal generalized displacements dX_i. Then the generalized work done *on* the system is

$$\delta W' = \sum_i J_i \, dX_i \tag{4.11}$$

Here, J_i are called generalized forces and X_i are the corresponding generalized displacements.

System	Generalized Force	Generalized Displacement	Work Term
Stretched wire	Tension F	Length dL	$F\,dL$
Liquid film	Surface tension γ	Area dA	$\gamma\,dA$
Magnetic material	Magnetic field H	Magnetization dM	$H\,dM$
Dielectric	Electric field E	Polarization dP	$E\,dP$

Table 4.1. Examples of generalized work terms

The First Law including generalized work terms may be written as

$$\delta Q + \delta W' = dU + \delta W \tag{4.12}$$

where the left-hand side represents energy supplied to the system (heat and work done on the system), while the right-hand side represents the increase in internal energy and the work done by the system.

Thus, the First Law takes the generalized form

$$dU = \delta Q - P\,dV + \sum_i J_i \, dX_i \tag{4.13}$$

We can include the term $-P\,dV$ in $J_k\,dX_k$ by identifying the pressure as a generalised force and the volume as the corresponding generalised displacement. Thus, taking

$$J_k = -P, \qquad X_k = V,$$

the work term is written as

$$\delta W = J_k\,dX_k = -P\,dV.$$

$$\boxed{dU = \delta Q + \sum_i J_i \, dX_i} \tag{4.14}$$

4.3.3 Chemical work and chemical potential

In systems where the number of particles is not conserved (such as open systems, chemical reactions, or particle exchange with a reservoir), an additional form of work appears known as *chemical work*.

If N_i denotes the number of particles of the i-th species and μ_i is the corresponding chemical potential, then the chemical work term is

$$\delta W_{\text{chem}} = -\sum_i \mu_i \, dN_i \tag{4.15}$$

The negative sign indicates that an increase in particle number increases the internal energy of the system.

Including chemical work, the most general form of the First Law becomes

$$\boxed{dU = \delta Q + \sum_i J_i \, dX_i + \sum_i \mu_i \, dN_i} \tag{4.16}$$

4.4 Applications of the First Law of Thermodynamics

4.4.1 Heat Capacity (C)

The heat capacity of a system is defined as the amount of heat required to raise the temperature of a substance containing n moles by one degree (1 K or 1°C). It is defined as

$$C = \frac{\delta Q}{dT} \qquad \Longrightarrow \qquad \delta Q = C \, dT \tag{4.17}$$

Since gases are compressible, the heat required to produce a given temperature change depends on the conditions under which heating takes place. Therefore, different heat capacities are defined depending on whether the volume or the pressure of the system is kept constant.

Heat Capacity at Constant Volume (C_V)

Consider a gas confined in a rigid container so that the volume remains constant ($dV = 0$). The work done by the system is then

$$\delta W = P \, dV = 0 \tag{4.18}$$

According to the First Law of Thermodynamics,

$$dU = \delta Q_V \tag{4.19}$$

Hence, the heat supplied at constant volume goes entirely into increasing the internal energy of the system. The heat capacity at constant volume is therefore defined by

$$\boxed{dU = C_V \, dT} \tag{4.20}$$

For an ideal gas, the internal energy U is a function of temperature only. Consequently, the relation $dU = C_V dT$ holds for an ideal gas irrespective of the nature of the process.

Heat Capacity at Constant Pressure (C_P)

If a gas is heated while maintaining constant pressure, the system is allowed to expand. In this case, the heat supplied to the system is partly used to increase the internal energy and partly to perform expansion work.

From the First Law,

$$\delta Q_P = dU + P\,dV \tag{4.21}$$

By definition,

$$\delta Q_P = C_P\,dT \tag{4.22}$$

Therefore,

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P \tag{4.23}$$

Using the definition of C_V, this may be written as

$$\boxed{C_P = C_V + P\left(\frac{\partial V}{\partial T}\right)_P} \tag{4.24}$$

Since the system performs work during expansion at constant pressure, a greater amount of heat is required to produce the same temperature rise than in the constant volume case. Hence,

$$C_P > C_V \tag{4.25}$$

Insight

Mayer's relation For an ideal gas obeying the equation of state $PV = nRT$,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P} \tag{4.26}$$

Substituting into the previous equation, we obtain Mayer's relation.

$$\boxed{C_P - C_V = nR} \tag{4.27}$$

Insight

Enthalpy: Second thermodynamic potential
we defined, Heat capacity at constant volume to be:

$$C_V = \left(\frac{\delta Q}{\delta T}\right)_V = \left(\frac{\delta U + PdV}{\delta T}\right)_V = \frac{dU}{dT} \tag{4.28}$$

For heat capacity at constant pressure:

$$C_P = \left(\frac{\delta Q}{\delta T}\right)_P = \left(\frac{dU + PdV}{\delta T}\right)_P = \frac{d(U + PV)}{dT} = \frac{dH}{dT} \tag{4.29}$$

where

$$H = U + PV \tag{4.30}$$

Known as the second thermodynamic potential, called enthalpy of the system.
For ideal gas, its only function of Temperature (similar to the Internal energy).

4.4.2 Adiabatic Process: Derivation of $PV^\gamma = \text{constant}$

For an adiabatic process, no heat is exchanged with the surroundings:

$$\delta Q = 0 \tag{4.31}$$

The First Law of Thermodynamics,

$$dU = \delta Q - P\,dV, \tag{4.32}$$

therefore reduces to

$$dU = -P\,dV. \tag{4.33}$$

For an ideal gas, the internal energy depends only on temperature:

$$dU = C_V\,dT. \tag{4.34}$$

Hence,

$$C_V\,dT = -P\,dV. \tag{4.35}$$

The equation of state of an ideal gas is

$$PV = nRT, \tag{4.36}$$

which implies

$$P\,dV + V\,dP = nR\,dT. \tag{4.37}$$

Using the definition of heat capacities:

At constant pressure,

$$\delta Q_P = C_P\,dT = dU + P\,dV, \tag{4.38}$$

which gives

$$C_P = C_V + P\left(\frac{\partial V}{\partial T}\right)_P. \tag{4.39}$$

For an ideal gas,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P}, \tag{4.40}$$

and therefore Mayer's relation follows:

$$C_P - C_V = nR. \tag{4.41}$$

Combining Eqs. (4.35) and (4.37):

Eliminating dT between the two equations and using $\gamma = C_P/C_V$, we obtain

$$\frac{dP}{P} + \gamma\frac{dV}{V} = 0. \tag{4.42}$$

Integrating,

$$\boxed{PV^\gamma = \text{constant}}. \tag{4.43}$$

4.4.3 Relation Between Adiabatic and Isothermal Elasticity

For a gas, the relevant elastic constant is the *bulk modulus*, defined as the ratio of stress to strain:

$$E = \frac{\text{stress}}{\text{strain}} = -V\left(\frac{\partial P}{\partial V}\right). \tag{4.44}$$

Depending on the thermodynamic condition, two elastic moduli can be defined.

Isothermal Elasticity

For an isothermal process,

$$PV = \text{constant}. \tag{4.45}$$

Differentiating,

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{P}{V}. \tag{4.46}$$

Hence, the isothermal elastic modulus is

$$E_T = -V\left(\frac{\partial P}{\partial V}\right)_T = P. \tag{4.47}$$

Adiabatic Elasticity

For an adiabatic process,

$$PV^\gamma = \text{constant}, \tag{4.48}$$

where $\gamma = C_P/C_V$.

Differentiating,

$$\left(\frac{\partial P}{\partial V}\right)_S = -\gamma\frac{P}{V}. \tag{4.49}$$

Thus, the adiabatic elastic modulus becomes

$$E_S = -V\left(\frac{\partial P}{\partial V}\right)_S = \gamma P. \tag{4.50}$$

Ratio of Elasticities

The ratio of adiabatic to isothermal elasticity is therefore

$$\boxed{\frac{E_S}{E_T} = \gamma = \frac{C_P}{C_V}} \tag{4.51}$$

Chapter 5

Second Law of Thermodynamics

The first law of thermodynamics establishes the principle of energy conservation—it tells us that energy can neither be created nor destroyed, only converted from one form to another. However, it places no restriction on the *direction* of energy transformations. A perfectly cold cup of tea spontaneously warming up by absorbing heat from its surroundings would not violate the first law, yet it never happens in nature.

The **Second Law of Thermodynamics** provides the missing directionality. It distinguishes between processes that are allowed by energy conservation and those that are actually observed in nature. In doing so it introduces two powerful concepts:

- The *efficiency limit* on converting heat into work.
- The concept of *entropy*, which always increases in any natural (irreversible) process.

This chapter develops the second law through its classical statements, the concept of reversibility, the ideal Carnot engine, the absolute temperature scale, and two practical engine cycles—the Otto and Diesel cycles.

5.1 Heat Engine

5.1.1 Working Principle

A **heat engine** is a device that converts thermal energy (heat) into useful mechanical work by operating in a cycle between two thermal reservoirs:

- A **hot reservoir** (source) at temperature T_H, which supplies heat Q_H to the working substance.
- A **cold reservoir** (sink) at temperature $T_C < T_H$, to which the engine rejects heat Q_C.

The working substance absorbs Q_H, converts a portion W into mechanical work, and rejects the remainder Q_C to the cold reservoir, completing one cycle.

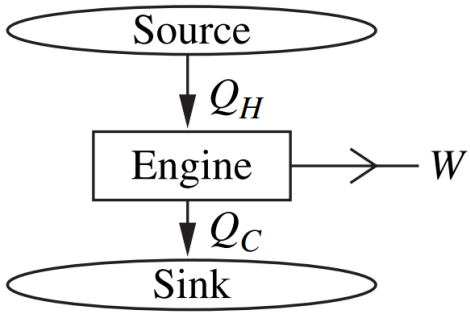

Figure 5.1. Schematic of an idealised heat engine operating between a hot reservoir at T_H and a cold reservoir at T_C.

5.1.2 Energy Balance — First Law Applied to a Cycle

Because the engine operates in a complete *cycle*, the internal energy of the working substance returns to its initial value after each cycle: $\Delta U_{\text{cycle}} = 0$. Applying the first law,

$$\Delta U_{\text{cycle}} = Q_{\text{net}} - W = 0 \implies W = Q_{\text{net}} = Q_H - Q_C. \tag{5.1}$$

Important

All heat and work quantities are treated as *positive* numbers. Q_H is the heat absorbed from the hot source, Q_C is the heat rejected to the cold sink, and W is the net work delivered by the engine. The sign convention is built into the energy-balance equation.

5.1.3 Thermal Efficiency

The thermal efficiency η of a heat engine measures how effectively the absorbed heat is converted into work:

$$\boxed{\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_C}{Q_H}.} \tag{5.2}$$

Since $Q_C > 0$ always (some heat must be rejected), we have $\eta < 1$, confirming that a *perfect* heat engine with $\eta = 1$ is impossible.

5.2 Refrigerator

5.2.1 Working Principle

A **refrigerator** (or heat pump) is a device that operates in a cycle in the *reverse* sense to a heat engine. External work W is supplied to transfer heat Q_C from a cold body to a hot body—a process that does not occur spontaneously.

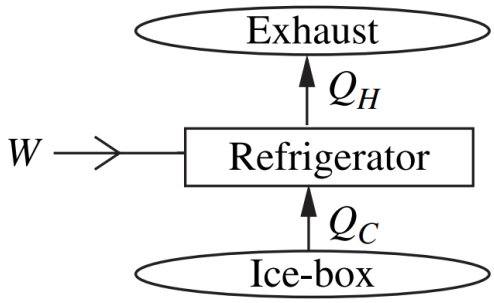

Figure 5.2. Schematic of an idealised refrigerator. Work W is supplied to extract heat Q_C from the cold reservoir and deliver Q_H to the hot reservoir.

5.2.2 Energy Balance

Again using $\Delta U_{\text{cycle}} = 0$:

$$W + Q_C = Q_H. \tag{5.3}$$

5.2.3 Coefficient of Performance

For a refrigerator the relevant figure of merit is the **coefficient of performance** (COP), denoted ω, which measures how much heat is extracted per unit of work input:

$$\boxed{\omega = \frac{Q_C}{W} = \frac{Q_C}{Q_H - Q_C}.} \tag{5.4}$$

Unlike efficiency, ω can be greater than 1.

Insight

A heat pump is the same device as a refrigerator but optimised for delivering heat Q_H to the warm side. Its COP is $\omega_{\text{HP}} = Q_H/W = \omega + 1 > \omega$. An electric heater has COP $= 1$, while a heat pump can deliver several units of heat per unit of electrical work—making it far more energy-efficient.

5.3 Second Law of Thermodynamics

The second law is expressed through two historically important statements, both of which are empirical (based on observation) and logically equivalent.

Second Law — Classical Statements

Kelvin–Planck Statement:
No process is possible whose *sole* result is the complete conversion of heat absorbed from a *single* thermal reservoir into work.

Clausius Statement:
No process is possible whose *sole* result is the transfer of heat from a *colder* body to a *hotter* body without any external work input.

In other words:

- A *perfect* heat engine ($\eta = 100\%$) is impossible (Kelvin–Planck).
- A *perfect* refrigerator (zero work input) is impossible (Clausius).

5.3.1 Equivalence of the Two Statements

The two statements are not independent—they are logically equivalent. We prove this by showing that violating one necessarily implies violating the other.

Violation of Clausius ⇒ Violation of Kelvin–Planck

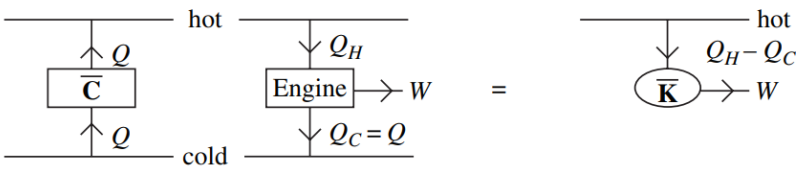

Figure 5.3. A Clausius-violating engine combined with a normal heat engine produces a Kelvin–Planck-violating engine.

Assumption: Suppose the Clausius statement is false. Then there exists an *anti-Clausius device* (ACD) that transfers heat Q from a cold reservoir to a hot reservoir *without any work input.*

Now let a normal heat engine (HE) operate between the *same* two reservoirs, absorbing Q_H from the hot reservoir, performing work W, and rejecting $Q_C = Q$ to the cold reservoir:

$$W = Q_H - Q_C = Q_H - Q. \tag{5.5}$$

Combined system (ACD + HE):

- **Cold reservoir:** receives Q from HE and loses Q to ACD ⇒ *zero net exchange.*
- **Hot reservoir:** supplies Q_H to HE and receives Q from ACD ⇒ net heat drawn from the hot reservoir is $Q_H - Q$.
- **Work output:** $W = Q_H - Q$.

The combined system absorbs $(Q_H - Q)$ from a *single* reservoir and converts it *entirely* into work, with zero interaction with the cold reservoir. This is precisely what the Kelvin–Planck statement forbids.

$$\therefore \quad \text{Violation of Clausius} \implies \text{Violation of Kelvin–Planck.}$$

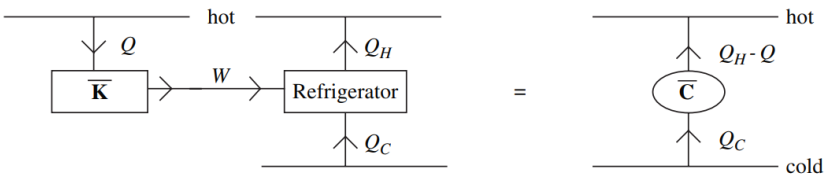

Figure 5.4. A Kelvin–Planck-violating engine driving a refrigerator produces a Clausius-violating device.

Violation of Kelvin–Planck ⇒ Violation of Clausius

Assumption: Suppose the Kelvin–Planck statement is false. Then there exists an *anti-Kelvin device* (AKD) that absorbs heat Q from a single hot reservoir and converts it *entirely* into work $W = Q$, with no heat rejection.

Let this work $W = Q$ drive a normal refrigerator (R) operating between the same two reservoirs. The refrigerator:

- Extracts heat Q_C from the cold reservoir,
- Receives work input $W = Q$,
- Rejects heat $Q_H = Q_C + W = Q_C + Q$ to the hot reservoir.

Combined system (AKD + R):

- **Hot reservoir:** loses Q to AKD and receives $Q_H = Q_C + Q$ from the refrigerator $\Rightarrow$ net gain $= Q_C$.
- **Cold reservoir:** loses Q_C to refrigerator.
- **Net work:** zero (work from AKD drives refrigerator exactly).

The combined system transfers heat Q_C from the cold reservoir to the hot reservoir *without any net work input*—exactly what the Clausius statement forbids.

$$\therefore \quad \text{Violation of Kelvin–Planck} \implies \text{Violation of Clausius.}$$

Summary

Conclusion: Since each statement implies the other on violation, the Kelvin–Planck and Clausius statements of the second law are *logically equivalent*. Any device that violates one necessarily violates the other.

5.4 Reversible and Irreversible Processes

The second law leads naturally to a fundamental classification of all thermodynamic processes.

5.4.1 Reversible Process

Definition

A thermodynamic process is **reversible** if, after the process has been completed, both the *system* and the *surroundings* can be restored *exactly* to their respective initial states by an infinitesimal change in external conditions, leaving *no net change anywhere in the universe.*

For a process to be reversible, the following conditions must be satisfied simultaneously:

1. **Quasi-static:** The process must proceed infinitely slowly through a continuous succession of equilibrium states.
2. **No dissipation:** There must be no friction, viscosity, electrical resistance, inelastic deformation, or any other dissipative mechanism.
3. **No finite temperature differences:** Heat transfer must occur across an infinitesimally small temperature gradient.
4. **No unrestrained expansion:** Pressure differences driving volume changes must be infinitesimal.

Common Misconception

A reversible process is an *idealisation*—a theoretical limiting case that real processes can approach but never exactly attain. Any real process involves some degree of irreversibility.

Examples of reversible processes

- Infinitely slow (quasi-static) isothermal expansion of an ideal gas.
- Heat transfer across an infinitesimally small temperature difference.
- Frictionless, quasi-static compression or extension of a spring.

5.4.2 Irreversible Process

Definition

A thermodynamic process is **irreversible** if the system and surroundings *cannot* both be restored to their original states once the process has occurred. Some permanent change is left in the universe.

Characteristics of irreversible processes:

- The system passes through *non-equilibrium* states during the process.
- Finite gradients of temperature, pressure, or chemical potential are present and drive the process.
- Dissipative effects (friction, viscosity, Joule heating) convert ordered energy into disordered internal energy.

Examples of irreversible processes

- Free (unrestrained) expansion of a gas into a vacuum.
- Heat conduction across a finite temperature difference.
- Mechanical motion with friction.
- Mixing of two different gases or liquids.
- Chemical reactions proceeding spontaneously.
- Joule heating (current through a resistor).

Physical Origins of Irreversibility

Irreversibility is rooted in microscopic disorder:

- **Friction and viscosity:** Directed kinetic energy is converted to random thermal motion of molecules.
- **Finite temperature gradients:** Heat flows spontaneously from hot to cold; reversing this requires external work.
- **Unrestrained expansion:** The gas does no work on a piston; the capacity for doing work is irretrievably lost.
- **Entropy generation:** All irreversible processes generate entropy within the system plus surroundings; the total entropy of the universe increases.

Insight

The second law may be restated in terms of entropy: for any isolated system, $\Delta S \geq 0$, with equality holding only for reversible processes. Irreversible processes *always* produce entropy. This is the microscopic meaning of irreversibility.

5.5 Carnot Engine

Definition

A **Carnot engine** is any engine that

1. operates in a *cycle*,
2. is *completely reversible*, and
3. exchanges heat *only* with a hot reservoir at fixed temperature T_H and a cold reservoir at fixed temperature $T_C < T_H$.

Insight

Reversibility means time-reversal symmetry: a reversible process can be run *backward in time* by simply reversing all heat and work exchanges. A Carnot

engine running in reverse becomes a Carnot refrigerator.

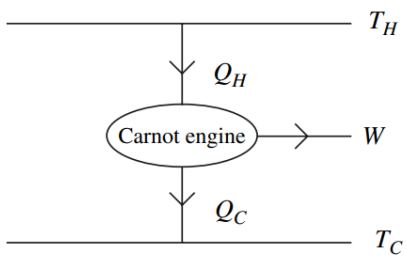

Figure 5.5. A Carnot engine operating between hot reservoir T_H and cold reservoir T_C.

5.5.1 Carnot's Theorem

Theorem: Carnot's Theorem

1. No engine operating between two heat reservoirs can be more efficient than a Carnot engine operating between the *same* two reservoirs.
2. All reversible engines operating between the same two reservoirs have *equal* efficiency, regardless of the working substance.

Proof of Carnot's Theorem

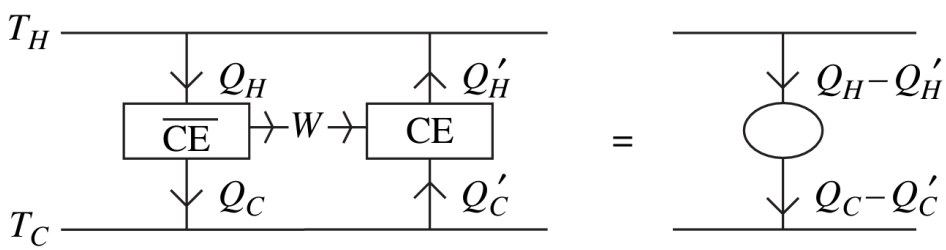

Figure 5.6. Proof of Carnot's theorem: an arbitrary engine $\overline{\text{CE}}$ (acting as a heat engine) combined with a Carnot engine CE (running in reverse as a refrigerator).

Consider two engines operating between the same hot reservoir (T_H) and cold reservoir (T_C):

1. An **arbitrary engine** $\overline{\text{CE}}$ (not necessarily reversible) acting as a heat engine, absorbing Q_H from the hot reservoir, performing work W, and rejecting Q_C to the cold reservoir:
$$\eta_{\overline{\text{CE}}} = \frac{W}{Q_H}, \quad W = Q_H - Q_C. \tag{5.6}$$

2. A **Carnot engine** CE running *in reverse* as a refrigerator, absorbing *exactly* the same work W from $\overline{\text{CE}}$, extracting Q'_C from the cold reservoir, and delivering Q'_H to the hot reservoir:

$$\eta_{\text{CE}} = \frac{W}{Q'_H}, \quad W = Q'_H - Q'_C. \tag{5.7}$$

The work output of $\overline{\text{CE}}$ exactly drives CE (reversed), so no net work crosses the system boundary.

Analysis of the Combined System

For the **hot reservoir**, the net heat drawn is:

$$Q_{\text{net,hot}} = Q_H - Q'_H. \tag{5.8}$$

For the **cold reservoir**, the net heat drawn is:

$$Q_{\text{net,cold}} = Q'_C - Q_C. \tag{5.9}$$

Energy conservation for the combined system (no net work):

$$Q_{\text{net,hot}} = Q_{\text{net,cold}} \implies Q_H - Q'_H = Q'_C - Q_C. \tag{5.10}$$

Applying the Second Law

By the Clausius statement, heat *cannot* flow spontaneously from the cold reservoir to the hot reservoir without net work input. Since the combined system does no net work, any net heat transfer must be *from hot to cold*:

$$Q_{\text{net,hot}} \geq 0 \implies Q_H \geq Q'_H. \tag{5.11}$$

Dividing by $W > 0$:

$$\frac{Q_H}{W} \geq \frac{Q'_H}{W} \implies \frac{1}{\eta_{\overline{\text{CE}}}} \geq \frac{1}{\eta_{\text{CE}}}. \tag{5.12}$$

Therefore,

$$\boxed{\eta_{\text{CE}} \geq \eta_{\overline{\text{CE}}}} \tag{5.13}$$

with equality if and only if $\overline{\text{CE}}$ is itself reversible (i.e., is also a Carnot engine). This proves both parts of Carnot's theorem. □

5.5.2 Thermodynamic Temperature Scale

Carnot's theorem shows that the efficiency of a reversible engine depends *only* on the temperatures of the two reservoirs—not on the working substance. We can use this fact to define an absolute (thermodynamic) temperature scale that is independent of any particular material.

Consider three reservoirs at temperatures $T_1 > T_2 > T_3$ and two Carnot engines:

- Engine CE_{12}: operates between T_1 and T_2, absorbs Q_1, rejects Q_2, performs work W_{12}.

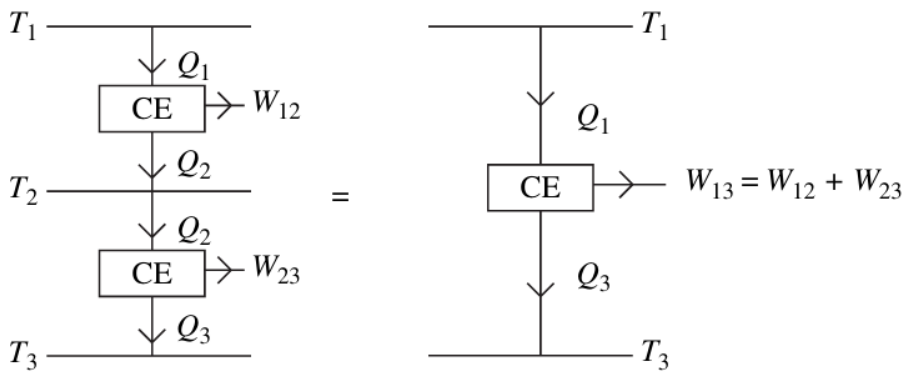

Figure 5.7. Three Carnot engines connected in series to derive the functional form of Carnot efficiency.

- Engine CE_{23}: operates between T_2 and T_3, absorbs Q_2, rejects Q_3, performs work W_{23}.

The heat rejected by CE_{12} is exactly absorbed by CE_{23}—the intermediate reservoir at T_2 has zero net exchange. The *combined* engine is therefore equivalent to a single Carnot engine CE_{13} operating between T_1 and T_3, absorbing Q_1 and rejecting Q_3.

Step 1: Express heat ratios in terms of efficiency

The efficiency of a Carnot engine between T_a and T_b is some function $\eta(T_a, T_b)$ of temperatures alone. Writing $1 - \eta = Q_{\text{out}}/Q_{\text{in}}$:

$$\frac{Q_2}{Q_1} = 1 - \eta(T_1, T_2), \tag{5.14}$$

$$\frac{Q_3}{Q_2} = 1 - \eta(T_2, T_3). \tag{5.15}$$

For the combined engine:

$$\frac{Q_3}{Q_1} = 1 - \eta(T_1, T_3). \tag{5.16}$$

Step 2: Functional equation

Multiplying (5.14) and (5.15):

$$\frac{Q_3}{Q_1} = \frac{Q_2}{Q_1} \cdot \frac{Q_3}{Q_2} = [1 - \eta(T_1, T_2)]\,[1 - \eta(T_2, T_3)]. \tag{5.17}$$

Equating with (5.16):

$$[1 - \eta(T_1, T_2)]\,[1 - \eta(T_2, T_3)] = 1 - \eta(T_1, T_3). \tag{5.18}$$

Step 3: Solving the functional equation

Equation (5.18) must hold for *all* T_1, T_2, T_3. The most general solution is:

$$1 - \eta(T_a, T_b) = \frac{f(T_b)}{f(T_a)}, \tag{5.19}$$

for some function f of temperature alone. (One can verify this by direct substitution into (5.18).)

Step 4: Defining the absolute temperature scale

We are free to *choose* the function f. Lord Kelvin's convention defines:

$$f(T) \equiv T \quad \text{(the Kelvin scale)}, \tag{5.20}$$

so that

$$1 - \eta(T_H, T_C) = \frac{T_C}{T_H}. \tag{5.21}$$

This yields the celebrated result:

$$\boxed{\eta_{\text{Carnot}}(T_H, T_C) = 1 - \frac{T_C}{T_H} = 1 - \frac{Q_C}{Q_H}.} \tag{5.22}$$

Important

Equation (5.22) does *double duty*: it defines the Kelvin temperature scale and gives the maximum possible efficiency of any engine operating between T_H and T_C. Note that temperatures here are in *absolute* Kelvin—never in Celsius or Fahrenheit.

5.5.3 Unattainability of Absolute Zero

As $T_C \to 0\,\text{K}$, equation (5.22) gives $\eta \to 1$. This means $Q_C \to 0$: the engine exchanges no heat with the cold reservoir, and *all* of the absorbed heat Q_H is converted into work. But this is exactly what the Kelvin–Planck statement forbids.

Summary

Third Law Connection: The impossibility of reaching $T_C = 0$ via any cyclic engine follows directly from the second law. This is consistent with the **Third Law of Thermodynamics**: absolute zero is unattainable by any finite sequence of thermodynamic operations.

5.6 The Carnot Cycle

The Carnot engine can be realised with an ideal gas as the working substance. The **Carnot cycle** consists of four reversible processes, forming a closed loop on the P–V diagram:

1. $A \to B$: Isothermal expansion at T_H

2. $B \to C$: Adiabatic expansion ($T_H \to T_C$)
3. $C \to D$: Isothermal compression at T_C
4. $D \to A$: Adiabatic compression ($T_C \to T_H$)

Process $A \to B$: Isothermal Expansion at T_H

The gas expands isothermally and reversibly at the hot-reservoir temperature T_H from (P_A, V_A, T_H) to (P_B, V_B, T_H). Since the internal energy of an ideal gas depends only on temperature, $\Delta U_{AB} = 0$, and by the first law all the heat absorbed equals the work done by the gas:

$$W_{AB} = nRT_H \ln\left(\frac{V_B}{V_A}\right) > 0, \tag{5.23}$$

$$Q_H = W_{AB} = nRT_H \ln\left(\frac{V_B}{V_A}\right) > 0. \tag{5.24}$$

This is the *heat intake* of the engine from the hot reservoir.

Process $B \to C$: Adiabatic Expansion ($T_H \to T_C$)

The gas expands adiabatically ($Q_{BC} = 0$) from (P_B, V_B, T_H) to (P_C, V_C, T_C). The adiabatic relation gives:

$$T_H V_B^{\gamma-1} = T_C V_C^{\gamma-1}. \tag{5.25}$$

Work done by the gas:

$$W_{BC} = \frac{nR(T_H - T_C)}{\gamma - 1} > 0. \tag{5.26}$$

Process $C \to D$: Isothermal Compression at T_C

The gas is compressed isothermally at T_C from V_C back to $V_D < V_C$. Again $\Delta U_{CD} = 0$, so heat rejected equals work done *on* the gas:

$$W_{CD} = nRT_C \ln\left(\frac{V_D}{V_C}\right) < 0, \tag{5.27}$$

$$Q_C = -W_{CD} = nRT_C \ln\left(\frac{V_C}{V_D}\right) > 0. \tag{5.28}$$

Process $D \to A$: Adiabatic Compression ($T_C \to T_H$)

The gas is compressed adiabatically ($Q_{DA} = 0$) from (P_D, V_D, T_C) back to the initial state (P_A, V_A, T_H):

$$T_C V_D^{\gamma-1} = T_H V_A^{\gamma-1}. \tag{5.29}$$

Work done *on* the gas:

$$W_{DA} = \frac{nR(T_C - T_H)}{\gamma - 1} < 0. \tag{5.30}$$

Adiabatic Constraint: Volume Ratios

Dividing equation (5.25) by equation (5.29):

$$\frac{V_B^{\gamma-1}}{V_A^{\gamma-1}} = \frac{V_C^{\gamma-1}}{V_D^{\gamma-1}} \implies \left(\frac{V_B}{V_A}\right)^{\gamma-1} = \left(\frac{V_C}{V_D}\right)^{\gamma-1}, \tag{5.31}$$

and since $\gamma \neq 1$:

$$\boxed{\frac{V_B}{V_A} = \frac{V_C}{V_D}.} \tag{5.32}$$

This is the key geometric property of the Carnot cycle.

5.6.1 Net Work Done

Since $W_{BC} + W_{DA} = 0$ (the adiabatic works exactly cancel):

$$\begin{aligned} W_{\text{net}} &= W_{AB} + W_{CD} \\ &= nRT_H \ln\left(\frac{V_B}{V_A}\right) + nRT_C \ln\left(\frac{V_D}{V_C}\right) \\ &= nRT_H \ln\left(\frac{V_B}{V_A}\right) - nRT_C \ln\left(\frac{V_C}{V_D}\right). \end{aligned} \tag{5.33}$$

Using the volume-ratio result (5.32), $\ln(V_B/V_A) = \ln(V_C/V_D)$, so:

$$\boxed{W_{\text{net}} = nR(T_H - T_C)\ln\left(\frac{V_B}{V_A}\right).} \tag{5.34}$$

5.6.2 Efficiency of the Carnot Cycle

$$\eta = \frac{W_{\text{net}}}{Q_H} = \frac{nR(T_H - T_C)\ln(V_B/V_A)}{nRT_H \ln(V_B/V_A)}$$

$$\implies \tag{5.35}$$

$$\boxed{\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H},} \tag{5.36}$$

confirming (5.22) from first principles.

Insight

Key insight: The efficiency of the Carnot cycle

- depends *only* on the reservoir temperatures T_H and T_C,
- is *independent* of the nature of the working substance (ideal gas, van der Waals gas, steam, etc.),
- increases as T_H increases or T_C decreases,
- equals 1 only if $T_C = 0\,\text{K}$—unattainable by the third law.

5.7 Otto Cycle (Petrol Engine)

The **Otto cycle** is the ideal thermodynamic model for a *spark-ignition* (petrol/gasoline) internal combustion engine. The working substance is treated as an ideal gas with constant specific heats.

Define the **compression ratio**:

$$r = \frac{V_1}{V_2} > 1, \tag{5.37}$$

where V_1 is the maximum (bottom-dead-centre) volume and V_2 is the minimum (top-dead-centre) volume. Let $\gamma = C_P/C_V$.

The four processes are:

Process $1 \to 2$: Adiabatic Compression

Air–fuel mixture is compressed adiabatically ($Q_{12} = 0$):

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \implies T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1} = T_1\, r^{\gamma-1}. \tag{5.38}$$

Work done *on* the gas (negative sign for work done by gas):

$$W_{12} = nC_V(T_1 - T_2) = nC_V T_1(1 - r^{\gamma-1}). \tag{5.39}$$

Process $2 \to 3$: Constant-Volume Heat Addition

At top-dead-centre, the spark ignites the mixture. Heat Q_{in} is added rapidly at *constant volume* ($V_2 = V_3$):

$$W_{23} = 0 \quad \text{(constant volume} \Rightarrow \text{no boundary work)}. \tag{5.40}$$

$$Q_{\text{in}} = nC_V(T_3 - T_2). \tag{5.41}$$

The pressure rises in proportion to the temperature:

$$\frac{P_3}{P_2} = \frac{T_3}{T_2}. \tag{5.42}$$

Process $3 \to 4$: Adiabatic Expansion (Power Stroke)

The high-pressure combustion gas expands adiabatically ($Q_{34} = 0$):

$$T_3 V_3^{\gamma-1} = T_4 V_4^{\gamma-1}. \tag{5.43}$$

Since $V_3 = V_2$ and $V_4 = V_1$:

$$T_4 = T_3 \left(\frac{V_3}{V_4}\right)^{\gamma-1} = T_3 \left(\frac{V_2}{V_1}\right)^{\gamma-1} = T_3\, r^{1-\gamma}. \tag{5.44}$$

Work done *by* the gas:

$$W_{34} = nC_V(T_3 - T_4). \tag{5.45}$$

Process $4 \to 1$: Constant-Volume Heat Rejection

The exhaust valve opens and heat Q_{out} is rejected at constant volume ($V_4 = V_1$):

$$W_{41} = 0. \tag{5.46}$$

$$Q_{\text{out}} = nC_V(T_4 - T_1). \tag{5.47}$$

5.7.1 Net Work Done

$$W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = nC_V[(T_3 - T_2) - (T_4 - T_1)]. \tag{5.48}$$

5.7.2 Efficiency of the Otto Cycle

$$\eta_{\text{Otto}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}} = 1 - \frac{T_4 - T_1}{T_3 - T_2}. \tag{5.49}$$

We now simplify using the adiabatic relations (5.38) and (5.44):

$$T_2 = T_1 r^{\gamma-1}, \qquad T_4 = T_3 r^{1-\gamma} = \frac{T_3}{r^{\gamma-1}}. \tag{5.50}$$

Substituting into the numerator and denominator of (5.49):

$$T_4 - T_1 = \frac{T_3}{r^{\gamma-1}} - T_1, \tag{5.51}$$

$$T_3 - T_2 = T_3 - T_1 r^{\gamma-1}. \tag{5.52}$$

Factoring out $1/r^{\gamma-1}$:

$$\frac{T_4 - T_1}{T_3 - T_2} = \frac{T_3 - T_1 r^{\gamma-1}}{r^{\gamma-1}(T_3 - T_1 r^{\gamma-1})} = \frac{1}{r^{\gamma-1}}. \tag{5.53}$$

Therefore:

$$\boxed{\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}}.} \tag{5.54}$$

Insight

Efficiency of the Otto Cycle:

- Increases with compression ratio r: higher r means the gas is more strongly heated during compression, so more work is extracted on expansion.
- Increases with γ: gases with higher γ (monatomic > diatomic) give higher efficiency for the same r.
- Is *independent* of the amount of heat supplied Q_{in} or the temperatures T_3, T_1.
- Typical petrol engines have $r \approx 8$–10, giving $\eta_{\text{Otto}} \approx 56\%$–$60\%$ for $\gamma = 1.4$ (air). In practice, efficiencies are 25%–35% due to friction, heat loss, and incomplete combustion.

5.8 Diesel Cycle (Diesel Engine)

The **Diesel cycle** is the ideal thermodynamic model for a *compression-ignition* (diesel) engine. Fuel is ignited by the high temperature produced by compressing air—no spark plug is needed. The key thermodynamic difference from the Otto cycle is that heat is added at *constant pressure* rather than constant volume.

Define:

$$r = \frac{V_1}{V_2} \quad \text{(compression ratio)}, \qquad \rho = \frac{V_3}{V_2} \quad \text{(cut-off ratio, } \rho > 1\text{)}, \qquad \gamma = \frac{C_P}{C_V}. \tag{5.55}$$

The cut-off ratio ρ is the ratio of the volume after fuel injection to the volume at the end of compression. It characterises how much fuel is injected: larger ρ means more fuel and more heat input.

Process $1 \to 2$: Adiabatic Compression

Air is compressed adiabatically ($Q = 0$) from V_1 to V_2:

$$T_2 = T_1 r^{\gamma-1}, \qquad P_2 = P_1 r^{\gamma}. \tag{5.56}$$

Work done *on* the gas:

$$\begin{aligned} W_{12} &= \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} \\ &= \frac{nRT_1}{\gamma - 1}\left(1 - r^{\gamma-1}\right) < 0. \end{aligned} \tag{5.57}$$

Process $2 \to 3$: Constant-Pressure Heat Addition

Fuel is injected and burns at constant pressure $P_2 = P_3$, expanding the gas from V_2 to $V_3 = \rho V_2$:

$$\frac{T_3}{T_2} = \frac{V_3}{V_2} = \rho \implies T_3 = \rho T_2 = \rho T_1 r^{\gamma-1}. \tag{5.58}$$

Heat supplied (at constant pressure):

$$Q_{\text{in}} = nC_P(T_3 - T_2) = nC_P T_2(\rho - 1) = nC_P T_1 r^{\gamma-1}(\rho - 1). \tag{5.59}$$

Work done by the gas at constant pressure:

$$W_{23} = P_2(V_3 - V_2) = nR(T_3 - T_2) = nRT_1 r^{\gamma-1}(\rho - 1) > 0. \tag{5.60}$$

Process $3 \to 4$: Adiabatic Expansion (Power Stroke)

Gas expands adiabatically ($Q = 0$) from V_3 to $V_4 = V_1$:

$$T_3 {V_3}^{\gamma-1} = T_4 {V_4}^{\gamma-1}. \tag{5.61}$$

Substituting $V_3 = \rho V_2 = \rho V_1/r$ and $V_4 = V_1$:

$$T_4 = T_3\left(\frac{V_3}{V_4}\right)^{\gamma-1} = T_3\left(\frac{\rho}{r}\right)^{\gamma-1}. \tag{5.62}$$

Using (5.58):

$$T_4 = \rho T_1 r^{\gamma-1} \cdot \frac{\rho^{\gamma-1}}{r^{\gamma-1}} = T_1 \rho^{\gamma}. \tag{5.63}$$

Work done by the gas:

$$W_{34} = \frac{nR(T_3 - T_4)}{\gamma - 1} = \frac{nRT_1}{\gamma - 1}\left(r^{\gamma-1}\rho - \rho^{\gamma}\right) > 0. \tag{5.64}$$

Process $4 \to 1$: Constant-Volume Heat Rejection

The exhaust valve opens at constant volume ($V_4 = V_1$):

$$W_{41} = 0. \tag{5.65}$$

$$Q_{\text{out}} = nC_V(T_4 - T_1) = nC_V T_1(\rho^{\gamma} - 1). \tag{5.66}$$

5.8.1 Efficiency of the Diesel Cycle

$$\eta_{\text{Diesel}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}} = 1 - \frac{nC_V T_1(\rho^{\gamma} - 1)}{nC_P T_1 r^{\gamma-1}(\rho - 1)}. \tag{5.67}$$

Since $C_P/C_V = \gamma$:

$$\boxed{\eta_{\text{Diesel}} = 1 - \frac{1}{r^{\gamma-1}} \cdot \frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}.} \tag{5.68}$$

5.8.2 Comparison: Otto vs. Diesel Cycle

The factor $\dfrac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}$ in (5.68) is always > 1 for $\rho > 1$ (which can be shown by the inequality of arithmetic and geometric means, or by L'Hôpital's rule as $\rho \to 1$). Therefore, for the *same compression ratio* r:

$$\eta_{\text{Diesel}} < \eta_{\text{Otto}} \quad \text{for the same } r. \tag{5.69}$$

Common Misconception

Common Misconception: Diesel engines are *not* less efficient than petrol engines in practice. Diesel engines operate at *much higher* compression ratios ($r \approx 14$–22) than petrol engines ($r \approx 8$–10) because the fuel is injected *after* compression (avoiding pre-ignition knock). At the same Q_{in}, a higher r more than compensates for the efficiency penalty of the cut-off factor.

Insight

Limiting case: As $\rho \to 1$ (infinitesimally small fuel injection), the Diesel cycle approaches the Otto cycle at the same compression ratio:

$$\lim_{\rho \to 1} \frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)} \overset{\text{L'H}}{=} \lim_{\rho \to 1} \frac{\gamma \rho^{\gamma-1}}{\gamma} = 1,$$

giving $\eta_{\text{Diesel}} \to \eta_{\text{Otto}}$.

Summary

Chapter Summary

- A heat engine converts $Q_H \to W + Q_C$ with $\eta = W/Q_H = 1 - Q_C/Q_H$.
- A refrigerator moves Q_C against the temperature gradient using work W, with COP $= Q_C/W$.
- The second law forbids $\eta = 1$ (Kelvin–Planck) and spontaneous heat flow from cold to hot (Clausius); the two statements are equivalent.
- Reversible processes pass through equilibrium states with no dissipation; all real processes are irreversible.
- Carnot's theorem: $\eta_{\text{rev}} \geq \eta_{\text{irrev}}$ for engines between the same reservoirs.
- Carnot efficiency: $\eta_{\text{Carnot}} = 1 - T_C/T_H$, defining the Kelvin temperature scale.
- Absolute zero is unattainable (second and third laws).
- Otto efficiency: $\eta = 1 - r^{1-\gamma}$ (spark-ignition engines).
- Diesel efficiency: $\eta = 1 - r^{1-\gamma}(\rho^\gamma - 1)/[\gamma(\rho - 1)]$ (compression-ignition engines).

Chapter 6

Entropy

6.1 Clausius Theorem

Clausius Theorem For any cyclic transformation (reversible or irreversible),

$$\oint \frac{\delta Q}{T} \leq 0 \tag{6.1}$$

where δQ is the infinitesimal heat supplied to the system at absolute temperature T.

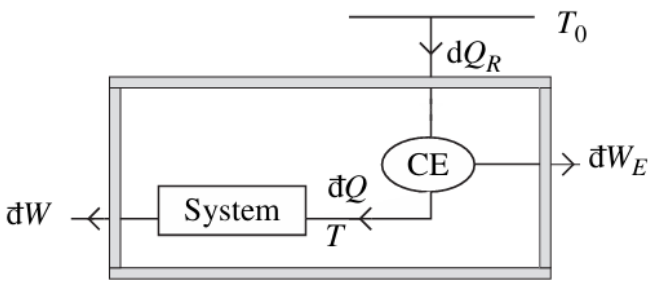

Figure 6.1. Illustration for Clausius theorem using a Carnot engine

Let us consider a Carnot engine working between two reservoirs at temperatures T_0 and T. The Carnot engine absorbs heat δQ_R from the reservoir at temperature T_0 and rejects heat δQ to a system at temperature T. Since the Carnot engine is reversible, it performs work δW_E.

For a reversible Carnot engine operating between temperatures T_0 and T,

$$\frac{\delta Q_R}{\delta Q} = \frac{T_0}{T}. \tag{6.2}$$

Hence,

$$\delta Q_R = T_0 \frac{\delta Q}{T}. \tag{6.3}$$

The system at temperature T absorbs heat δQ from the Carnot engine and performs work δW.

Over a complete cycle, the total heat absorbed by the Carnot engine is

$$Q_R = T_0 \oint \frac{\delta Q}{T}. \tag{6.4}$$

Now consider the *combined system*, consisting of the original system and the Carnot engine together. This combined system absorbs heat Q_R from the single reservoir at temperature T_0 and converts it entirely into work W', where

$$Q_R = W'. \tag{6.5}$$

This appears to violate the second law of thermodynamics (Kelvin–Planck statement), which forbids the complete conversion of heat from a single reservoir into work.

To ensure consistency with the second law, we must have

$$Q_R \leq 0. \tag{6.6}$$

Using Eqs. (6.4) and (6.6), and noting that $T_0 > 0$, we obtain

$$\boxed{\oint \frac{\delta Q}{T} \leq 0} \tag{6.7}$$

This proves Clausius theorem.

6.2 Consequences of Clausius Theorem

Clausius theorem states that for any cyclic process,

$$\oint \frac{\delta Q}{T} \leq 0. \tag{6.8}$$

This result has several important consequences.

6.2.1 Reversible Cyclic Process

For a *reversible* cyclic process, there is no dissipation and no entropy production. Hence the inequality becomes an equality:

$$\oint \frac{\delta Q_{\text{rev}}}{T} = 0. \tag{6.9}$$

This result is crucial, because it suggests that for reversible processes the quantity $\delta Q_{\text{rev}}/T$ behaves in a special way.

6.2.2 Irreversible Cyclic Process

For an *irreversible* cyclic process, dissipation is always present (for example due to friction, finite temperature gradients, etc.). In this case,

$$\oint \frac{\delta Q}{T} < 0. \tag{6.10}$$

Thus, the inequality in Clausius theorem distinguishes between reversible and irreversible processes.

6.2.3 Path Independence for Reversible Processes

Consider two equilibrium states A and B, connected by two different reversible paths R_1 and R_2. Form a closed cycle by going from A to B along R_1 and returning from B to A along R_2 (in reverse direction).

Applying Eq. (6.9) to this reversible cycle,

$$\oint \frac{\delta Q_{\text{rev}}}{T} = \int_A^B \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_1} + \int_B^A \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_2} = 0. \tag{6.11}$$

Therefore,

$$\int_A^B \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_1} = \int_A^B \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_2}. \tag{6.12}$$

This shows that for reversible processes, the integral $\int \delta Q_{\text{rev}}/T$ depends only on the initial and final states, and not on the path.

6.3 Entropy

The result obtained above suggests the existence of a new thermodynamic state function.

Since the integral

$$\int_A^B \frac{\delta Q_{\text{rev}}}{T} \tag{6.13}$$

has the same value for all reversible paths connecting two equilibrium states A and B, it must correspond to the change of some state function.

This state function is called *entropy*.

6.3.1 Definition of Entropy

Entropy, denoted by S, is defined such that its infinitesimal change is

$$dS \equiv \frac{\delta Q_{\text{rev}}}{T}. \tag{6.14}$$

For a finite reversible process between states A and B,

$$S(B) - S(A) = \int_A^B \frac{\delta Q_{\text{rev}}}{T}. \tag{6.15}$$

Thus, entropy is a *state function*, depending only on the equilibrium state of the system.

6.4 Entropy Change for Irreversible Processes

Consider an irreversible process that takes a system from an equilibrium state A to another equilibrium state B. Let the system return from B to A along a reversible path.

Together, these two processes form a closed thermodynamic cycle. According to Clausius theorem, for any cyclic process,

$$\oint \frac{\delta Q}{T} \leq 0. \tag{6.16}$$

The cyclic integral may be expressed as the sum of the integrals over the two paths:

$$\oint \frac{\delta Q}{T} = \int_A^B \frac{\delta Q}{T} + \int_B^A \frac{\delta Q_{\text{rev}}}{T}. \tag{6.17}$$

Since reversing the limits of integration changes the sign,

$$\int_B^A \frac{\delta Q_{\text{rev}}}{T} = -\int_A^B \frac{\delta Q_{\text{rev}}}{T}, \tag{6.18}$$

we obtain

$$\int_A^B \frac{\delta Q}{T} \leq \int_A^B \frac{\delta Q_{\text{rev}}}{T}. \tag{6.19}$$

The equality holds only if the entire cycle is reversible. Since the path from A to B is irreversible, the inequality is strict.

Using the definition of entropy,

$$S(B) - S(A) = \int_A^B \frac{\delta Q_{\text{rev}}}{T}, \tag{6.20}$$

we finally obtain

$$\int_A^B \frac{\delta Q}{T} < S(B) - S(A). \tag{6.21}$$

In differential form, this may be written as

$$\boxed{dS > \frac{\delta Q}{T}} \tag{6.22}$$

This inequality provides a general mathematical statement of the second law of thermodynamics for irreversible processes.

6.5 Entropy Change for an Isolated System

An isolated system does not exchange heat with its surroundings. Hence,

$$\delta Q = 0. \tag{6.23}$$

From the entropy inequality,

$$dS \geq \frac{\delta Q}{T}, \tag{6.24}$$

it follows immediately that

$$dS \geq 0. \tag{6.25}$$

Thus, the entropy of an isolated system can never decrease. It remains constant for reversible processes and increases for irreversible processes.

This result introduces a natural direction (or *arrow*) to thermodynamic processes.

6.5.1 Principle of Increase of Entropy

The behavior of entropy in isolated systems leads to a fundamental statement of the second law of thermodynamics.

Principle of Increase of Entropy The entropy of an isolated system can never decrease:

$$\Delta S \geq 0. \tag{6.26}$$

The equality holds for reversible processes, while the inequality holds for irreversible processes.

6.5.2 Reversible and Irreversible Processes

- **Reversible process:** For a reversible process occurring in an isolated system,

$$\Delta S = 0. \tag{6.27}$$

  The entropy of the system remains constant.

- **Irreversible process:** For an irreversible process occurring in an isolated system,

$$\Delta S > 0. \tag{6.28}$$

  The entropy of the system increases.

Hence, entropy production provides a quantitative measure of irreversibility in a thermodynamic process.

Insight

Arrow of Time The principle of increase of entropy introduces a preferred direction to physical processes. While the laws of mechanics are time-reversible, thermodynamic processes are not.
Spontaneous processes in an isolated system always proceed in the direction of increasing entropy.

6.5.3 Entropy and Equilibrium

For an isolated system in thermodynamic equilibrium,

$$dS = 0. \tag{6.29}$$

Any small spontaneous deviation from equilibrium leads to a decrease of entropy,

$$d^2S < 0. \tag{6.30}$$

Therefore, the equilibrium state of an isolated system corresponds to a state of *maximum entropy*.

Insight

Maximum Entropy Principle Thermodynamic equilibrium in an isolated system is characterized by maximum entropy. This principle provides the final criterion for equilibrium and stability.

6.6 First Law Revisited: Entropy Representation

We now return to the first law of thermodynamics and reformulate it using the concept of entropy.

The first law for an infinitesimal process is

$$\delta Q = dU + \delta W. \tag{6.31}$$

For a reversible process, the heat absorbed by the system can be written in terms of entropy as

$$\delta Q_{\text{rev}} = T\,dS. \tag{6.32}$$

Substituting this into the first law, we obtain

$$T\,dS = dU + \delta W_{\text{rev}}. \tag{6.33}$$

For a simple compressible system performing only $P\,dV$ work,

$$\delta W_{\text{rev}} = P\,dV. \tag{6.34}$$

Hence, the first law becomes

$$\boxed{dU = T\,dS - P\,dV} \tag{6.35}$$

This equation is the *fundamental thermodynamic relation* for a simple system.

6.6.1 Significance of the Entropy Formulation

The introduction of entropy allows heat, which is a path-dependent quantity, to be expressed in terms of exact differentials.

While δQ is not a state function, the quantity $\delta Q_{\text{rev}}/T$ is an exact differential:

$$dS = \frac{\delta Q_{\text{rev}}}{T}. \tag{6.36}$$

Thus, entropy provides a bridge between energy conservation (first law) and irreversibility (second law).

Insight

Unified View The equation

$$dU = T\,dS - P\,dV$$

combines the first and second laws into a single mathematical statement.
Energy conservation determines *how much* energy changes, while entropy determines *how* the process can occur.

- The term $T\,dS$ represents the reversible heat exchange.
- The term $P\,dV$ represents mechanical work.

Chapter 7

Thermodynamic Potentials

A thermodynamic system can be described by many different state functions. However, the choice of *which* state function to use depends critically on the physical constraints imposed on the system. The four fundamental thermodynamic potentials—internal energy U, enthalpy H, Helmholtz free energy F, and Gibbs free energy G—are related by **Legendre transformations**, each one optimally suited to a particular set of natural variables. This chapter develops these potentials systematically, derives the four Maxwell relations, and highlights their physical significance.

7.1 The Fundamental Relation

All thermodynamic potentials descend from a single master equation called the *fundamental relation of thermodynamics*. For a simple compressible system of N particles, the internal energy U satisfies:

$$\boxed{dU = T\,dS - P\,dV + \mu\,dN} \tag{7.1}$$

where:

- T is the *absolute temperature* (intensive),
- S is the *entropy* (extensive),
- P is the *pressure* (intensive),
- V is the *volume* (extensive),
- μ is the *chemical potential* (intensive),
- N is the *number of particles* (extensive).

7.1.1 Extensivity and the Euler Relation

Internal energy is an *extensive* quantity: if every extensive variable is scaled by λ, then U scales by the same factor,

$$U(\lambda S,\, \lambda V,\, \lambda N) = \lambda\, U(S, V, N). \tag{7.2}$$

Differentiating both sides with respect to λ and then setting $\lambda = 1$ (Euler's theorem for homogeneous functions) yields the **Euler relation**:

$$\boxed{U = TS - PV + \mu N.} \tag{7.3}$$

This expresses internal energy entirely in terms of conjugate products: *thermal* (TS), *mechanical* ($-PV$), and *chemical* (μN) contributions.

7.1.2 The Gibbs–Duhem Relation

Taking the total differential of the Euler relation (7.3):

$$dU = T\,dS + S\,dT - P\,dV - V\,dP + \mu\,dN + N\,d\mu. \tag{7.4}$$

Comparing with the fundamental relation (7.1), the surplus terms must vanish:

$$\boxed{S\,dT - V\,dP + N\,d\mu = 0.} \tag{7.5}$$

Insight

Physical meaning of Gibbs–Duhem The intensive variables T, P, and μ are *not independent*. For a single-component system, specifying any two determines the third. More generally, a system with c components has only $c + 1$ independent intensive variables.

7.1.3 The Legendre Transform: Changing Natural Variables

In practice, experiments are often performed under conditions of *controlled temperature* (not controlled entropy), or *controlled pressure* (not controlled volume). It is therefore useful to re-express U in terms of different natural variables.

The mathematical tool for changing natural variables while preserving all thermodynamic information is the **Legendre transformation**. If $f(x)$ is a convex function with slope $p = \partial f/\partial x$, its Legendre transform is:

$$g(p) = f(x) - p\,x, \qquad \text{so that} \quad dg = -x\,dp. \tag{7.6}$$

Applying Legendre transformations to $U(S, V, N)$ by exchanging extensive variables for their conjugate intensive variables generates the four standard thermodynamic potentials, summarised in Table 7.1.

Each potential is a *characteristic function* for its own set of natural variables: given it as a function of its natural variables, all equations of state follow by differentiation alone.

7.2 Internal Energy U

7.2.1 Definition and Physical Content

The **internal energy** U is the total energy stored within a thermodynamic system. It includes the kinetic energy of all microscopic constituents (translational, rotational, vibrational), the potential energy of intermolecular interactions, and any other internal degrees of freedom. It does *not* include the macroscopic kinetic or potential energy of the system as a whole.

Table 7.1. The four thermodynamic potentials, natural variables, and differentials (fixed N implied for brevity in the last column).

Potential	**Defn.**	**Nat. vars.**	**Differential**
Internal energy U	—	S, V	$dU = T\,dS - P\,dV$
Enthalpy H	$U + PV$	S, P	$dH = T\,dS + V\,dP$
Helmholtz free energy F	$U - TS$	T, V	$dF = -S\,dT - P\,dV$
Gibbs free energy G	$U + PV - TS$	T, P	$dG = -S\,dT + V\,dP$

7.2.2 First Maxwell Relation

For a fixed number of particles ($dN = 0$), the fundamental relation reduces to:

$$dU = T\,dS - P\,dV. \tag{7.7}$$

This shows that U is a function of the two independent variables S and V:

$$U = U(S, V). \tag{7.8}$$

Setting $dV = 0$ (constant volume):

$$dU = \left(\frac{\partial U}{\partial S}\right)_V dS = T\,dS \qquad \Longrightarrow \qquad \boxed{T = \left(\frac{\partial U}{\partial S}\right)_V} \tag{7.9}$$

Setting $dS = 0$ (constant entropy):

$$dU = \left(\frac{\partial U}{\partial V}\right)_S dV = -P\,dV \qquad \Longrightarrow \qquad \boxed{-P = \left(\frac{\partial U}{\partial V}\right)_S} \tag{7.10}$$

Since $U(S, V)$ is a well-defined state function, its mixed second partial derivatives must commute. That is, at any point (S, V):

$$\left.\frac{\partial^2 U}{\partial V\,\partial S}\right|_{(S,V)} = \frac{\partial}{\partial V}\left[\left(\frac{\partial U}{\partial S}\right)_V\right]_S = \left.\frac{\partial T}{\partial V}\right|_S, \tag{7.11}$$

and equally:

$$\left.\frac{\partial^2 U}{\partial S\,\partial V}\right|_{(S,V)} = \frac{\partial}{\partial S}\left[\left(\frac{\partial U}{\partial V}\right)_S\right]_V = \left.\frac{\partial(-P)}{\partial S}\right|_V = -\left.\frac{\partial P}{\partial S}\right|_V. \tag{7.12}$$

Since the two mixed derivatives must be equal:

$$\boxed{\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V} \tag{M1}$$

This is the **first Maxwell relation**.

7.3 Enthalpy H

7.3.1 Motivation and Definition

Many physical and chemical processes occur at *constant pressure* (open laboratory vessels, atmospheric reactions, biological systems). Under constant pressure, volume is free to change, and it becomes inconvenient to keep tracking V separately. We want a potential whose natural variable is P (instead of V).

This is achieved by a Legendre transformation: we add the product PV to U, trading the natural variable V for P:

$$\boxed{H \equiv U + PV.} \tag{7.13}$$

Insight

Physical meaning of H Enthalpy is the energy required to *create* the system and also to push back the surroundings (make room for it) against an external pressure P. In a constant-pressure process, the heat absorbed equals ΔH.

7.3.2 Enthalpy and Heat at Constant Pressure

At constant pressure ($dP = 0$) and fixed N:

$$dH\Big|_P = T\,dS = \delta Q_{\text{rev}}. \tag{7.14}$$

Therefore, the heat absorbed reversibly at constant pressure equals the change in enthalpy:

$$\boxed{Q_P = \Delta H.} \tag{7.15}$$

This is why enthalpy is ubiquitous in calorimetry and chemistry: the *enthalpy of reaction* ΔH_{rxn} directly gives the heat released or absorbed in a reaction at constant pressure.

7.3.3 Second Maxwell Relation

Taking the total differential of $H = U + PV$:

$$dH = dU + P\,dV + V\,dP. \tag{7.16}$$

Substituting the fundamental relation $dU = T\,dS - P\,dV$:

$$\begin{aligned} dH &= \big(T\,dS - P\,dV\big) + P\,dV + V\,dP \\ &= T\,dS - P\,dV + P\,dV + V\,dP \end{aligned}$$

$$\boxed{dH = T\,dS + V\,dP.} \tag{7.17}$$

Setting $dP = 0$ (constant pressure):

$$dH = \left(\frac{\partial H}{\partial S}\right)_P dS = T\,dS \qquad \Longrightarrow \qquad \boxed{T = \left(\frac{\partial H}{\partial S}\right)_P} \tag{7.18}$$

Setting $dS = 0$ (constant entropy):

$$dH = \left(\frac{\partial H}{\partial P}\right)_S dP = V\,dP \qquad \Longrightarrow \qquad \boxed{V = \left(\frac{\partial H}{\partial P}\right)_S} \tag{7.19}$$

Since $H(S, P)$ is a well-defined state function, its mixed second partial derivatives must commute. At any point (S, P):

$$\left.\frac{\partial^2 H}{\partial P\,\partial S}\right|_{(S,P)} = \frac{\partial}{\partial P}\left[\left(\frac{\partial H}{\partial S}\right)_P\right]_S = \left.\frac{\partial T}{\partial P}\right|_S, \tag{7.20}$$

and equally:

$$\left.\frac{\partial^2 H}{\partial S\,\partial P}\right|_{(S,P)} = \frac{\partial}{\partial S}\left[\left(\frac{\partial H}{\partial P}\right)_S\right]_P = \left.\frac{\partial V}{\partial S}\right|_P. \tag{7.21}$$

Since the two mixed derivatives must be equal:

$$\boxed{\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P} \tag{M2}$$

This is the **second Maxwell relation**.

7.4 Helmholtz Free Energy F

7.4.1 Motivation and Definition

Consider a system held at constant temperature T by contact with a heat reservoir, at constant volume. It is natural to use (T, V) as control variables. The Legendre transformation that trades the extensive variable S for its conjugate intensive variable T gives:

$$\boxed{F \equiv U - TS.} \tag{7.22}$$

Insight

Physical meaning of F Of the total internal energy U, the portion TS is "thermally bound" — it cannot be converted into work at temperature T. The remainder, $F = U - TS$, is the energy *free* to do work. More precisely, $-\Delta F$ equals the maximum work extractable from the system in any isothermal process.

7.4.2 Third Maxwell Relation

Taking the total differential of $F = U - TS$:

$$dF = dU - T\,dS - S\,dT. \tag{7.23}$$

Substituting the fundamental relation $dU = T\,dS - P\,dV$:

$$\begin{aligned} dF &= \big(T\,dS - P\,dV\big) - T\,dS - S\,dT \\ &= T\,dS - P\,dV - T\,dS - S\,dT \end{aligned}$$

$$\boxed{dF = -S\,dT - P\,dV.} \tag{7.24}$$

The $T\,dS$ terms cancel exactly. This shows that F is naturally a function of T and V:

$$F = F(T, V). \tag{7.25}$$

Writing the total differential of $F(T, V)$ formally:

$$dF = \left(\frac{\partial F}{\partial T}\right)_V dT + \left(\frac{\partial F}{\partial V}\right)_T dV. \tag{7.26}$$

Comparing with Eq. (7.24) term by term:

Setting $dV = 0$ (constant volume):

$$dF = \left(\frac{\partial F}{\partial T}\right)_V dT = -S\,dT \qquad \Longrightarrow \qquad \boxed{-S = \left(\frac{\partial F}{\partial T}\right)_V} \tag{7.27}$$

Setting $dT = 0$ (constant temperature):

$$dF = \left(\frac{\partial F}{\partial V}\right)_T dV = -P\,dV \qquad \Longrightarrow \qquad \boxed{-P = \left(\frac{\partial F}{\partial V}\right)_T} \tag{7.28}$$

Since $F(T, V)$ is a well-defined state function, its mixed second partial derivatives must commute. At any point (T, V):

$$\left.\frac{\partial^2 F}{\partial V\,\partial T}\right|_{(T,V)} = \frac{\partial}{\partial V}\left[\left(\frac{\partial F}{\partial T}\right)_V\right]_T = \left.\frac{\partial(-S)}{\partial V}\right|_T = -\left.\frac{\partial S}{\partial V}\right|_T, \tag{7.29}$$

and equally:

$$\left.\frac{\partial^2 F}{\partial T\,\partial V}\right|_{(T,V)} = \frac{\partial}{\partial T}\left[\left(\frac{\partial F}{\partial V}\right)_T\right]_V = \left.\frac{\partial(-P)}{\partial T}\right|_V = -\left.\frac{\partial P}{\partial T}\right|_V. \tag{7.30}$$

Setting the two mixed derivatives equal:
The minus signs cancel on both sides, giving:

$$\boxed{\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V} \tag{M3}$$

This is the **third Maxwell relation**. It connects the entropy–volume variation (not directly measurable) to the pressure–temperature variation (readily available from any equation of state).

Insight

Example: Ideal gas check For an ideal gas $PV = nRT$, so $(\partial P/\partial T)_V = nR/V$. The third Maxwell relation then predicts $(\partial S/\partial V)_T = nR/V > 0$: entropy increases as an ideal gas expands isothermally. This is consistent with the statistical picture of more accessible microstates at larger volume.

Insight

Equilibrium criterion for F At constant T and V, spontaneous processes decrease F, and equilibrium is reached at the *minimum* of F.

7.5 Gibbs Free Energy G

7.5.1 Motivation and Definition

The Gibbs free energy is tailored for systems at constant temperature *and* constant pressure — the conditions most common in chemistry, biology, and materials science. Starting from $F = U - TS$, we perform a second Legendre transformation trading the extensive variable V for P:

$$\boxed{G \equiv U + PV - TS = H - TS = F + PV.} \tag{7.31}$$

Insight

Physical meaning of G G is the maximum *non-PV* (useful) work obtainable from the system at constant T and P. At constant T and P, spontaneous processes decrease G, and chemical or phase equilibrium is reached when G is minimised.

Chemical potential. For a single-component system, the chemical potential μ is the Gibbs free energy per particle:

$$\mu = \left(\frac{\partial G}{\partial N}\right)_{T,P} = \frac{G}{N}. \tag{7.32}$$

7.5.2 The Work Interpretation

At constant T and P:

$$W_{\text{useful, max}} = -\Delta G, \tag{7.33}$$

where *useful work* excludes the $P\,\Delta V$ work done against the atmosphere. This is why G governs electrochemical cells (electrical work), osmotic processes, and biochemical reactions.

7.5.3 Fourth Maxwell Relation

Taking the total differential of $G = H - TS$:

$$dG = dH - T\,dS - S\,dT. \tag{7.34}$$

Substituting $dH = T\,dS + V\,dP$ from Eq. (7.17):

$$\begin{aligned} dG &= \big(T\,dS + V\,dP\big) - T\,dS - S\,dT \\ &= T\,dS + V\,dP - T\,dS - S\,dT \end{aligned}$$

$$\boxed{dG = -S\,dT + V\,dP.} \tag{7.35}$$

The $T\,dS$ terms cancel exactly. This shows that G is naturally a function of T and P:

$$G = G(T,\, P). \tag{7.36}$$

Setting $dP = 0$ (constant pressure):

$$dG = \left(\frac{\partial G}{\partial T}\right)_P dT = -S\,dT \qquad \Longrightarrow \qquad \boxed{-S = \left(\frac{\partial G}{\partial T}\right)_P} \tag{7.37}$$

Setting $dT = 0$ (constant temperature):

$$dG = \left(\frac{\partial G}{\partial P}\right)_T dP = V\,dP \qquad \Longrightarrow \qquad \boxed{V = \left(\frac{\partial G}{\partial P}\right)_T} \tag{7.38}$$

We have established that $G = G(T, P)$ is a state function with:

$$-S = \left(\frac{\partial G}{\partial T}\right)_P, \qquad V = \left(\frac{\partial G}{\partial P}\right)_T. \tag{7.39}$$

Since $G(T, P)$ is a well-defined state function, its mixed second partial derivatives must commute. At any point (T, P):

$$\left.\frac{\partial^2 G}{\partial P\,\partial T}\right|_{(T,P)} = \frac{\partial}{\partial P}\left[\left(\frac{\partial G}{\partial T}\right)_P\right]_T = \left.\frac{\partial(-S)}{\partial P}\right|_T = -\left.\frac{\partial S}{\partial P}\right|_T, \tag{7.40}$$

and equally:

$$\left.\frac{\partial^2 G}{\partial T\,\partial P}\right|_{(T,P)} = \frac{\partial}{\partial T}\left[\left(\frac{\partial G}{\partial P}\right)_T\right]_P = \left.\frac{\partial V}{\partial T}\right|_P. \tag{7.41}$$

Setting the two mixed derivatives equal:

$$-\left.\frac{\partial S}{\partial P}\right|_T = \left.\frac{\partial V}{\partial T}\right|_P. \tag{7.42}$$

Rearranging (multiply both sides by -1):

$$\boxed{\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P} \tag{M4}$$

This is the **fourth Maxwell relation**.

+ Extra Reading

The right-hand side is directly related to the **isobaric thermal expansivity**:

$$\alpha_P \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P, \tag{7.43}$$

so the fourth Maxwell relation becomes:

$$\left(\frac{\partial S}{\partial P}\right)_T = -V\alpha_P. \tag{7.44}$$

Since $\alpha_P > 0$ for most substances, entropy *decreases* as pressure increases at constant temperature — compression forces particles closer together, reducing the number of accessible microstates.

Insight

Equilibrium criterion for G At constant T and P, spontaneous processes decrease G, and equilibrium is reached at the *minimum* of G. This is the most relevant criterion for chemistry, phase transitions, and biology.

7.6 The Maxwell Relations: Unified Summary

Each thermodynamic potential, being a state function with an exact differential, yields one Maxwell relation from the commutation of its mixed second partial derivatives. All four are collected below.

Table 7.2. The four Maxwell relations.

Label	Potential	Maxwell relation
(M1)	$U(S,V)$	$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$
(M2)	$H(S,P)$	$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$
(M3)	$F(T,V)$	$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$
(M4)	$G(T,P)$	$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$

Chapter 8

Thermodynamic relations

8.1 The Gibbs–Helmholtz Equation

We want a relation connecting the Gibbs free energy G to the enthalpy H, expressed purely in terms of measurable quantities.

Starting from the definition $G = H - TS$ and using $S = -(\partial G/\partial T)_P$, we substitute S:

$$G = H - T\left(-\frac{\partial G}{\partial T}\bigg|_P\right) = H + T\left(\frac{\partial G}{\partial T}\right)_P. \tag{8.1}$$

Rearranging:

$$H = G - T\left(\frac{\partial G}{\partial T}\right)_P. \tag{8.2}$$

Now consider the derivative of G/T with respect to T at constant P:

$$\begin{aligned}\left(\frac{\partial(G/T)}{\partial T}\right)_P &= \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_P + G\cdot\frac{\partial(1/T)}{\partial T}\\ &= \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_P - \frac{G}{T^2}.\end{aligned} \tag{8.3}$$

Multiplying through by T^2:

$$T^2\left(\frac{\partial(G/T)}{\partial T}\right)_P = T\left(\frac{\partial G}{\partial T}\right)_P - G. \tag{8.4}$$

Comparing with Eq. (8.2), the right-hand side equals $-H$:

$$T^2\left(\frac{\partial(G/T)}{\partial T}\right)_P = -H. \tag{8.5}$$

Dividing both sides by T^2:

$$\boxed{\left(\frac{\partial(G/T)}{\partial T}\right)_P = -\frac{H}{T^2}.} \tag{8.6}$$

This is the **Gibbs–Helmholtz equation**.

Insight

Practical use If $G(T)$ is known at one temperature (e.g. from electrochemical measurements), integration of the Gibbs–Helmholtz equation gives G at any other temperature, provided H is known:

$$\frac{G(T_2)}{T_2} - \frac{G(T_1)}{T_1} = -\int_{T_1}^{T_2} \frac{H}{T^2}\, dT.$$

8.2 $C_P - C_V$ Relations

$$C_V = \left(\frac{\delta Q}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V \tag{8.7}$$

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_P \tag{8.8}$$

Taking Entropy S: as a function of T and V

$$S = S(T, V) \tag{8.9}$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \tag{8.10}$$

Which implies:

$$\left(\frac{\partial S}{\partial T}\right)_P = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \tag{8.11}$$

from eqn 8.8 and 8.11:

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_V + T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \tag{8.12}$$

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = C_V + T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \tag{8.13}$$

Using maxwell's relation:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V \tag{8.14}$$

$$\boxed{C_P - C_V = T\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_P} \tag{8.15}$$

8.3 The $T\,dS$ Equations

Entropy is a state function and may be expressed in terms of different pairs of independent thermodynamic variables. Depending on the choice of variables, different forms of the $T\,dS$ equation are obtained. We derive all three standard forms below.

8.3.1 Entropy as a Function of T and V

Let the entropy be expressed as

$$S = S(T, V). \tag{8.16}$$

The total differential of entropy is

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV. \tag{8.17}$$

Multiplying throughout by T,

$$T\,dS = T\left(\frac{\partial S}{\partial T}\right)_V dT + T\left(\frac{\partial S}{\partial V}\right)_T dV. \tag{8.18}$$

From the definition of heat capacity at constant volume,

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V, \tag{8.19}$$

and using the Maxwell relation

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V, \tag{8.20}$$

we obtain

$$\boxed{T\,dS = C_V\,dT + T\left(\frac{\partial P}{\partial T}\right)_V dV.} \tag{8.21}$$

This is the first $T\,dS$ equation.

8.3.2 Entropy as a Function of T and P

Let entropy be written as

$$S = S(T, P). \tag{8.22}$$

The total differential is

$$dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP. \tag{8.23}$$

Multiplying by T,

$$T\,dS = T\left(\frac{\partial S}{\partial T}\right)_P dT + T\left(\frac{\partial S}{\partial P}\right)_T dP. \tag{8.24}$$

Using the definition of heat capacity at constant pressure,

$$C_P = T\left(\frac{\partial S}{\partial T}\right)_P, \tag{8.25}$$

and the Maxwell relation

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P, \tag{8.26}$$

we obtain

$$\boxed{T\,dS = C_P\,dT - T\left(\frac{\partial V}{\partial T}\right)_P dP.} \tag{8.27}$$

This is the second $T\,dS$ equation.

8.3.3 Entropy as a Function of P and V

Let entropy be expressed as

$$S = S(P, V). \tag{8.28}$$

The total differential is

$$dS = \left(\frac{\partial S}{\partial P}\right)_V dP + \left(\frac{\partial S}{\partial V}\right)_P dV. \tag{8.29}$$

Multiplying by T,

$$T\,dS = T\left(\frac{\partial S}{\partial P}\right)_V dP + T\left(\frac{\partial S}{\partial V}\right)_P dV. \tag{8.30}$$

Using thermodynamic identities,

$$T\left(\frac{\partial S}{\partial P}\right)_V = C_V\left(\frac{\partial T}{\partial P}\right)_V, \tag{8.31}$$

$$T\left(\frac{\partial S}{\partial V}\right)_P = C_P\left(\frac{\partial T}{\partial V}\right)_P, \tag{8.32}$$

we obtain

$$\boxed{T\,dS = C_V\left(\frac{\partial T}{\partial P}\right)_V dP + C_P\left(\frac{\partial T}{\partial V}\right)_P dV.} \tag{8.33}$$

This is the third $T\,dS$ equation.

8.4 Thermodynamic Stability Conditions

For a state to be a true equilibrium (a minimum of the appropriate potential, not just an extremum), the second derivatives of that potential must have the correct signs.

Mechanical stability: from $F(T, V)$

At constant T, the Helmholtz free energy must be a *minimum* with respect to volume fluctuations. This requires:

$$\left(\frac{\partial^2 F}{\partial V^2}\right)_T \geq 0. \tag{8.34}$$

Since $-P = (\partial F/\partial V)_T$, differentiating again:

$$\left(\frac{\partial^2 F}{\partial V^2}\right)_T = -\left(\frac{\partial P}{\partial V}\right)_T \geq 0, \tag{8.35}$$

which means:

$$\left(\frac{\partial P}{\partial V}\right)_T \leq 0. \tag{8.36}$$

In terms of the **isothermal compressibility** $\kappa_T \equiv -\frac{1}{V}(\partial V/\partial P)_T$, this is equivalent to:

$$\boxed{\kappa_T \geq 0.} \tag{8.37}$$

A substance must become denser when compressed isothermally. Violation of this condition ($\kappa_T < 0$) implies mechanical instability and signals a phase transition (e.g. spinodal decomposition in a van der Waals gas).

Thermal stability: from $G(T, P)$

At constant P, the Gibbs free energy must be a *minimum* with respect to temperature fluctuations. Since G is a concave function of T (entropy must be positive), we require:

$$\left(\frac{\partial^2 G}{\partial T^2}\right)_P \leq 0. \tag{8.38}$$

Since $-S = (\partial G/\partial T)_P$, differentiating again:

$$\left(\frac{\partial^2 G}{\partial T^2}\right)_P = -\left(\frac{\partial S}{\partial T}\right)_P = -\frac{C_P}{T} \leq 0, \tag{8.39}$$

which gives:

$$\boxed{C_P \geq 0.} \tag{8.40}$$

Heat capacity at constant pressure must be non-negative: adding heat must raise temperature, never lower it. Violation signals a thermal instability and a phase transition.

8.5 Clapeyron Equation

The Clapeyron equation describes the slope of a phase boundary in the pressure–temperature (P–T) diagram. It is derived from the condition of thermodynamic equilibrium between two coexisting phases.

Thermodynamic Condition for Phase Equilibrium

Consider a system in which two phases, denoted by α and β, coexist in equilibrium. The necessary and sufficient condition for phase equilibrium is the equality of chemical potentials:

$$\mu^{(\alpha)}(T, P) = \mu^{(\beta)}(T, P). \tag{8.41}$$

Along the phase coexistence curve, this equality must hold for infinitesimal changes in T and P. Hence,

$$d\mu^{(\alpha)} = d\mu^{(\beta)}. \tag{8.42}$$

Differential of Chemical Potential

For a simple compressible system, the differential of chemical potential is given by Gibbs Dulhem Equation:

$$d\mu = -S_m\, dT + V_m\, dP, \tag{8.43}$$

where S_m and V_m are the molar entropy and molar volume, respectively.

Applying this to the two phases,

$$d\mu^{(\alpha)} = -S_m^{(\alpha)}\, dT + V_m^{(\alpha)}\, dP, \tag{8.44}$$

$$d\mu^{(\beta)} = -S_m^{(\beta)}\, dT + V_m^{(\beta)}\, dP. \tag{8.45}$$

Since $d\mu^{(\alpha)} = d\mu^{(\beta)}$, we obtain

$$\left[S_m^{(\beta)} - S_m^{(\alpha)}\right] dT = \left[V_m^{(\beta)} - V_m^{(\alpha)}\right] dP. \tag{8.46}$$

Clapeyron Equation

Rearranging the above relation,

$$\boxed{\frac{dP}{dT} = \frac{S_m^{(\beta)} - S_m^{(\alpha)}}{V_m^{(\beta)} - V_m^{(\alpha)}}} \tag{8.47}$$

Using the definition of latent heat per mole,

$$L = T\left(S_m^{(\beta)} - S_m^{(\alpha)}\right), \tag{8.48}$$

the Clapeyron equation can be written in the standard form

$$\boxed{\frac{dP}{dT} = \frac{L}{T\left(V_m^{(\beta)} - V_m^{(\alpha)}\right)}} \tag{8.49}$$

This is the *Clapeyron equation*. It is valid for all first-order phase transitions such as solid–liquid, liquid–vapour, and solid–vapour equilibrium.

8.5.1 Clausius–Clapeyron Equation (Liquid–Vapour Transition)

For a liquid–vapour phase transition,

$$V_m^{(v)} \gg V_m^{(l)}. \tag{8.50}$$

Hence,

$$V_m^{(v)} - V_m^{(l)} \approx V_m^{(v)}. \tag{8.51}$$

Assuming the vapour behaves as an ideal gas,

$$V_m^{(v)} = \frac{RT}{P}. \tag{8.52}$$

Substituting into the Clapeyron equation,

$$\frac{dP}{dT} = \frac{L}{T} \cdot \frac{P}{RT} \tag{8.53}$$

$$= \frac{LP}{RT^2}. \tag{8.54}$$

Rearranging,

$$\boxed{\frac{dP}{P} = \frac{L}{RT^2}\, dT} \tag{8.55}$$

This is the *Clausius–Clapeyron equation* in differential form.

Integrated Clausius–Clapeyron Equation

If the latent heat L is assumed to be temperature independent, integration yields

$$\int_{P_1}^{P_2} \frac{dP}{P} = \frac{L}{R} \int_{T_1}^{T_2} \frac{dT}{T^2}. \tag{8.56}$$

Performing the integrations,

$$\boxed{\ln\left(\frac{P_2}{P_1}\right) = -\frac{L}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)} \tag{8.57}$$

This form is widely used to determine latent heat from vapour pressure data.

8.6 Joule–Thomson Expansion

The Joule–Thomson effect refers to the change in temperature of a real gas when it is forced to expand from a region of high pressure to a region of low pressure through a porous plug or throttle valve under adiabatic conditions. Since the system is thermally insulated, no heat is exchanged with the surroundings during the process.

Such an expansion is commonly known as *throttling expansion*. The Joule–Thomson effect is of fundamental importance in refrigeration, liquefaction of gases, and cryogenic applications.

For most real gases at ordinary temperatures, the gas cools upon expansion. However, gases such as hydrogen and helium exhibit heating under similar conditions unless they are cooled below a characteristic temperature called the *inversion temperature*.

8.6.1 Isoenthalpic Nature of Joule–Thomson Expansion

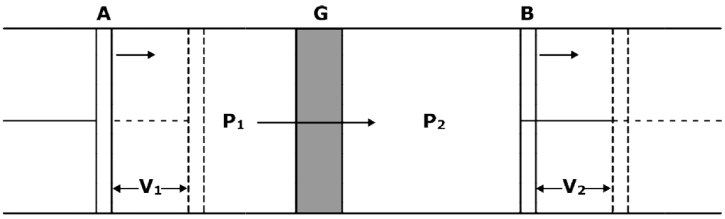

Figure 8.1. The porous experiment

Consider a gas flowing steadily through a porous plug from an initial state (P_1, V_1, T_1) to a final state (P_2, V_2, T_2) in an insulated tube.

The work done on the gas at the inlet is

$$W_1 = P_1 V_1, \tag{8.58}$$

and the work done by the gas at the outlet is

$$W_2 = P_2 V_2. \tag{8.59}$$

Since the process is adiabatic and no macroscopic kinetic energy change is involved, the first law of thermodynamics gives

$$U_2 - U_1 = P_1 V_1 - P_2 V_2. \tag{8.60}$$

Rearranging,

$$U_1 + P_1 V_1 = U_2 + P_2 V_2. \tag{8.61}$$

Using the definition of enthalpy $H = U + PV$, we obtain

$$H_1 = H_2, \tag{8.62}$$

or

$$\boxed{\Delta H = 0.} \tag{8.63}$$

Thus, Joule–Thomson expansion is an *isoenthalpic* process.

8.6.2 Joule–Thomson Coefficient

The *Joule–Thomson coefficient* μ_{JT} is defined as

$$\boxed{\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_H} \tag{8.64}$$

The quantity μ_{JT} is called the *Joule–Thomson coefficient*. It represents the rate of change of temperature with pressure during an isoenthalpic process.

- $\mu_{JT} > 0$: cooling on expansion
- $\mu_{JT} < 0$: heating on expansion
- $\mu_{JT} = 0$: no temperature change

Starting from the thermodynamic identity,

$$dH = T\,dS + V\,dP, \tag{8.65}$$

Since we are interested in change in temperature with pressure, we write entropy as

$$S = S(T, P)$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP.$$

Substituting,

$$dH = T\left(\frac{\partial S}{\partial T}\right)_P dT + \left[V + T\left(\frac{\partial S}{\partial P}\right)_T\right] dP. \tag{8.66}$$

Using the Maxwell relation,

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P, \tag{8.67}$$

and Heat capacity at constant Pressure:

$$C_P = \left(\frac{\delta Q}{\delta T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_P \tag{8.68}$$

we get

$$dH = C_P dT + \left[V - T\left(\frac{\partial V}{\partial T}\right)_P\right] dP. \tag{8.69}$$

For Joule-Thomson Expansion, $dH = 0$, implies:

$$C_P dT + \left[V - T\left(\frac{\partial V}{\partial T}\right)_P\right] dP = 0 \tag{8.70}$$

the Joule–Thomson coefficient may be written as

$$\boxed{\mu_{JT} \equiv \left(\frac{\partial T}{\partial P}\right)_H = -\frac{1}{C_P}\left(\frac{\partial H}{\partial P}\right)_T = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right].} \tag{8.71}$$

8.6.3 Ideal Gas Case

For an ideal gas,

$$V = \frac{nRT}{P}, \qquad \left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P}.$$

Substituting,

$$\mu_{JT} = \frac{1}{C_P}\left(\frac{nRT}{P} - \frac{nRT}{P}\right) = 0.$$

8.6.4 Joule–Thomson Coefficient for van der Waals

For one mole of a van der Waals gas,

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT. \tag{8.72}$$

For low pressures, neglecting higher-order terms,

$$V \approx \frac{RT}{P} + b - \frac{a}{RT}. \tag{8.73}$$

Differentiating with respect to T at constant P,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P} + \frac{a}{RT^2}. \tag{8.74}$$

Substituting into the expression for μ_{JT},

$$\mu_{JT} = \frac{1}{C_P}\left(\frac{2a}{RT} - b\right). \tag{8.75}$$

Inversion Temperature

The inversion temperature T_i is defined as the temperature at which the Joule–Thomson coefficient changes sign, i.e.,

$$\mu_{JT} = 0. \tag{8.76}$$

From the above expression,

$$\frac{2a}{RT_i} = b, \tag{8.77}$$

which gives

$$\boxed{T_i = \frac{2a}{Rb}.} \tag{8.78}$$

For $T < T_i$, the gas cools on expansion ($\mu_{JT} > 0$), while for $T > T_i$, the gas heats up ($\mu_{JT} < 0$).

Table 8.1. Cooling techniques and the temperature ranges they access.

Method	Range	Substance	Limiting principle
Vapour-compression	200–273 K	Freon, ammonia	Compressor efficiency
Dry ice (solid CO_2)	~ 195 K	CO_2	Sublimation point
Liquid nitrogen	~ 77 K	N_2	Normal boiling point
Liquid hydrogen	~ 20 K	H_2	Normal boiling point
Liquid helium-4	1.5–4.2 K	^{4}He	Pumped λ-point
Liquid helium-3	0.3–3 K	^{3}He	Vapour pressure
^{3}He/^{4}He dilution	5–100 mK	^{3}He/^{4}He	Mixing enthalpy
Adiabatic demag.	**1 mK–1 K**	Paramagnetic salt	Electronic spin entropy
Nuclear adiabatic demag.	μK–mK	Cu, $PrNi_5$	Nuclear spin heat capacity
Laser cooling	$\sim \mu$K	Alkali atoms	Recoil limit
Evaporative cooling (BEC)	$\sim$ nK	Dilute atomic gases	Collisional thermalisation

8.7 Cooling by Adiabatic Demagnetization

8.7.1 The Quest for Ultralow Temperatures

Reaching temperatures far below those achievable by ordinary refrigeration is one of the great challenges of experimental physics. Each successive technique extends human reach into progressively colder territory. Table 8.1 summarises the principal methods, the approximate temperature range each covers, and the physical principle that limits it.

A paramagnetic salt contains ions whose unpaired electrons carry magnetic moments. In the absence of an external field these moments point in random directions, contributing a large *magnetic entropy* to the system. When a strong magnetic field is applied *isothermally*, the moments align partially with the field and the magnetic entropy decreases; the released heat flows into a surrounding bath. If the system is then thermally isolated and the magnetic field is *slowly* (quasi-statically) reduced to zero, the total entropy must remain constant. To compensate for the decrease in magnetic order that accompanies the vanishing field, the thermal (lattice) disorder must decrease — which means the temperature must fall.

This is **adiabatic demagnetization**, first proposed independently by Debye and Giauque in 1926.

Insight

Physical picture Think of the magnetic entropy as a "reservoir" of disorder. Isothermal magnetization drains this reservoir (at fixed T, ordered spins carry less entropy). Adiabatic demagnetization then refills it at the expense of thermal energy — the

lattice cools to provide the energy needed to re-disorder the spins.

8.7.2 Thermodynamics of a Magnetic System

For a simple magnetic system in which the only work done is magnetic work, the first law of thermodynamics reads:

$$dU = T\,dS + \mu_0 H\,dM, \tag{8.79}$$

where H is the applied magnetic field intensity, M is the total magnetisation of the sample, and μ_0 is the permeability of free space. (Compare with the mechanical case: $-P\,dV$ is replaced by $+\mu_0 H\,dM$.) For clarity we absorb μ_0 into the definition of H in what follows, writing simply:

$$\boxed{dU = T\,dS + H\,dM.} \tag{8.80}$$

The magnetic Helmholtz free energy

Exactly as in the mechanical case, we introduce the **magnetic Helmholtz free energy**

$$F = U - TS, \tag{8.81}$$

whose total differential, using Eq. (8.79), is:

$$\begin{aligned} dF &= dU - T\,dS - S\,dT \\ &= (T\,dS + H\,dM) - T\,dS - S\,dT \end{aligned}$$

$$\boxed{dF = -S\,dT + H\,dM.} \tag{8.82}$$

The natural variables of F are therefore (T, M), and the thermodynamic derivatives are:

$$S = -\left(\frac{\partial F}{\partial T}\right)_M, \qquad H = \left(\frac{\partial F}{\partial M}\right)_T. \tag{8.83}$$

Maxwell relation for the magnetic system

Since $F(T, M)$ is a state function, its mixed second partial derivatives must commute. At any point (T, M):

$$\left.\frac{\partial^2 F}{\partial M\,\partial T}\right|_{(T,M)} = \frac{\partial}{\partial M}\left[\left(\frac{\partial F}{\partial T}\right)_M\right]_T = \left.\frac{\partial(-S)}{\partial M}\right|_T = -\left(\frac{\partial S}{\partial M}\right)_T, \tag{8.84}$$

and equally:

$$\left.\frac{\partial^2 F}{\partial T\,\partial M}\right|_{(T,M)} = \frac{\partial}{\partial T}\left[\left(\frac{\partial F}{\partial M}\right)_T\right]_M = \left(\frac{\partial H}{\partial T}\right)_M. \tag{8.85}$$

Setting the two mixed derivatives equal:

$$\boxed{\left(\frac{\partial S}{\partial H}\right)_T = \left(\frac{\partial M}{\partial T}\right)_H.} \tag{8.86}$$

This is the **Maxwell relation for the magnetic system**. Note carefully that we have converted the left side from a derivative at constant M to one at constant H; the two are equal because for a paramagnetic system M and H are in one-to-one correspondence at fixed T (via Curie's law, below).

8.7.3 The Adiabatic Condition

Treating entropy as a function of temperature and field, $S = S(T, H)$, the total differential is:

$$dS = \left(\frac{\partial S}{\partial T}\right)_H dT + \left(\frac{\partial S}{\partial H}\right)_T dH. \tag{8.87}$$

We now identify each partial derivative.

First term — heat capacity at constant field. At constant H, the heat absorbed reversibly is $\delta Q = T\,dS\Big|_H = C_H\,dT$, hence:

$$\left(\frac{\partial S}{\partial T}\right)_H = \frac{C_H}{T}, \tag{8.88}$$

where C_H is the *heat capacity at constant applied field.*

Second term — Maxwell relation. From Eq. (8.86):

$$\left(\frac{\partial S}{\partial H}\right)_T = \left(\frac{\partial M}{\partial T}\right)_H. \tag{8.89}$$

Substituting Eqs. (8.88) and (8.89) into Eq. (8.87):

$$dS = \frac{C_H}{T}\,dT + \left(\frac{\partial M}{\partial T}\right)_H dH. \tag{8.90}$$

For a **reversible adiabatic** (isentropic) process, $dS = 0$:

$$\frac{C_H}{T}\,dT = -\left(\frac{\partial M}{\partial T}\right)_H dH. \tag{8.91}$$

This is the governing equation of adiabatic demagnetization. The sign is instructive: if $(\partial M/\partial T)_H < 0$ (magnetisation decreases with temperature, as in a paramagnet) and $dH < 0$ (field is reduced), then the right-hand side is positive, giving $dT > 0$... wait, this would imply heating. But that contradicts experiment!

The resolution lies in the sign of $(\partial M/\partial T)_H$ for a *paramagnet*: from Curie's law $M = CH/T$, we have $(\partial M/\partial T)_H = -CH/T^2 < 0$. Substituting:

$$\frac{C_H}{T}\,dT = -\left(-\frac{C_\mathrm{C} H}{T^2}\right) dH = \frac{C_\mathrm{C} H}{T^2}\,dH.$$

Now, when $dH < 0$, the right-hand side is negative, so $dT < 0$: the salt *cools*. This is consistent.

8.7.4 Curie's Law and the Cooling Formula

A paramagnetic salt obeys **Curie's law**:

$$M = \frac{C_{\mathrm{C}}\, H}{T}, \tag{8.92}$$

where C_{C} is *Curie's constant* (a material property, not to be confused with the heat capacity). Therefore:

$$\left(\frac{\partial M}{\partial T}\right)_H = -\frac{C_{\mathrm{C}}\, H}{T^2}. \tag{8.93}$$

Substituting into the adiabatic condition (8.91):

$$\frac{C_H}{T}\, dT = \frac{C_{\mathrm{C}}\, H}{T^2}\, dH. \tag{8.94}$$

Rearranging:

$$C_H\, T\, dT = C_{\mathrm{C}}\, H\, dH. \tag{8.95}$$

Assuming C_H is approximately constant (valid at low temperatures where the dominant contribution comes from the spin system itself), we integrate both sides:

$$C_H \int_{T_i}^{T_f} T\, dT = C_{\mathrm{C}} \int_{H_i}^{H_f} H\, dH, \tag{8.96}$$

$$C_H \cdot \frac{T_f^2 - T_i^2}{2} = C_{\mathrm{C}} \cdot \frac{H_f^2 - H_i^2}{2}. \tag{8.97}$$

Therefore:

$$\boxed{T_f^2 - T_i^2 = \frac{C_{\mathrm{C}}}{C_H}\left(H_f^2 - H_i^2\right).} \tag{8.98}$$

This is the **exact integrated result** for adiabatic demagnetization under Curie's law with constant C_H.

The simple ratio law

In practice one starts with a large initial field H_i and reduces it to $H_f \ll H_i$ (often to zero). We can write Eq. (8.98) as:

$$\frac{T_f^2}{T_i^2} = 1 + \frac{C_{\mathrm{C}}}{C_H T_i^2}(H_f^2 - H_i^2). \tag{8.99}$$

But there is a cleaner route. Equation (8.95), $C_H\, T\, dT = C_{\mathrm{C}}\, H\, dH$, can be rewritten as:

$$d(T^2) = \frac{C_{\mathrm{C}}}{C_H}\, d(H^2). \tag{8.100}$$

This shows that along any adiabat under Curie's law, $T^2 - (C_{\mathrm{C}}/C_H)H^2 = \text{constant}$. In the **deep cooling limit** where the spin heat capacity dominates and C_H is itself proportional to H^2/T^2 (from the two-level spin system result $C_H = R(CH/T)^2$), the combination $T^2/(C_{\mathrm{C}}H^2/C_H)$ becomes constant, which reduces simply to:

$$\frac{T}{H} = \text{constant}, \tag{8.101}$$

giving the celebrated result:

$$\boxed{T_f = T_i\, \frac{H_f}{H_i}.} \tag{8.102}$$

Insight

What this says The final temperature is simply proportional to the final field. To reach 1 mK starting from $T_i = 1\,\mathrm{K}$ at $H_i = 5\,\mathrm{T}$, one needs only to reduce the field to $H_f = 5\,\mathrm{mT}$ — well within experimental reach.

8.7.5 The S–T Diagram

Figure 8.2 shows the entropy–temperature diagram for a paramagnetic salt and the three-step cooling cycle.

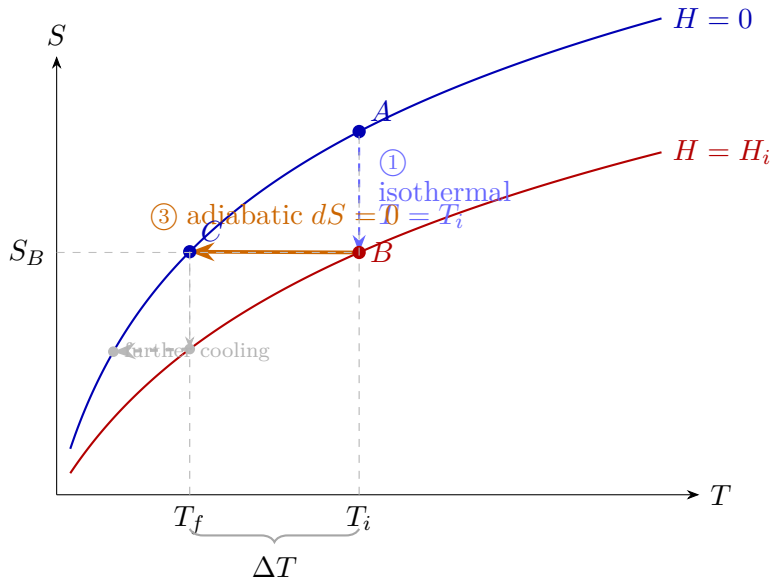

Figure 8.2. Entropy–temperature diagram for adiabatic demagnetization. The upper curve ($H = 0$) lies above the lower curve ($H = H_i$) because entropy is higher when the field is absent (disordered spins). **Step 1** ($A \to B$): isothermal magnetisation at $T = T_i$ reduces entropy as heat flows to the bath. **Step 3** ($B \to C$): adiabatic demagnetisation at constant entropy S_B moves the system left to a lower temperature T_f. Repeating the cycle reaches progressively lower temperatures.

8.7.6 Practical Aspects and Limitations

Choice of salt. Commonly used paramagnetic salts include cerium magnesium nitrate (CMN, $T_{\min} \approx 2\,\mathrm{mK}$) and chromium potassium alum (CPA, $T_{\min} \approx 10\,\mathrm{mK}$).

Nuclear adiabatic demagnetization. Using nuclear spins (which are far more weakly coupled to the lattice) instead of electronic spins pushes the technique to the microkelvin range. Temperatures below $100\,\mu\mathrm{K}$ have been achieved in copper.

Why a true $T = 0$ is unattainable. As $H \to 0$ and $T \to 0$, the entropy of both curves (Figure 8.2) approaches the same minimum value (the third law: $S \to 0$). The two curves merge, and the "step" $B \to C$ shrinks to zero. No matter how many cycles are performed, $T = 0\,\mathrm{K}$ is never reached in a finite number of steps — in perfect agreement with the **third law of thermodynamics**.

Insight

Connection to the Third Law The convergence of the $H = 0$ and $H = H_i$ entropy curves as $T \to 0$ is not a practical inconvenience — it is a manifestation of the third law. The unattainability of absolute zero by adiabatic demagnetization (or any other process) is *equivalent* to Nernst's statement that the entropy of all systems vanishes at $T = 0$.

Chapter 9

The Third Law of Thermodynamics

> *"The first law says you cannot win. The second law says you cannot break even. The third law says you cannot quit the game — you can never reach absolute zero."*

The first and second laws introduce energy conservation and entropy, but they say nothing about the *absolute value* of entropy — only about its changes. The third law supplies this missing reference by specifying the behaviour of entropy as $T \to 0$. Far from being a mere bookkeeping rule, the third law has deep consequences: it governs the behaviour of heat capacities, chemical equilibria, thermoelectric effects, and sets a fundamental limit on the attainability of low temperatures. It is also the gateway into the quantum world — without quantum mechanics, the third law would be inexplicable.

9.1 The Need for an Absolute Entropy Reference

From the second law, entropy change along a reversible path is

$$dS = \frac{\delta Q_{\text{rev}}}{T}. \tag{9.1}$$

This determines only *differences* in entropy. To assign a unique value $S(T)$ to every state of a system, one needs a reference state. The third law provides this by anchoring entropy at $T = 0$.

The **absolute entropy** at temperature T is then

$$S(T) = S(0) + \int_0^T \frac{C(T')}{T'}\, dT' + \sum_i \frac{\Delta H_i}{T_i} \tag{9.2}$$

where the sum runs over any phase transitions (latent heats ΔH_i at transition temperatures T_i) occurring between 0 and T. The third law asserts $S(0) = 0$ for a perfect crystal, making (9.2) an absolute quantity.

9.2 Statements of the Third Law

Three logically distinct (but equivalent) statements have been proposed.

9.2.1 Planck's Statement

Theorem: Third Law — Planck (1911)

The entropy of every perfect crystalline substance is zero at absolute zero temperature:

$$\lim_{T\to 0} S_{\text{perfect crystal}}(T) = 0. \tag{9.3}$$

A *perfect crystal* is one whose ground state is non-degenerate ($W = 1$). The statistical foundation is Boltzmann's formula:

$$S = k_B \ln W \implies S(0) = k_B \ln 1 = 0. \tag{9.4}$$

9.2.2 Nernst Heat Theorem (1906)

Walther Nernst proposed his heat theorem eight years before the quantum revolution was complete, based purely on calorimetric observations of chemical reactions at low temperatures.

Theorem: Nernst Heat Theorem

For any isothermal reversible process,

$$\lim_{T\to 0} \Delta S_{\text{isothermal}} = 0. \tag{9.5}$$

Mathematical justification

Consider a reaction $A \to B$ at constant temperature T. The Gibbs energy change is $\Delta G = \Delta H - T\Delta S$. Nernst observed experimentally that

$$\lim_{T\to 0} \Delta G = \lim_{T\to 0} \Delta H \tag{9.6}$$

and moreover that both approach the same limit with the *same slope*:

$$\lim_{T\to 0} \frac{\partial(\Delta G)}{\partial T} = \lim_{T\to 0} \frac{\partial(\Delta H)}{\partial T} = 0. \tag{9.7}$$

Since $(\partial\Delta G/\partial T)_P = -\Delta S$, equation (9.7) implies $\Delta S \to 0$, consistent with (9.5).

9.2.3 The Unattainability Principle (Simon, 1927)

Theorem: Unattainability of Absolute Zero

It is impossible to reduce the temperature of any system to absolute zero in a finite number of operations.

The equivalence of this statement with Nernst's theorem is non-trivial. The argument runs as follows: each cooling step changes the temperature by

$$dT = -\frac{T}{C_X}\left(\frac{\partial S}{\partial X}\right)_T dX \tag{9.8}$$

where X is the external parameter being varied (magnetic field, pressure, etc.). If $\Delta S \to 0$ as $T \to 0$ (Nernst's theorem), then successive steps remove ever-smaller amounts of entropy and the temperature change per step shrinks to zero — absolute zero is approached asymptotically, never reached.

9.3 Heat Capacity Must Vanish at Absolute Zero

For the entropy integral (9.2) to converge at the lower limit, we need

$$\int_0^{T_0} \frac{C(T')}{T'}\, dT' < \infty \quad \text{for any finite } T_0. \tag{9.9}$$

This requires

$$\lim_{T\to 0} C(T) = 0. \tag{9.10}$$

The heat capacity must vanish at $T = 0$ — a result that is *impossible* to obtain from classical physics (equipartition always gives a finite C) and was one of the great mysteries of nineteenth-century thermodynamics.

Insight

Why classical physics fails. The classical equipartition theorem assigns energy $\frac{1}{2}k_B T$ per quadratic degree of freedom regardless of temperature. Every mode contributes a fixed amount to C down to $T = 0$. The observed vanishing of C is therefore a purely quantum effect: at low T the thermal energy $k_B T$ falls below the spacing $\hbar\omega$ between energy levels, modes "freeze out," and their contribution to C is exponentially suppressed.

9.4 Debye Theory and the Low-Temperature Behaviour of Solids

The Debye model provides the most important and widely used description of heat capacity at low temperatures.

9.4.1 The Debye Model

Peter Debye (1912) modelled a solid as a collection of coupled harmonic oscillators with a continuous distribution of frequencies up to a maximum, the *Debye frequency* ω_D. The density of states is

$$g(\omega) = \frac{9N}{\omega_D^3}\,\omega^2, \quad 0 \le \omega \le \omega_D. \tag{9.11}$$

The Debye temperature is defined as

$$\Theta_D \equiv \frac{\hbar\omega_D}{k_B}. \tag{9.12}$$

Heat capacity

The full Debye heat capacity is

$$C_V = 9Nk_B \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x-1)^2}\, dx, \qquad x = \frac{\hbar\omega}{k_B T}. \tag{9.13}$$

In the low-temperature limit $T \ll \Theta_D$, the upper limit of the integral goes to infinity, giving the celebrated **Debye T^3 law**:

$$C_V \xrightarrow{T \ll \Theta_D} \frac{12\pi^4}{5} N k_B \left(\frac{T}{\Theta_D}\right)^3 \propto T^3. \tag{9.14}$$

This vanishes as $T \to 0$, consistent with the third law.

The entropy at low temperature follows by integration:

$$S(T) = \int_0^T \frac{C_V(T')}{T'}\, dT' = \frac{4\pi^4 N k_B}{5} \left(\frac{T}{\Theta_D}\right)^3 \propto T^3. \tag{9.15}$$

Both C_V and S vanish as T^3 — a clean third-law signature.

9.4.2 TikZ: Debye Heat Capacity

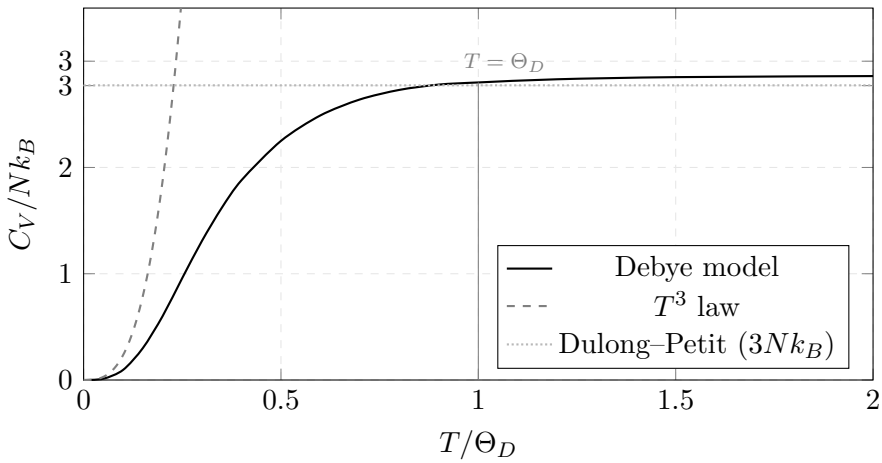

Figure 9.1. Debye heat capacity as a function of reduced temperature T/Θ_D. At low T the curve follows the T^3 law (dashed); at high T it saturates at the classical Dulong–Petit value $3Nk_B$ (dotted).

9.5 Residual Entropy and Frozen Disorder

The third law strictly applies to *perfect* crystals. Many real substances cool into disordered states from which further equilibration is kinetically impossible. They carry a non-zero **residual entropy** at $T = 0$:

$$S_{\text{res}} = k_B \ln W_0 \tag{9.16}$$

where $W_0 > 1$ is the number of degenerate ground-state configurations.

9.5.1 Ice: A Classic Example (Pauling, 1935)

Water ice has a disordered proton arrangement even at 0 K. Each oxygen atom has two O–H covalent bonds and two O$\cdots$H hydrogen bonds, but the assignment of which two of the four neighbouring hydrogens are covalently attached can be done in $\binom{4}{2} = 6$ ways per oxygen, subject to the "ice rule" that exactly one hydrogen sits on each bond.

Pauling estimated the residual entropy by noting that each H sits on a bond between two oxygens, and the probability that it obeys the ice rule is $\frac{1}{2}$:

$$W_0 = \left(\frac{3}{2}\right)^N \tag{9.17}$$

for N water molecules, giving

$$S_{\text{res}}(\text{ice}) = Nk_B \ln\frac{3}{2} = 3.37\ \text{J mol}^{-1}\,\text{K}^{-1}. \tag{9.18}$$

The experimentally measured residual entropy of ice is 3.41 J mol^{-1} K^{-1} — in remarkable agreement.

9.5.2 Carbon Monoxide: Orientational Disorder

CO molecules are almost symmetric (the C and O ends have very similar sizes), so at low temperatures they can freeze in either orientation (CO or OC) with equal probability. For N molecules with 2 orientations each:

$$W_0 = 2^N \implies S_{\text{res}}(\text{CO}) = Nk_B \ln 2 = 5.76\ \text{J mol}^{-1}\,\text{K}^{-1}. \tag{9.19}$$

The measured value is 4.2 J mol^{-1} K^{-1}, somewhat less due to a weak preference for one orientation.

9.6 Adiabatic Demagnetisation and Reaching $T \approx 0$

The most powerful technique for approaching absolute zero is **adiabatic demagnetisation** (Debye 1926; Giauque 1927), which exploits the entropy of a paramagnetic spin system.

9.6.1 Entropy of a Spin System

Consider N spin-$\frac{1}{2}$ ions in a magnetic field B. Each spin has two states with energies $\pm\mu_B B$. The partition function per spin is

$$z = e^{+\mu_B B/k_B T} + e^{-\mu_B B/k_B T} = 2\cosh\left(\frac{\mu_B B}{k_B T}\right). \tag{9.20}$$

The Helmholtz free energy per spin is $f = -k_B T \ln z$, and the entropy per spin is

$$s = -\frac{\partial f}{\partial T} = k_B\left[\ln 2\cosh\left(\frac{\mu_B B}{k_B T}\right) - \frac{\mu_B B}{k_B T}\tanh\left(\frac{\mu_B B}{k_B T}\right)\right]. \tag{9.21}$$

Let $x \equiv \mu_B B/k_B T$. Then:

- $x \to 0$ ($B \to 0$ or $T \to \infty$): $s \to k_B \ln 2$ (maximum disorder, both states equally populated).
- $x \to \infty$ ($T \to 0$ at fixed B): $s \to 0$ (fully polarised, unique ground state).

9.6.2 The Demagnetisation Cycle

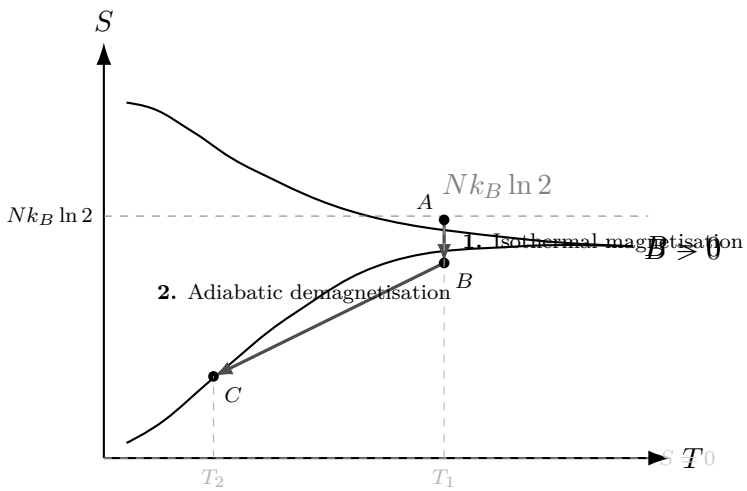

Figure 9.2. Entropy–temperature diagram for adiabatic demagnetisation. Step **1** ($A \to B$): isothermal magnetisation reduces entropy at constant T_1 (field switched on). Step **2** ($B \to C$): adiabatic demagnetisation reduces temperature to $T_2 \ll T_1$ at constant entropy (field switched off). Repeated cycles bring the system progressively closer to $T = 0$ but never reach it.

Protocol

1. **Isothermal magnetisation** ($A \to B$, field on): The spin system is coupled to a heat reservoir at T_1 and a large field B is applied. Spins align, entropy decreases from the $B = 0$ curve to the $B > 0$ curve at constant T_1.

2. **Adiabatic demagnetisation** ($B \to C$, field off): The spin system is thermally isolated from the reservoir, and the field is slowly reduced to zero. The process is isentropic. Since entropy on the $B = 0$ curve at some T_2 equals the entropy on the $B > 0$ curve at T_1, we get $T_2 \ll T_1$.

The temperature achieved after one step is approximately

$$T_2 \approx T_1 \cdot \frac{B_f}{B_i} \tag{9.22}$$

where B_i is the initial field and B_f is the residual (stray) field. Modern nuclear adiabatic demagnetisation (using nuclear magnetic moments, which are $\sim 1000\times$ smaller than electronic moments) has achieved temperatures below $1\,\mu$K.

Insight

Why absolute zero is unreachable. At each demagnetisation step the two entropy curves ($B = 0$ and $B > 0$) approach each other as $T \to 0$. In the limit $T = 0$ both curves must meet at $S = 0$ (third law). This means the entropy drop achievable per step vanishes, and an infinite number of steps would be required to reach $T = 0$ — a precise quantitative expression of the unattainability principle.

9.7 Third Law and Chemical Equilibrium

The third law has a profound practical consequence: it makes the equilibrium constant $K(T)$ calculable from calorimetric data alone, without needing to measure the reaction directly.

9.7.1 The van 't Hoff Equation and the Integration Constant

The temperature dependence of the equilibrium constant is given by the van 't Hoff equation:

$$\frac{d\ln K}{dT} = \frac{\Delta H^\circ}{RT^2}. \tag{9.23}$$

Integrating:

$$\ln K(T) = -\frac{\Delta G^\circ(T)}{RT} = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} + \text{const.} \tag{9.24}$$

Before Nernst's theorem, the integration constant was unknowable. The third law fixes it: since $\Delta S^\circ \to 0$ as $T \to 0$, the constant is determined, and $K(T)$ becomes fully calculable from heat-capacity data via:

$$\Delta S^\circ(T) = \int_0^T \frac{\Delta C_p(T')}{T'}\, dT' + \sum_i \frac{\Delta H_i}{T_i} \tag{9.25}$$

where $\Delta C_p = C_p(\text{products}) - C_p(\text{reactants})$.

This was Nernst's original motivation: equation (9.25) allows chemists to compute equilibrium constants — and thus predict whether a reaction will go — from purely thermal measurements.

9.8 Third-Law Entropy and Spectroscopic Entropy

A beautiful consistency check of the third law is the comparison between two independent measurements of absolute entropy.

1. **Calorimetric (third-law) entropy**: integrate C_p/T from 0 to T using measured heat capacities, adding latent heats at phase transitions.

2. **Spectroscopic entropy**: use quantum mechanical partition functions computed from molecular energy levels measured by spectroscopy:

$$S = Nk_B \left[\ln z + T\frac{\partial \ln z}{\partial T}\right] \tag{9.26}$$

   where z is the molecular partition function including translational, rotational, vibrational, and electronic contributions.

For ideal gases these two methods agree to within experimental uncertainty (typically $< 0.1\,\%$) — except for substances with residual entropy, where the calorimetric value is lower by exactly $k_B \ln W_0$ per molecule. This discrepancy is itself a precise measurement of the residual entropy.

Substance	S_{cal}	S_{spec}	$\Delta S = k_B \ln W_0$
HCl(g)	186.2	186.8	0 $\text{J mol}^{-1}\,\text{K}^{-1}$ (good agreement)
CO(g)	193.0	197.9	$4.9 \approx R \ln 2$ (orientational disorder)
H_2O(g)	185.4	188.7	$3.3 \approx R \ln(3/2)$ (Pauling ice)

9.9 Behaviour of Other Thermodynamic Quantities at $T = 0$

The third law has consequences for many thermodynamic quantities beyond C_V.

9.9.1 Thermal Expansion Vanishes

The volume expansivity is

$$\alpha_V \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P. \tag{9.27}$$

Using a Maxwell relation:

$$\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T. \tag{9.28}$$

Since $S \to 0$ as $T \to 0$ for all values of P, the derivative $(\partial S/\partial P)_T \to 0$, hence:

$$\lim_{T\to 0} \alpha_V = 0. \tag{9.29}$$

The thermal expansion coefficient vanishes at absolute zero. This is confirmed experimentally: helium, which remains liquid at low pressure, has a measurable $\alpha_V(T) \propto T^3$ at low temperatures.

9.9.2 Thermoelectric Effects Vanish

The Seebeck coefficient (thermopower) $\mathcal{S}$ of a thermoelectric junction satisfies, from the Kelvin relation,

$$\mathcal{S} = \frac{1}{T}\int_0^T \frac{\Pi(T')}{T'}\,dT' \tag{9.30}$$

where Π is the Peltier coefficient. Because all heat-related coefficients vanish at $T = 0$:

$$\lim_{T\to 0} \mathcal{S}(T) = 0. \tag{9.31}$$

This is observed in all metals and semiconductors.

9.9.3 Isothermal Compressibility

For an ideal quantum gas, the isothermal compressibility $\kappa_T = -V^{-1}(\partial V/\partial P)_T$ approaches a finite non-zero limit at $T = 0$ (it is set by the zero-point pressure), in contrast to classical gases where $\kappa_T = 1/P \to \infty$ as $T \to 0$. This illustrates that the third law governs the *thermal* response functions but not purely mechanical ones.

9.10 The Third Law in Quantum Field Theory and Cosmology

9.10.1 Black Hole Thermodynamics

Black holes obey laws formally analogous to thermodynamics (Bardeen, Carter, Hawking, 1973). The analogy is:

Law	Thermodynamics	Black Hole
Zeroth	$T = \text{const}$ in equilibrium	Surface gravity $\kappa = \text{const}$ on horizon
First	$dE = T\,dS - P\,dV$	$dM = \frac{\kappa}{8\pi}\,dA + \Omega\,dJ + \Phi\,dQ$
Second	$dS \geq 0$	$dA \geq 0$ (area theorem)
Third	$T = 0$ unattainable	$\kappa = 0$ unattainable

The Hawking temperature of a black hole of mass M is

$$T_{\text{BH}} = \frac{\hbar c^3}{8\pi G M k_B}, \tag{9.32}$$

and the Bekenstein–Hawking entropy is

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar}\,A = \frac{k_B A}{4\,l_P^2} \tag{9.33}$$

where A is the horizon area and $l_P = \sqrt{G\hbar/c^3}$ is the Planck length. As $M \to 0$ (the black hole evaporates), $T_{\text{BH}} \to \infty$ and the analogy with an ordinary thermodynamic system breaks down — the third law in this context is still an active research question.

9.10.2 The Cosmological Significance of the Third Law

In the early universe, at temperatures far above $k_B T \gg m_p c^2$ (the proton rest energy), matter and radiation were in thermal equilibrium. As the universe expanded and cooled, species "froze out" sequentially. The third law asserts that in the far future, as the universe approaches a heat death at $T \to 0$, all thermal fluctuations cease and entropy production stops. The maximum entropy state of the universe — a cold, dark, uniformly distributed gas of photons and black holes slowly evaporating — is approached asymptotically but never quite reached.

9.11 Summary

Summary

Key results of this chapter.

- The third law anchors entropy at $S(0) = 0$ for perfect crystals, making absolute entropy a well-defined quantity.

- Three equivalent formulations: Planck's statement, Nernst's heat theorem, and the unattainability principle.
- The third law *requires* all heat capacities to vanish at $T = 0$ — a result that demands quantum mechanics.
- The Debye T^3 law, $C_V \propto T^3$, is the quantitative realisation of this requirement for solids.
- Residual entropy ($S_{\text{res}} = k_B \ln W_0$) occurs in systems with frozen orientational disorder (ice, CO).
- Adiabatic demagnetisation provides the most powerful route to ultralow temperatures; the third law itself explains why $T = 0$ can never be reached.
- The third law determines integration constants in chemical equilibrium and enables the calculation of $K(T)$ from calorimetry alone.
- Thermal expansion α_V and thermoelectric coefficients both vanish at $T = 0$ as direct consequences.

Chapter 10

Calorimetry and Thermometry

"Temperature is not a measure of how much heat a body contains, but of how fiercely it wants to give that heat away."

Heat and temperature are among the most familiar concepts in physics, yet their precise measurement and the laws governing heat flow demand considerable care. This chapter develops the subject in three interlocking parts: the *laws of heat transfer* (conduction, convection, radiation), the *instruments of temperature measurement* (from liquid thermometers to quantum-noise thermometry), and the *thermoelectric effects* that couple heat flow to electric current. The chapter closes with the Onsager framework, which reveals these effects as two faces of a single, irreversible thermodynamic process.

10.1 Fourier's Law of Heat Conduction

The rate of heat flow through a material is directly proportional to the cross-sectional area and the temperature gradient, provided steady-state conditions hold.

$$\frac{dQ}{dt} = -kA\frac{dT}{dx} \tag{10.1}$$

The negative sign expresses the physical fact that heat flows *down* the temperature gradient. The quantities are: dQ/dt (W), k ($\mathrm{W\,m^{-1}\,K^{-1}}$), A ($\mathrm{m^2}$), dT/dx ($\mathrm{K\,m^{-1}}$).

10.1.1 Vector Form

In three dimensions:

$$\mathbf{q} = -k\,\nabla T \tag{10.2}$$

where $\mathbf{q}$ is the heat flux vector ($\mathrm{W\,m^{-2}}$).

10.1.2 The Heat Equation

Combining Fourier's law (10.2) with energy conservation for a volume element gives the **heat (diffusion) equation**:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k\,\nabla T) + \dot{Q}_{\mathrm{gen}} \tag{10.3}$$

---

where ρ is mass density, c_p specific heat, and $\dot{Q}_{\text{gen}}$ any volumetric heat generation rate. For a homogeneous material with no internal generation:

$$\frac{\partial T}{\partial t} = \alpha_d \nabla^2 T, \qquad \alpha_d \equiv \frac{k}{\rho c_p} \tag{10.4}$$

where α_d ($\mathrm{m^2\,s^{-1}}$) is the **thermal diffusivity** — the ratio of how quickly a material conducts heat to how much heat it can store.

Material	k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	α_d (10^{-6} $\mathrm{m^2\,s^{-1}}$)
Diamond	2200	1000
Copper	385	117
Aluminium	205	84
Steel	50	14
Glass	1.0	0.34
Water	0.60	0.14
Air (300 K)	0.026	22

10.1.3 Thermal Resistance

By analogy with Ohm's law, define the **thermal resistance** of a slab of thickness L, area A, conductivity k:

$$R_{\text{th}} = \frac{L}{kA} \quad (\mathrm{K\,W^{-1}}) \tag{10.5}$$

so that $\dot{Q} = \Delta T / R_{\text{th}}$. For layers in series and parallel:

$$R_{\text{series}} = R_1 + R_2 + \cdots \tag{10.6}$$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots \tag{10.7}$$

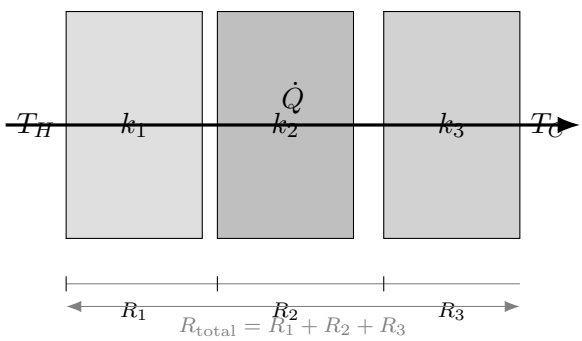

Figure 10.1. Composite wall: three layers in series. Each layer has its own conductivity k_i and thermal resistance $R_i = L_i / k_i A$.

10.2 Specific Heat and Calorimetry

$$Q = mc\,\Delta T \tag{10.8}$$

The **molar heat capacity** $C_m = Mc$ (where M is the molar mass) is the more fundamental quantity from the standpoint of statistical mechanics.

10.2.1 Water Equivalent

The water equivalent W of a calorimeter is the mass of water that has the same heat capacity as the calorimeter vessel:

$$W = \frac{C_{\text{vessel}}}{c_w} = \frac{m_{\text{v}} c_{\text{v}}}{c_w} \tag{10.9}$$

In an experiment, the effective thermal mass is $(m_{\text{water}} + W)c_w$.

10.2.2 Bomb Calorimetry

A **bomb calorimeter** measures heats of combustion at constant volume. A sample is placed in a sealed steel vessel (the "bomb") surrounded by water in an insulated container. Since $V = \text{const}$, the heat evolved equals the internal energy change:

$$Q_V = \Delta U = -C_{\text{cal}}\,\Delta T_{\text{cal}} \tag{10.10}$$

The heat of combustion at constant pressure (the enthalpy change, more practically relevant) is:

$$\Delta H = \Delta U + \Delta n_g RT \tag{10.11}$$

where Δn_g is the change in moles of gas during combustion.

10.3 Searle's Method: Thermal Conductivity of Good Conductors

10.3.1 Apparatus and Principle

A long metal rod is maintained in steady state: one end heated by steam and the other cooled by a flowing water stream.

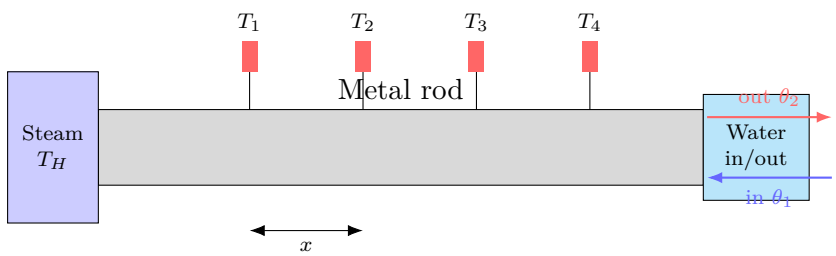

Figure 10.2. Searle's apparatus. Thermocouples at positions $T_1, \ldots, T_4$ monitor the rod; inlet and outlet water temperatures θ_1, θ_2 measure heat removal.

10.3.2 Derivation

At steady state, heat conducted along the rod equals heat removed by the cooling water per unit time:

$$kA\frac{T_1 - T_2}{x} = \dot{m}\, c_w\, (\theta_2 - \theta_1) \tag{10.12}$$

Solving:

$$k = \frac{\dot{m}\, c_w\, (\theta_2 - \theta_1)\, x}{A\,(T_1 - T_2)} \tag{10.13}$$

where $\dot{m}$ is the mass flow rate of water ($\mathrm{kg\,s^{-1}}$).

10.4 Lee's Disc Method: Poor Conductors

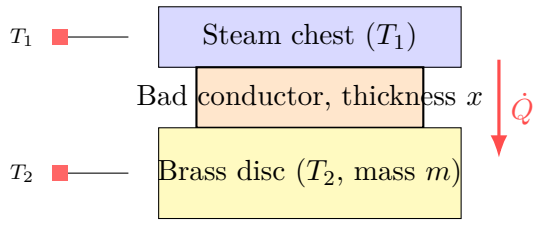

Figure 10.3. Lee's disc apparatus for poor conductors. The bad-conductor sample is sandwiched between the steam chest and the brass disc.

At steady state, heat conducted through the disc equals heat lost by the brass disc to surroundings:

$$kA\frac{T_1 - T_2}{x} = mc\left(\frac{dT}{dt}\right)_{T=T_2} \tag{10.14}$$

The cooling rate $(dT/dt)_{T_2}$ is measured by removing the heat source and recording the rate of temperature fall of the brass disc alone at T_2. Then:

$$k = \frac{mcx}{A(T_1 - T_2)}\left(\frac{dT}{dt}\right)_{T=T_2} \tag{10.15}$$

Insight

Why measure the cooling rate separately? During the experiment the disc loses heat from *all* its exposed surfaces, not just the top face. By removing the sample and measuring free cooling, one automatically accounts for the total surface area through which heat escapes, making the method self-correcting.

10.5 Newton's Law of Cooling and its Limitations

Newton's law states that the rate of heat loss is proportional to the excess temperature over the surroundings:

$$\frac{dT}{dt} = -K(T - T_s) \tag{10.16}$$

10.5.1 Solution and Exponential Decay

Separating variables and integrating:

$$T(t) = T_s + (T_0 - T_s)\, e^{-Kt} \tag{10.17}$$

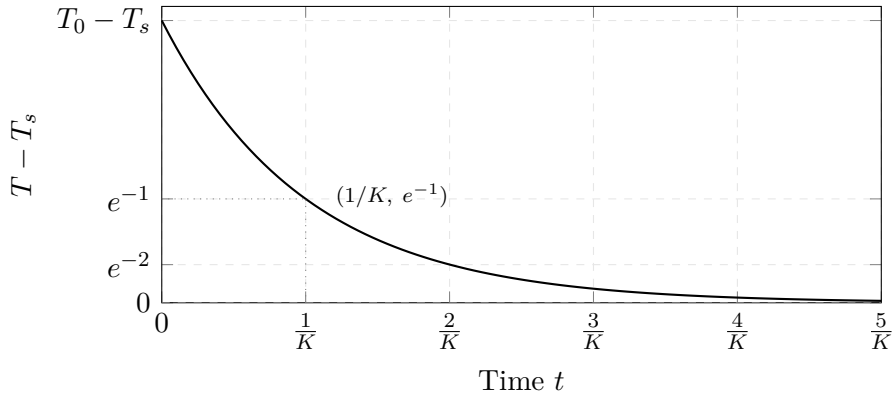

Figure 10.4. Exponential decay of excess temperature under Newton's law of cooling. The time constant $\tau = 1/K$ is the e-folding time.

10.5.2 Stefan–Boltzmann Correction at High Temperatures

Newton's law is only an approximation valid for small excess temperatures. The exact radiative heat loss from a body of emissivity ε, area A, at temperature T to surroundings at T_s is given by the **Stefan–Boltzmann law**:

$$\dot{Q}_{\text{rad}} = \varepsilon\sigma A(T^4 - T_s^4) \tag{10.18}$$

For $T = T_s + \Delta T$ with $\Delta T \ll T_s$:

$$T^4 - T_s^4 \approx 4T_s^3 \Delta T \tag{10.19}$$

so Newton's law is recovered with $K = 4\varepsilon\sigma A T_s^3/mc$. At high temperatures the full T^4 behaviour dominates and exponential cooling no longer holds.

10.6 Thermal Waves and Periodic Heating

A beautiful consequence of equation (10.4) is the existence of **thermal waves**. If the surface of a semi-infinite solid is subjected to a periodic temperature variation $T(0,t) = T_0 \cos(\omega t)$, the solution is:

$$T(x,t) = T_0\, e^{-x/\delta} \cos\!\left(\omega t - \frac{x}{\delta}\right) \tag{10.20}$$

where the **thermal skin depth** is:

$$\delta = \sqrt{\frac{2\alpha_d}{\omega}} \tag{10.21}$$

- The amplitude decays exponentially with depth.
- The wave travels with phase velocity $v = \omega\delta = \sqrt{2\alpha_d\omega}$.
- There is a phase lag of x/δ between surface and depth x.

Insight

Geophysical application. The daily temperature cycle ($\omega = 2\pi/86400\ \mathrm{rad\,s^{-1}}$) penetrates soil (typical $\alpha_d \approx 0.4 \times 10^{-6}\ \mathrm{m^2\,s^{-1}}$) to a skin depth of only $\delta \approx 10$ cm. The annual cycle ($\omega = 2\pi/\mathrm{yr}$) penetrates to $\delta \approx 2$ m — which is why underground cellars maintain nearly constant temperature year-round. These same equations describe eddy-current losses in transformer cores (electromagnetic skin effect) with α_d replaced by the magnetic diffusivity.

10.7 Thermometers: Principles and Types

10.7.1 Liquid-in-Glass Thermometers

Operation relies on differential thermal expansion. For a liquid column of initial volume V_0 in a tube of bore A:

$$V_t = V_0(1 + \gamma_{\mathrm{app}}\,\Delta T), \qquad \gamma_{\mathrm{app}} = \gamma_l - \gamma_g \tag{10.22}$$

The column height:

$$h = \frac{V_0\,\gamma_{\mathrm{app}}}{A}\,(T - T_0) \tag{10.23}$$

Mercury is used above $-39\,^\circ\mathrm{C}$; alcohol-based fluids extend the range to $-100\,^\circ\mathrm{C}$.

10.7.2 Constant-Volume Gas Thermometer (CVGT)

The CVGT is the primary thermometer on which the International Temperature Scale is based, because its readings are connected to the Kelvin scale through the ideal gas law alone:

$$T = T_{\mathrm{ref}}\left(\frac{P}{P_{\mathrm{ref}}}\right) \tag{10.24}$$

The triple point of water (273.16 K exactly) serves as the reference. Real gas corrections:

$$T = \lim_{P\to 0} \frac{273.16\,\mathrm{K}}{P_{\mathrm{tp}}}\,P \tag{10.25}$$

ensure that residual non-ideal behaviour is extrapolated away.

10.8 Electrical Thermometry

10.8.1 Platinum Resistance Thermometer (PRT)

The resistance of platinum follows:

$$R_T = R_0(1 + \alpha T + \beta T^2) \qquad \text{(Callendar–Van Dusen)} \tag{10.26}$$

For the range 0–660 °C, the quadratic term is negligible and the linear form suffices: $R_T \approx R_0(1 + \alpha T)$ with $\alpha \approx 3.85 \times 10^{-3}\,^\circ\mathrm{C}^{-1}$.

The sensitivity $S = dR/dT = R_0\alpha \approx 0.385\,\Omega/^\circ\mathrm{C}$ for a standard $100\,\Omega$ PRT (Pt100).

Four-wire (Kelvin) Measurement

Contact and lead resistances corrupt two-wire measurements. The four-wire method eliminates this:

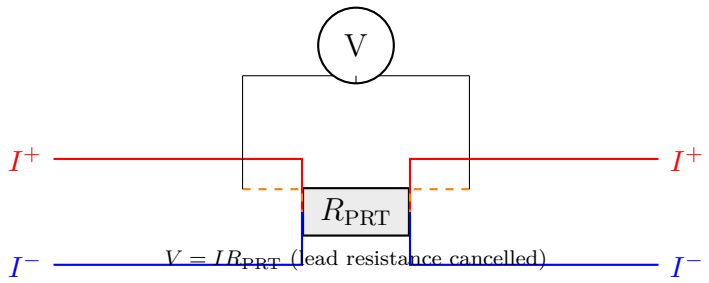

Figure 10.5. Four-wire resistance measurement. Current is supplied through the outer pair; voltage is sensed through the inner pair with negligible current, so lead resistances do not appear in the measurement.

10.8.2 Thermistors

Thermistors are semiconductor devices with a strongly temperature-dependent resistance. The physical basis is thermally activated conduction:

$$\sigma = \sigma_0\, e^{-E_a/k_B T} \quad \Longrightarrow \quad R = R_0\, e^{B(1/T - 1/T_0)} \tag{10.27}$$

where $B = E_a/k_B$ (typically 3000–5000 K).

The temperature coefficient:

$$\alpha = \frac{1}{R}\frac{dR}{dT} = -\frac{B}{T^2} \tag{10.28}$$

is *negative* (NTC thermistor) and large (~ -3 to $-6\%\,^\circ\mathrm{C}^{-1}$), making thermistors far more sensitive than PRTs.

The more accurate **Steinhart–Hart equation**:

$$\frac{1}{T} = A + B\ln R + C(\ln R)^3 \tag{10.29}$$

fits most thermistors to better than 0.01 °C over a 200 °C span.

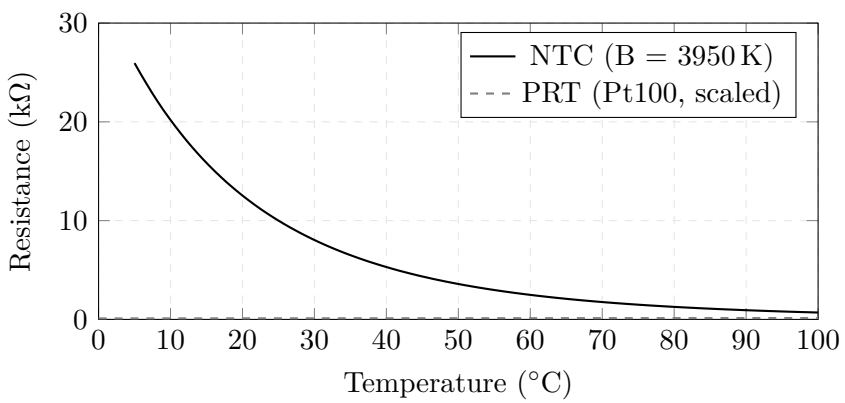

Figure 10.6. Resistance vs temperature for an NTC thermistor ($B = 3950$ K, $R_0 = 10\,\mathrm{k\Omega}$ at 25 °C) compared with a Pt100 PRT (scaled for visibility). The thermistor's exponential sensitivity is immediately apparent.

10.8.3 Noise Thermometry (Johnson–Nyquist)

A resistor R in thermal equilibrium at temperature T generates a voltage noise with spectral density:

$$S_V(f) = 4k_B TR \tag{10.30}$$

(valid for $hf \ll k_B T$, which holds to $f \sim 10^{12}$ Hz at room temperature). The RMS noise voltage in bandwidth B:

$$V_{\text{rms}} = \sqrt{4k_B TRB} \tag{10.31}$$

Measuring S_V with a known R gives T directly — no calibration, no reference material. This is a *primary* thermometer that realises the Kelvin from fundamental constants alone. It is used at NIST and PTB as a key bridge between electrical standards and temperature metrology.

10.9 The Seebeck Effect and Thermocouples

When two dissimilar metals A and B are joined in a loop with junctions at T_1 (hot) and T_2 (cold), an emf drives a current. This is the **Seebeck effect**.

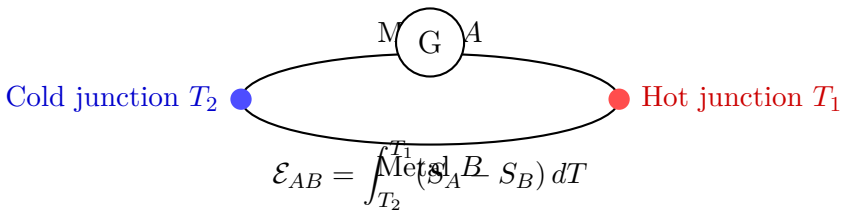

Figure 10.7. Thermocouple circuit. The net emf depends on both the material combination and the temperature difference between junctions.

10.9.1 Seebeck Coefficient

$$S = \frac{d\mathcal{E}}{dT} \qquad (\text{units: V K}^{-1}) \tag{10.32}$$

The net thermocouple emf is:

$$\mathcal{E}_{AB} = \int_{T_2}^{T_1} (S_A - S_B)\, dT \tag{10.33}$$

For small ΔT, the empirical quadratic:

$$\mathcal{E} = a(T - T_{\text{ref}}) + \tfrac{1}{2} b(T - T_{\text{ref}})^2 \tag{10.34}$$

is used for calibration tables.

10.9.2 Microscopic Origin: Mott Formula

From linear response theory (Sommerfeld expansion), the Seebeck coefficient of a metal is:

$$S = -\frac{\pi^2 k_B^2 T}{3e} \left. \frac{d \ln \sigma(E)}{dE} \right|_{E=E_F} \tag{10.35}$$

where $\sigma(E)$ is the energy-dependent conductivity. This is the **Mott formula**. It tells us:

- $S \propto T$ for metals (small at room temperature, $\sim \mu$V/K).
- S is large where $\sigma(E)$ varies steeply near E_F — the design principle for thermoelectric materials.
- Sign of S reflects whether carriers are electrons ($S < 0$) or holes ($S > 0$).

10.9.3 Standard Thermocouple Types

Type	Materials	Range (°C)	Sensitivity (μV/°C)
K	Chromel/Alumel	−200 to 1372	41
J	Iron/Constantan	−40 to 750	51
T	Copper/Constantan	−200 to 350	43
E	Chromel/Constantan	−200 to 900	68
S	Pt/Pt-10%Rh	0 to 1768	10
B	Pt-30%Rh/Pt-6%Rh	100 to 1820	6

10.10 The Peltier Effect

When a current I passes through a junction of metals A and B, heat is absorbed or released at the rate:

$$\dot{Q}_{\text{Peltier}} = \Pi_{AB}\, I = (S_A - S_B) T\, I \tag{10.36}$$

where $\Pi_{AB} = (S_A - S_B)T$ is the **Peltier coefficient** (W/A = V).

Insight

Peltier vs Joule heating. Joule heating $I^2 R$ is *irreversible*, distributed throughout the conductor, and always positive regardless of current direction. Peltier heating ΠI is *reversible*, localised at the junction, and changes sign when I reverses. In a Peltier cooler the two effects compete: the cold junction absorbs Peltier heat but is also heated by Joule dissipation conducted back from the hot junction. The net cooling is:

$$\dot{Q}_{\text{cold}} = \Pi I - \tfrac{1}{2} I^2 R - k\Delta T$$

The factor $\frac{1}{2}$ arises because half the Joule heat flows toward the cold junction.

10.11 The Thomson Effect

A less well-known but equally fundamental effect: when current flows through a *single* homogeneous conductor with a temperature gradient, heat is absorbed or released throughout the bulk at a rate per unit length:

$$\frac{d\dot{Q}}{dx} = \mu_T \, J \, \frac{dT}{dx} \tag{10.37}$$

where μ_T is the **Thomson coefficient** ($\mathrm{V\,K^{-1}}$). This is distinct from Joule heating because it is proportional to J (not J^2) and changes sign with current direction.

10.11.1 Kelvin's Relations

William Thomson (later Lord Kelvin) derived two fundamental relations connecting all three thermoelectric coefficients:

$$\Pi_{AB} = S_{AB} \, T \tag{10.38}$$

$$\frac{dS_{AB}}{dT} = \frac{\mu_{T,A} - \mu_{T,B}}{T} \tag{10.39}$$

These are not merely empirical: equation (10.38) is equivalent to the Onsager reciprocity relation (see Section 10.13) and has been confirmed to extraordinary precision. Equation (10.39) allows the Seebeck coefficient to be computed by integrating Thomson coefficients from 0 to T.

10.12 Figure of Merit and Thermoelectric Efficiency

The efficiency of a thermoelectric device depends on all three transport properties. The key dimensionless parameter is the **thermoelectric figure of merit**:

$$ZT = \frac{S^2 \sigma T}{k} = \frac{S^2 T}{\rho \, k} \tag{10.40}$$

where S is the Seebeck coefficient, σ electrical conductivity, $\rho = 1/\sigma$ electrical resistivity, and k thermal conductivity.

The maximum efficiency of a thermoelectric generator operating between T_H and T_C is:

$$\eta_{\max} = \eta_{\text{Carnot}} \cdot \frac{\sqrt{1 + ZT_{\text{mean}}} - 1}{\sqrt{1 + ZT_{\text{mean}}} + T_C/T_H} \tag{10.41}$$

where $T_{\text{mean}} = (T_H + T_C)/2$. For $ZT \to \infty$, $\eta \to \eta_{\text{Carnot}}$.

Insight

The ZT design problem. A good thermoelectric needs large S (poor conductor behaviour), large σ (good conductor behaviour), and small k (poor thermal conductor behaviour). These requirements conflict: the Wiedemann–Franz law states $k_{\text{el}}/\sigma T = L_0 = 2.44 \times 10^{-8}\,\mathrm{W\,\Omega\,K^{-2}}$ (Lorenz number), so reducing electrical resistivity increases electronic thermal conductivity. Modern strategies to circum-

vent this constraint include phonon engineering in nanostructures, rattler atoms in cage compounds (skutterudites, clathrates), and grain-boundary scattering of phonons in nanocomposites. The best current materials (Bi_2Te_3 alloys, GeTe, half-Heusler compounds) achieve $ZT \approx 1$–3.

10.13 Onsager Reciprocity and the Unified Transport Theory

10.13.1 Generalised Forces and Fluxes

Near equilibrium, Onsager's formalism writes any linear irreversible process as:

$$J_i = \sum_j L_{ij} X_j \tag{10.42}$$

where J_i are fluxes and X_j are thermodynamic forces. The **Onsager reciprocal relations** state:

$$L_{ij} = L_{ji} \tag{10.43}$$

a consequence of time-reversal symmetry of microscopic dynamics.

10.13.2 Thermoelectric Transport Matrix

For coupled electrical and heat transport, the forces are the electric field $\mathbf{E}$ and $-\nabla T/T$, giving:

$$\begin{pmatrix} \mathbf{J} \\ \mathbf{J}_Q \end{pmatrix} = \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ -\nabla T/T \end{pmatrix} \tag{10.44}$$

where $\mathbf{J}$ is electric current density and $\mathbf{J}_Q$ is heat current density. Identifying coefficients with measurable quantities:

$$\sigma = L_{11} \qquad \text{(electrical conductivity)} \tag{10.45}$$

$$S = \frac{L_{12}}{TL_{11}} = \frac{L_{21}}{TL_{11}} \qquad \text{(Seebeck coefficient)} \tag{10.46}$$

$$k = \frac{1}{T}\left(L_{22} - \frac{L_{12}L_{21}}{L_{11}}\right) \qquad \text{(thermal conductivity at } J = 0\text{)} \tag{10.47}$$

The symmetry $L_{12} = L_{21}$ immediately gives Kelvin's first relation $\Pi = ST$.

10.13.3 Physical Interpretation

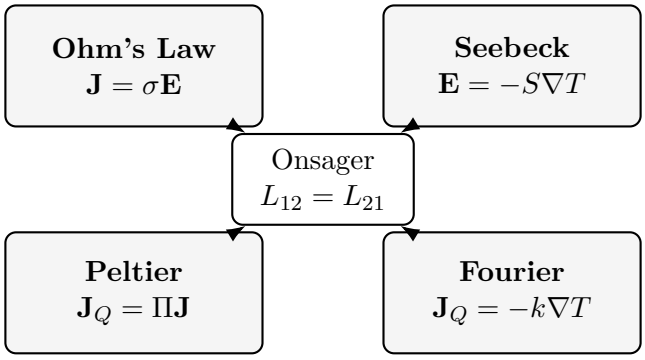

Figure 10.8. The four transport laws as manifestations of the single Onsager matrix. The reciprocity $L_{12} = L_{21}$ (centre) is equivalent to Kelvin's relation $\Pi = ST$.

10.14 The International Temperature Scale (ITS-90)

Temperature is one of the seven SI base quantities, but it cannot be measured directly from first principles in most practical situations. The **International Temperature Scale of 1990 (ITS-90)** defines a set of fixed points and interpolation instruments that reproduce the thermodynamic temperature T to within measurement uncertainty.

Fixed point	T_{90} (K)	t_{90} (°C)
He vapour pressure	3–5	—
Triple point of equilibrium H_2	13.8033	−259.3467
Triple point of Ne	24.5561	−248.5939
Triple point of O_2	54.3584	−218.7916
Triple point of Ar	83.8058	−189.3442
Triple point of Hg	234.3156	−38.8344
Triple point of H_2O	273.16	0.01
Melting point of Ga	302.9146	29.7646
Freezing point of In	429.7485	156.5985
Freezing point of Sn	505.078	231.928
Freezing point of Zn	692.677	419.527
Freezing point of Al	933.473	660.323
Freezing point of Ag	1234.93	961.78
Freezing point of Au	1337.33	1064.18
Freezing point of Cu	1357.77	1084.62

Between fixed points, interpolation is performed by:

- 0.65–5 K: ^{3}He and ^{4}He vapour pressure equations.
- 13.8–961.78 °C: Standard Platinum Resistance Thermometer (SPRT).
- Above 961.78 °C: Planck's radiation law (optical pyrometry) using the silver freezing point as reference.

10.15 Summary

Summary

Key results of this chapter.

- Fourier's law $\mathbf{q} = -k\nabla T$ leads to the heat equation and admits wave-like solutions with thermal skin depth $\delta = \sqrt{2\alpha_d/\omega}$.
- Thermal resistance $R_{\text{th}} = L/kA$ provides an analogy with Ohm's law for composite structures.
- Newton's law of cooling is valid only for small ΔT; the exact radiative law $\propto T^4$ applies at large temperatures.
- Searle's method (good conductors) and Lee's disc (poor conductors) are the standard calorimetric techniques for measuring k.
- PRTs offer stability and linearity; thermistors offer sensitivity; noise thermometry offers a primary measurement from fundamental constants.
- The Seebeck, Peltier, and Thomson effects are three aspects of thermoelectric coupling; Kelvin's relations $\Pi = ST$ and $dS/dT = \mu_T/T$ connect them.
- The figure of merit $ZT = S^2\sigma T/k$ controls thermoelectric efficiency; $ZT \to \infty$ gives Carnot efficiency.
- The Onsager matrix unifies all four transport laws (Ohm, Fourier, Seebeck, Peltier) and the reciprocity $L_{12} = L_{21}$ is Kelvin's first relation.
- ITS-90 realises the Kelvin scale through 15 fixed points and three interpolating instruments.

Chapter 11

Black Body Radiation

11.1 Basic Definitions

Before studying black body radiation, we introduce some important radiative quantities.

Total Energy Density

The total energy density u is defined as the total radiant energy per unit volume inside an enclosure:

$$u = \frac{\text{Total radiant energy}}{\text{Volume}} \tag{11.1}$$

Spectral Energy Density

The spectral energy density u_λ is the energy per unit volume per unit wavelength range:

$$u_\lambda = \frac{du}{d\lambda} \tag{11.2}$$

such that

$$u = \int_0^\infty u_\lambda \, d\lambda \tag{11.3}$$

Emissive Power

The emissive power E is defined as the total radiant energy emitted per unit area per unit time:

$$E = \frac{\text{Energy emitted}}{\text{Area} \times \text{time}} \tag{11.4}$$

Spectral Emissive Power

The spectral emissive power E_λ is the energy emitted per unit area per unit time per unit wavelength:

$$E_\lambda = \frac{dE}{d\lambda} \tag{11.5}$$

and

$$E = \int_0^\infty E_\lambda \, d\lambda \tag{11.6}$$

Emissivity

Emissivity e of a surface is defined as the ratio of emissive power of the surface to that of a perfect black body at the same temperature:

$$e = \frac{E}{E_b} \tag{11.7}$$

Emissivity of a Black Body

For a perfect black body,

$$e = 1 \tag{11.8}$$

A black body is thus defined as an ideal system that absorbs all incident radiation and emits the maximum possible radiation at a given temperature.

Spectral Absorptivity

Spectral absorptivity a_λ is defined as the fraction of incident radiation of wavelength λ that is absorbed:

$$a_\lambda = \frac{\text{Energy absorbed at } \lambda}{\text{Energy incident at } \lambda} \tag{11.9}$$

Reflectivity

Reflectivity r is the fraction of incident radiation that is reflected:

$$r = \frac{\text{Reflected energy}}{\text{Incident energy}} \tag{11.10}$$

Transmissivity

Transmissivity t is the fraction of incident radiation that passes through the material:

$$t = \frac{\text{Transmitted energy}}{\text{Incident energy}} \tag{11.11}$$

11.2 Conservation of Radiant Energy

When radiation is incident on a surface, it can be partially absorbed, reflected, or transmitted. By conservation of energy, the sum of all fractions must be unity:

$$r + a + t = 1 \tag{11.12}$$

For an opaque body, transmission is zero ($t = 0$), hence:

$$r + a = 1 \tag{11.13}$$

For a perfect black body:

$$a = 1, \quad r = 0, \quad t = 0 \tag{11.14}$$

Thus, a black body absorbs all incident radiation and reflects or transmits none.

11.3 Kirchhoff's Law of Thermal Radiation

Historical Note

Gustav Kirchoff Gustav Kirchhoff was a German physicist who made fundamental contributions to thermal radiation and spectroscopy. In 1859, he formulated Kirchhoff's law of thermal radiation, which states that for a body in thermal equilibrium, the ratio of emissive power to absorptive power is the same for all bodies and depends only on temperature and wavelength.
This insight led to the concept of an ideal emitter known as a *black body*, which absorbs and emits radiation perfectly.
Kirchhoff's work laid the foundation for the entire study of black body radiation, making it one of the starting points of modern theoretical physics.

Kirchhoff's law establishes a fundamental relation between the emissive and absorptive properties of a body in thermal equilibrium.

11.3.1 Statement of Kirchhoff's Law

At a given temperature and wavelength, the ratio of the spectral emissive power to the spectral absorptivity is the same for all bodies and is equal to that of a perfect black body. Mathematically,

$$\frac{E_\lambda}{a_\lambda} = E_\lambda^{(b)} \tag{11.15}$$

where E_λ is the spectral emissive power of the body, a_λ is its spectral absorptivity, and $E_\lambda^{(b)}$ is the spectral emissive power of a black body at the same temperature and wavelength.

11.3.2 Implication of the Law

From the above relation, we can write:

$$E_\lambda = a_\lambda \, E_\lambda^{(b)} \tag{11.16}$$

This shows that a good absorber is also a good emitter. In particular, for a perfect black body,

$$a_\lambda = 1 \quad \Rightarrow \quad E_\lambda = E_\lambda^{(b)} \tag{11.17}$$

Thus, a black body is the most efficient emitter of radiation.

11.3.3 Proof

Consider two bodies placed inside an enclosure maintained at a constant temperature T, such that thermal equilibrium is established. Let radiation of wavelength λ be incident on both bodies.

Let:

- a_λ be the spectral absorptivity,
- E_λ be the spectral emissive power.

In thermal equilibrium, the energy absorbed must be equal to the energy emitted; otherwise, the temperature would change.

If the incident radiation has spectral energy density u_λ, then the energy absorbed per unit time is proportional to $a_\lambda u_\lambda$, and the energy emitted per unit time is proportional to E_λ.

Thus, for equilibrium:

$$E_\lambda \propto a_\lambda u_\lambda \tag{11.18}$$

Since u_λ is the same for all bodies in the enclosure at equilibrium, we obtain:

$$\frac{E_\lambda}{a_\lambda} = \text{constant} \tag{11.19}$$

This constant depends only on temperature and wavelength, and is identified as the spectral emissive power of a black body:

$$\frac{E_\lambda}{a_\lambda} = E_\lambda^{(b)} \tag{11.20}$$

11.3.4 Black Body

- Kirchhoff's law shows that emission and absorption are not independent properties.
- It provides a method to define black body radiation independent of the nature of materials.
- It implies that no body can emit more radiation than a black body at the same temperature.
- It forms the basis for the development of Planck's radiation law.

All bodies at a temperature above absolute zero emit electromagnetic radiation due to the thermal motion of their charged particles. This radiation, known as **thermal radiation**, does not require a material medium for its propagation and can occur even in vacuum.

The nature and intensity of radiation emitted by a body depend on its temperature and surface properties. The idealized concept used to study thermal radiation is that of a **blackbody**.

A **blackbody** is an ideal body that absorbs all the radiation incident upon it, regardless of wavelength and direction. Since it absorbs perfectly, it is also the most efficient emitter of radiation.

In practice, a blackbody can be approximated by a cavity with a small hole. Radiation entering the hole undergoes multiple reflections inside the cavity and is almost completely absorbed.

The radiation emitted by a blackbody in thermal equilibrium is called **blackbody radiation**. The distribution of energy with wavelength depends only on the temperature of the blackbody and not on its material composition.

11.4 Wien's Displacement Law

Historical Note

Wilhelm Wien's Legacy Wilhelm Wien was a German physicist known for his pioneering work on thermal radiation. He formulated Wien's displacement law, which describes how the peak wavelength of black body radiation shifts with temperature.
Wien was awarded the Nobel Prize in Physics in 1911 for his discoveries regarding the laws governing the radiation of heat.

Wien's displacement law describes how the wavelength corresponding to the maximum emission of a black body radiation spectrum depends on temperature.

The wavelength λ_{max} at which the spectral emissive power of a black body is maximum is inversely proportional to its absolute temperature T:

$$\lambda_{\text{max}} T = b \tag{11.21}$$

where b is Wien's displacement constant, given by

$$b \approx 2.898 \times 10^{-3}\,\text{m K} \tag{11.22}$$

As the temperature of a black body increases, the peak of its radiation curve shifts toward shorter wavelengths. This implies:

- At low temperatures, radiation is mainly in the infrared region.
- At higher temperatures, radiation shifts toward the visible region.
- At very high temperatures, radiation extends into the ultraviolet region.

Thus, hotter bodies appear to emit radiation of shorter wavelengths.

- It explains the change in color of hot objects (e.g., red → yellow → white).
- It is used to estimate the surface temperature of stars.
- It confirms that black body radiation depends only on temperature.

11.5 Stefan–Boltzmann Law

Historical Note

J. Stefan and His student Josef Stefan, an Austrian physicist, experimentally discovered that the total energy radiated by a black body is proportional to the fourth power of its temperature.
His student, Ludwig Boltzmann, later provided the theoretical derivation of this law using thermodynamics and statistical mechanics.
Boltzmann's work was crucial in establishing the connection between microscopic atomic behavior and macroscopic physical laws, making him one of the founders of statistical mechanics.

The Stefan–Boltzmann law gives the total energy emitted per unit area of a black body across all wavelengths as a function of its temperature.

The total emissive power of a perfect black body is directly proportional to the fourth power of its absolute temperature:

$$E = \sigma T^4 \tag{11.23}$$

where σ is the Stefan–Boltzmann constant, given by

$$\sigma = 5.67 \times 10^{-8}\,\mathrm{W\ m^{-2}K^{-4}} \tag{11.24}$$

For a Real Body

For a real surface, the emissive power is less than that of a black body and is given by:

$$E = e\,\sigma T^4 \tag{11.25}$$

where e is the emissivity of the surface.

Net Radiative Heat Loss

If a body at temperature T is surrounded by an enclosure at temperature T_0, the net rate of heat loss per unit area is:

$$E_{\text{net}} = \sigma\left(T^4 - T_0^4\right) \tag{11.26}$$

For a real body:

$$E_{\text{net}} = e\,\sigma\left(T^4 - T_0^4\right) \tag{11.27}$$

Applications

- Estimation of energy output of stars.
- Calculation of radiative heat loss in engineering systems.
- Determination of surface temperatures in thermal physics.

11.6 Rayleigh–Jeans Law

Historical Note

Lord Rayleigh and Sir James Jeans Lord Rayleigh (John William Strutt) and Sir James Jeans were British physicists who developed the Rayleigh–Jeans law based on classical physics.
Rayleigh was awarded the Nobel Prize in Physics in 1904 for his discovery of argon.
Jeans made significant contributions to astrophysics and the theory of radiation.

11.6.1 Derivation

The Rayleigh–Jeans law is a classical result that describes the spectral distribution of black body radiation at long wavelengths.

The derivation is based on the following assumptions:

- Radiation inside a cavity is treated as standing electromagnetic waves.
- The number of modes in the frequency range ν to $\nu + d\nu$ is:

$$g(\nu)\, d\nu = \frac{8\pi\nu^2}{c^3}\, d\nu \tag{11.28}$$

- By the equipartition theorem, each mode has an average energy ($\frac{1}{2}k_BT$ per degree of freedom, and Electromagnetic wave has degree of freedom = 2):

$$\langle E \rangle = kT \tag{11.29}$$

Thus, the spectral energy density becomes:

$$\boxed{u_\nu d\nu = g(\nu)\, \langle E \rangle d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu} \tag{11.30}$$

Limitations

- The law agrees well with experimental results at long wavelengths (low frequencies).
- However, it fails completely at short wavelengths (high frequencies).
- As $\lambda \to 0$, the energy density $u_\lambda \to \infty$, which is physically unacceptable.

11.6.2 Ultraviolet Catastrophe

The failure of the Rayleigh–Jeans law at short wavelengths is known as the **ultraviolet catastrophe**.

According to classical theory:

$$u_\lambda \propto \frac{1}{\lambda^4} \tag{11.31}$$

Thus, as $\lambda \to 0$ (ultraviolet region), the energy density tends to infinity:

$$u_\lambda \to \infty \tag{11.32}$$

This implies that a black body would emit infinite energy in the ultraviolet region, which contradicts experimental observations.

- The ultraviolet catastrophe revealed the failure of classical physics in explaining black body radiation.
- It indicated the need for a new theoretical framework.
- This problem was resolved by Planck, who introduced the concept of energy quantization.

11.7 Planck's Radiation Law

Historical Note

Planck's Legacy
Max Planck was a German physicist and the founder of quantum theory. In 1900, he proposed that energy is not emitted continuously but in discrete packets called quanta.
Using this revolutionary idea, Planck successfully derived the correct formula for black body radiation, resolving the ultraviolet catastrophe.
He was awarded the Nobel Prize in Physics in 1918 for his discovery of energy quanta. Planck's work marked the beginning of modern physics and fundamentally changed our understanding of nature.

11.7.1 Planck's Quantum Hypothesis

Planck made the following fundamental assumptions:

- Energy is not emitted or absorbed continuously, but in discrete packets called **quanta**.
- The energy of each quantum is proportional to the frequency of radiation:

$$E = h\nu \tag{11.33}$$

  where h is Planck's constant.
- The oscillators in the walls of a black body can have only discrete energy values:

$$E_n = nh\nu \quad (n = 0, 1, 2, 3, \dots) \tag{11.34}$$

11.7.2 Average Energy of an Oscillator

According to statistical mechanics, the probability of an oscillator having energy E_n is given by the Boltzmann distribution:

$$P_n \propto e^{-E_n/kT} \tag{11.35}$$

Thus, the average energy is:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} E_n e^{-E_n/kT}}{\sum_{n=0}^{\infty} e^{-E_n/kT}} \tag{11.36}$$

Substituting $E_n = nh\nu$:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} nh\nu\, e^{-nh\nu/kT}}{\sum_{n=0}^{\infty} e^{-nh\nu/kT}} \tag{11.37}$$

Let

$$x = e^{-h\nu/kT} \tag{11.38}$$

Then,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad \sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2} \tag{11.39}$$

Using these results:

$$\langle E \rangle = h\nu \frac{x}{1-x} \tag{11.40}$$

Substituting back $x = e^{-h\nu/kT}$:

$$\langle E \rangle = \frac{h\nu}{e^{h\nu/kT} - 1} \tag{11.41}$$

11.7.3 Planck's Radiation Law

The number of electromagnetic modes per unit volume in the frequency range ν to $\nu + d\nu$ is:

$$g(\nu)\, d\nu = \frac{8\pi\nu^2}{c^3}\, d\nu \tag{11.42}$$

Thus, the spectral energy density is:

$$u_\nu d\nu = g(\nu)\, \langle E \rangle d\nu \tag{11.43}$$

Substituting:

$$u_\nu d\nu = \frac{8\pi\nu^2}{c^3} \cdot \frac{h\nu}{e^{h\nu/kT} - 1} d\nu \tag{11.44}$$

Hence,

$$\boxed{u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} d\nu} \tag{11.45}$$

This is Planck's radiation law in frequency form.
Using the relation $\nu = \frac{c}{\lambda}$, we obtain:

$$\boxed{u_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} d\lambda} \tag{11.46}$$

- It successfully explains the entire black body spectrum.
- It resolves the ultraviolet catastrophe.
- It marks the origin of quantum theory.
- It shows that energy exchange is quantized.

11.7.4 Derivation of Classical Laws from Planck's Radiation Law

Planck's radiation law successfully explains black body radiation. It also reduces to previously known laws under appropriate limits. In this section, we derive Wien's law, Rayleigh–Jeans law, and Stefan–Boltzmann law from Planck's formula.

Rayleigh–Jeans Law (Low Frequency Limit)

For low frequencies ($h\nu \ll kT$), we use the approximation:

$$e^{h\nu/kT} \approx 1 + \frac{h\nu}{kT} \tag{11.47}$$

Thus,

$$e^{h\nu/kT} - 1 \approx \frac{h\nu}{kT} \tag{11.48}$$

Substituting into Planck's law:

$$u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \cdot \frac{kT}{h\nu} d\nu \tag{11.49}$$

Simplifying:

$$u_\nu d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu \tag{11.50}$$

This is the Rayleigh–Jeans law.

Wien's Law (High Frequency Limit)

For high frequencies ($h\nu \gg kT$), we use:

$$e^{h\nu/kT} \gg 1 \quad \Rightarrow \quad e^{h\nu/kT} - 1 \approx e^{h\nu/kT} \tag{11.51}$$

Substituting into Planck's law:

$$u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} e^{-h\nu/kT} d\nu \tag{11.52}$$

This is Wien's radiation law:

$$u_\nu d\nu \propto \nu^3 e^{-h\nu/kT} d\nu \tag{11.53}$$

Stefan–Boltzmann Law

To derive Stefan's law, we integrate Planck's distribution over all frequencies:

$$u = \int_0^\infty u_\nu \, d\nu \tag{11.54}$$

Substituting Planck's law:

$$u = \int_0^\infty \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} \, d\nu \tag{11.55}$$

Let

$$x = \frac{h\nu}{kT} \quad \Rightarrow \quad d\nu = \frac{kT}{h} dx \tag{11.56}$$

Then,

$$u = \frac{8\pi (kT)^4}{h^3 c^3} \int_0^\infty \frac{x^3}{e^x - 1} \, dx \tag{11.57}$$

Using the standard result:

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15} \tag{11.58}$$

Thus,

$$u = \frac{8\pi^5 k^4}{15 h^3 c^3} T^4 \tag{11.59}$$

Since emissive power $E = \frac{c}{4} u$, we get:

$$E = \sigma T^4 \tag{11.60}$$

where

$$\sigma = \frac{2\pi^5 k^4}{15 h^3 c^2} \tag{11.61}$$

This is the Stefan–Boltzmann law.

Appendix A

The Polytropic Process

A polytropic process is a general thermodynamic path followed by an ideal gas that satisfies the mathematical relationship:

$$PV^x = \text{constant} \tag{A.1}$$

where x is the polytropic index.

By varying the value of x, we recover the four fundamental processes:

Index (x)	**Process Type**	**Mathematical Result**	**Physical Meaning**
$x = 0$	Isobaric	$P = \text{const}$	Constant Pressure
$x = 1$	Isothermal	$PV = \text{const}$	Constant Temperature
$x = \gamma$	Adiabatic	$PV^\gamma = \text{const}$	No heat exchange ($Q = 0$)
$x = \infty$	Isochoric	$V = \text{const}$	Constant Volume

Table A.1. Polytropic process

Appendix B

Equations of State

"The equation of state is the fingerprint of a substance — it encodes everything thermodynamics needs to know about how matter responds to pressure, volume, and temperature."

An **equation of state** (EOS) is a mathematical relation connecting the thermodynamic state variables of a system — most commonly pressure P, molar volume V_m, and temperature T. The simplest example, the ideal gas law $PV = nRT$, assumes point-like, non-interacting molecules. Real substances deviate from this picture in two important ways: molecules occupy finite volume, and they exert attractive (and at very short range, repulsive) forces on each other. Every equation of state discussed in this appendix is, in essence, an attempt to account for these two effects with increasing sophistication.

The equations are arranged in a natural progression — from gentle empirical corrections to the ideal gas, through cubic equations widely used in chemical engineering, all the way to equations that describe quantum matter such as photon gases and degenerate electrons.

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1. The Virial Equation of State

The physical picture

The ideal gas law can be derived by assuming that gas molecules interact with each other not at all. The virial equation relaxes this assumption by treating intermolecular interactions as a perturbation, expanded in powers of the density $1/V_m$. At low densities (large V_m) only pairs of molecules encounter each other, at moderate densities triplets become important, and so on. The equation is therefore exact in principle — it is simply a power series, and including enough terms reproduces any real-gas behaviour.

Equation

$$\frac{PV_m}{RT} = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \cdots \tag{B.1}$$

The dimensionless ratio $Z = PV_m/RT$ is called the *compressibility factor*. For an ideal gas $Z = 1$ everywhere; real gases have $Z < 1$ at moderate pressures (attraction dominates) and $Z > 1$ at high pressures (repulsion dominates).

Parameters

- $B(T)$: **second virial coefficient** — encodes pairwise interaction. It is negative at low temperatures (attraction wins) and becomes positive at high temperatures.
- $C(T)$: **third virial coefficient** — encodes three-body interactions; significant only at high densities.

Applicability

The virial expansion works best for gases at low to moderate pressures. It is the preferred equation when one requires a theoretically rigorous connection between the EOS and intermolecular potentials — $B(T)$ can be computed directly from the pair potential via statistical mechanics.

2. The van der Waals Equation

The physical picture

Proposed by Johannes Diderik van der Waals in 1873, this was the first equation to incorporate both finite molecular size and intermolecular attraction in a physically motivated way. The pressure correction $-a/V_m^2$ reflects the inward pull experienced by a molecule near the boundary of the gas; the volume correction $-b$ excludes the space already occupied by the molecules themselves.

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT \tag{B.2}$$

Parameters

- a: **attraction parameter** ($\mathrm{Pa\,m^6\,mol^{-2}}$) — larger for strongly polar or polarisable molecules.
- b: **co-volume** ($\mathrm{m^3\,mol^{-1}}$) — roughly four times the actual volume of a single molecule.

Applicability

Despite its simplicity, van der Waals' equation predicts the existence of a *critical point*, the liquid–gas phase transition, and the law of corresponding states. It is the conceptual foundation on which all cubic equations of state are built.

3. The Berthelot Equation

The physical picture

Daniel Berthelot observed that the attractive forces between molecules weaken as temperature rises. He introduced a $1/T$ factor into the attractive term of the van der Waals equation, making the correction temperature-aware. This modest change significantly improves accuracy at elevated temperatures where the original van der Waals equation over-estimates the effect of attraction.

$$P = \frac{RT}{V_m - b} - \frac{a}{T\,V_m^2} \tag{B.3}$$

Parameters

- a: modified attraction constant (units differ from van der Waals because of the extra $1/T$).
- b: excluded volume, same physical meaning as in van der Waals.

Applicability

Useful for gases at moderate pressures and temperatures above the critical temperature. It performs better than van der Waals for polar gases and is often used in engineering estimates.

4. The Clausius Equation

The physical picture

Rudolf Clausius introduced a third parameter c to account for the fact that the attractive interaction between a molecule and its neighbours depends not only on the density of the gas but also on the local geometry of molecular clustering. The shifted denominator $(V_m + c)^2$ provides an extra degree of freedom to fit experimental data more accurately.

$$P = \frac{RT}{V_m - b} - \frac{a}{T\,(V_m + c)^2} \tag{B.4}$$

Parameters

- a: temperature-scaled attraction constant.
- b: excluded volume.
- c: cluster-correction parameter; when $c = 0$ the equation reduces to Berthelot's form.

5. The Redlich–Kwong Equation

The physical picture

Proposed by Otto Redlich and Joseph Nong Shong Kwong in 1949, this cubic equation substantially improves upon van der Waals by replacing the $1/V_m^2$ attractive term with $1/[\sqrt{T}\,V_m(V_m+b)]$. The $\sqrt{T}$ dependence captures the softening of intermolecular forces with temperature, while the product $V_m(V_m+b)$ in the denominator better models the geometry of molecular collisions at high densities.

$$P = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T}\,V_m(V_m+b)} \tag{B.5}$$

Parameters

- $a = 0.42748\,R^2T_c^{2.5}/P_c$
- $b = 0.08664\,RT_c/P_c$

where T_c and P_c are the critical temperature and pressure, making the equation entirely predictive once the critical constants are known.

Applicability

Widely used for non-polar gases and hydrocarbons at pressures up to several hundred atmospheres. It is the ancestor of the SRK and Peng–Robinson equations.

6. The Soave–Redlich–Kwong (SRK) Equation

The physical picture

Giorgio Soave's 1972 modification replaces the $1/\sqrt{T}$ factor with a general temperature-correction function $\alpha(T)$ fitted to vapour pressure data. This single change dramatically extends the accuracy of the Redlich–Kwong equation to subcritical temperatures and phase equilibrium calculations — tasks for which the original RK equation performs poorly.

$$P = \frac{RT}{V_m - b} - \frac{a\,\alpha(T)}{V_m(V_m+b)} \tag{B.6}$$

where

$$\alpha(T) = \left[1 + m\left(1 - \sqrt{T/T_c}\right)\right]^2, \qquad m = 0.480 + 1.574\,\omega - 0.176\,\omega^2$$

and ω is the *acentric factor* — a measure of how non-spherical and polar a molecule is.

Applicability

The SRK equation is the workhorse of the petrochemical industry. It is reliable for vapour–liquid equilibria of non-polar and mildly polar systems.

7. The Peng–Robinson Equation

The physical picture

Ding-Yu Peng and Donald Robinson (1976) noticed that the SRK equation systematically under-predicts liquid densities near the critical point. Their modification changes the denominator of the attractive term from $V_m(V_m + b)$ to $V_m(V_m + b) + b(V_m - b)$, which shifts the predicted critical compressibility factor Z_c closer to the experimentally observed average value of 0.307.

$$P = \frac{RT}{V_m - b} - \frac{a\,\alpha(T)}{V_m(V_m + b) + b(V_m - b)} \tag{B.7}$$

where $\alpha(T)$ has the same functional form as SRK but with

$$m = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2.$$

Applicability

The Peng–Robinson equation is the most widely used cubic EOS in chemical and petroleum engineering today. It gives excellent results for hydrocarbon systems and is suitable for both vapour and liquid phases, including near-critical conditions.

8. The Hard-Sphere Equation (Carnahan–Starling)

The physical picture

In the hard-sphere model, molecules are treated as rigid billiard balls: they interact only through perfectly elastic collisions and experience no attractive forces whatsoever. Despite its simplicity, the model captures the repulsive core of real interactions and provides the reference equation for building more elaborate theories. The Carnahan–Starling equation is the most accurate closed-form expression for the hard-sphere compressibility factor.

$$Z = \frac{PV_m}{RT} = \frac{1 + \eta + \eta^2 - \eta^3}{(1 - \eta)^3} \tag{B.8}$$

Parameters

- $\eta = \frac{\pi}{6}\rho\sigma^3$: **packing fraction** (volume fraction occupied by the spheres), where σ is the hard-sphere diameter and ρ the number density.
- $\eta = 0$ corresponds to a vacuum; the random close-packing limit is $\eta \approx 0.64$.

Applicability

Used as the repulsive reference term in perturbation theories of liquids, colloidal suspensions, and granular media. It is also a good approximation for noble gases at high pressures where attractive forces become secondary to excluded-volume effects.

9. The Benedict–Webb–Rubin (BWR) Equation

The physical picture

For applications requiring high accuracy over a wide range of temperatures and pressures — such as the design of liquefaction plants and cryogenic storage — the simple cubic equations are insufficient. The BWR equation, proposed in 1940, is a deliberately empirical multi-parameter relation that sacrifices theoretical elegance for practical accuracy. It was initially fitted to hydrocarbons and has since been extended (e.g., the modified BWR of Starling) to many industrially important fluids.

$$P = RT\rho + (B_0RT - A_0 - C_0/T^2)\rho^2 + (bRT - a)\rho^3 + a\alpha\,\rho^6 + \frac{c\,\rho^3}{T^2}(1+\gamma\rho^2)\,e^{-\gamma\rho^2} \quad \text{(B.9)}$$

where $\rho = 1/V_m$ is the molar density.

Parameters

A_0, B_0, C_0, a, b, c, α, γ — eight empirical constants fitted to experimental P-V-T data for each substance.

Applicability

Natural gas processing, refrigeration cycles, and any situation where accuracy over many decades of pressure and temperature is required. The exponential term is specifically designed to reproduce the rapid density increase at the onset of condensation.

10. Radiation Equation of State

The physical picture

Electromagnetic radiation in thermal equilibrium with matter — a *photon gas* — also exerts pressure. Unlike ordinary gases, a photon gas has no conserved particle number (photons are freely created and destroyed) and its pressure depends only on temperature. This equation follows directly from Maxwell's electromagnetic theory: the radiation pressure equals one-third of the energy density.

$$P_{\text{rad}} = \frac{1}{3}\,a\,T^4 \quad \text{(B.10)}$$

Parameter

- $a = 4\sigma/c = 7.566 \times 10^{-16}\ \mathrm{J\,m^{-3}\,K^{-4}}$: radiation (Stefan) constant, where σ is the Stefan–Boltzmann constant and c the speed of light.

Applicability

Radiation pressure is negligible in everyday conditions but becomes the dominant contribution to total pressure in the interiors of massive stars (where $T \sim 10^7\,\mathrm{K}$) and in the early universe. It is a central equation in stellar structure theory and cosmology.

11. Degenerate Fermi Gas

The physical picture

Fermions — particles with half-integer spin such as electrons, protons, and neutrons — obey the Pauli exclusion principle: no two fermions may occupy the same quantum state. At sufficiently low temperatures (or equivalently, at sufficiently high densities), this purely quantum mechanical effect generates a *degeneracy pressure* that is entirely independent of temperature. This is in stark contrast to classical gases, where pressure vanishes as $T \to 0$.

The degenerate limit is reached when $k_B T \ll E_F$, where $E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$ is the Fermi energy and n is the number density.

$$P = K\,\rho^{5/3} \tag{B.11}$$

Parameter

- $K = \dfrac{\hbar^2}{5m}\left(\dfrac{3\pi^2}{m_u}\right)^{2/3}$ for electrons of mass m, where m_u is the atomic mass unit.

Applicability

This equation governs the pressure inside **white dwarf stars**, where the outward degeneracy pressure of electrons balances gravitational collapse. It also applies to conduction electrons in metals at room temperature (which are in the degenerate limit because the Fermi energy of metals is of order 1–10 eV, far above $k_B T \approx 0.025$ eV at room temperature).

12. Bose Gas Equation of State

The physical picture

Bosons — particles with integer spin, such as photons, ^{4}He atoms, and Cooper pairs — are not subject to the Pauli exclusion principle. Instead, they have an inherent tendency to *cluster* in the same quantum state, a purely statistical effect with no classical analogue. Below a critical temperature T_c a macroscopic fraction of bosons condenses into the ground state — *Bose–Einstein condensation* (BEC) — and the equation of state changes character dramatically.

$$P = \frac{k_B T}{\lambda^3}\, g_{5/2}(z) \tag{B.12}$$

Parameters

- $\lambda = \sqrt{2\pi\hbar^2/mk_B T}$: **thermal de Broglie wavelength** — the quantum length scale below which wave-like behaviour of particles becomes important.
- $g_{5/2}(z) = \sum_{k=1}^{\infty} z^k/k^{5/2}$: **Bose–Einstein function** (polylogarithm of order 5/2).

- $z = e^{\mu/k_B T}$: **fugacity**, where μ is the chemical potential. For an ideal Bose gas $\mu \leq 0$, so $0 < z \leq 1$.

Applicability

Above T_c: describes dilute quantum gases such as ultra-cold alkali metal vapours (rubidium, sodium) in magnetic traps, where BEC was first achieved experimentally in 1995 (Cornell, Wieman, Ketterle — Nobel Prize 2001). The ^{4}He superfluid transition at 2.17 K is the oldest known bosonic phase transition in a real substance.

Summary

A hierarchy of equations of state. Moving through this appendix, each equation adds a layer of physical reality: volume exclusion, temperature-dependent attraction, empirical flexibility, and finally quantum statistics. No single EOS is universally best — the choice always depends on the substance, the thermodynamic conditions, and the accuracy required. Understanding *why* each correction is introduced is more valuable than memorising the formula itself.

Appendix C

Spherical Coordinate System

Definition of Spherical Coordinates

In the spherical coordinate system, the position of a point in space is specified by the coordinates (r, θ, ϕ), where:

- r is the radial distance from the origin,
- θ is the polar angle measured from the positive z-axis $(0 \leq \theta \leq \pi)$,
- ϕ is the azimuthal angle measured in the xy-plane from the positive x-axis $(0 \leq \phi < 2\pi)$.

The transformation between Cartesian and spherical coordinates is given by

$$x = r \sin\theta \cos\phi, \tag{C.1}$$

$$y = r \sin\theta \sin\phi, \tag{C.2}$$

$$z = r \cos\theta. \tag{C.3}$$

Conversely,

$$r = \sqrt{x^2 + y^2 + z^2}, \tag{C.4}$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right), \tag{C.5}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right). \tag{C.6}$$

Unit Vectors in Spherical Coordinates

The spherical coordinate system uses three mutually orthogonal unit vectors:

- $\hat{\mathbf{r}}$ in the radial direction,
- $\hat{\boldsymbol{\theta}}$ in the direction of increasing θ,
- $\hat{\boldsymbol{\phi}}$ in the direction of increasing ϕ.

These unit vectors vary with position and are related to the Cartesian unit vectors by

$$\hat{\mathbf{r}} = \sin\theta \cos\phi\, \hat{\mathbf{x}} + \sin\theta \sin\phi\, \hat{\mathbf{y}} + \cos\theta\, \hat{\mathbf{z}}, \tag{C.7}$$

$$\hat{\boldsymbol{\theta}} = \cos\theta \cos\phi\, \hat{\mathbf{x}} + \cos\theta \sin\phi\, \hat{\mathbf{y}} - \sin\theta\, \hat{\mathbf{z}}, \tag{C.8}$$

$$\hat{\boldsymbol{\phi}} = -\sin\phi\, \hat{\mathbf{x}} + \cos\phi\, \hat{\mathbf{y}}. \tag{C.9}$$

Differential Length, Area, and Volume Elements

In spherical coordinates, the differential displacement vector is

$$d\mathbf{l} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}. \tag{C.10}$$

The corresponding differential area elements are

$$d\mathbf{a}_r = r^2\sin\theta\,d\theta\,d\phi\,\hat{\mathbf{r}}, \tag{C.11}$$

$$d\mathbf{a}_\theta = r\sin\theta\,dr\,d\phi\,\hat{\boldsymbol{\theta}}, \tag{C.12}$$

$$d\mathbf{a}_\phi = r\,dr\,d\theta\,\hat{\boldsymbol{\phi}}. \tag{C.13}$$

The differential volume element is

$$d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi. \tag{C.14}$$

Gradient, Divergence, and Curl in Spherical Coordinates

For a scalar field $\phi(r,\theta,\phi)$, the gradient is

$$\boldsymbol{\nabla}\phi = \hat{\mathbf{r}}\frac{\partial\phi}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial\phi}{\partial\theta} + \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\phi}. \tag{C.15}$$

For a vector field $\mathbf{A} = A_r\hat{\mathbf{r}} + A_\theta\hat{\boldsymbol{\theta}} + A_\phi\hat{\boldsymbol{\phi}}$, the divergence is

$$\boldsymbol{\nabla}\cdot\mathbf{A} = \frac{1}{r^2}\frac{\partial}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}. \tag{C.16}$$

The curl of $\mathbf{A}$ is given by

$$\boldsymbol{\nabla}\times\mathbf{A} = \frac{1}{r\sin\theta}\begin{vmatrix} \hat{\mathbf{r}} & r\hat{\boldsymbol{\theta}} & r\sin\theta\,\hat{\boldsymbol{\phi}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}. \tag{C.17}$$

Laplacian in Spherical Coordinates

The Laplacian of a scalar field $\phi(r,\theta,\phi)$ in spherical coordinates is

$$\nabla^2\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\phi^2}. \tag{C.18}$$

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